

Document No. RDSO/SPN/257/2025	Version. 2.0	Date Effective: 31.12.2025
Document Title : Specification For Remote Diagnostic and Predictive Maintenance System (RDPMS).		

RESEARCH DESIGNS & STANDARDS ORGANISATION
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Specification For Remote Diagnostic and Predictive Maintenance System (RDPMS)

Document No. RDSO/SPN/257/2025, Version. 2.0

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FOREWORD:

This Specification requires reference to the latest version of following Specifications/ Documents: –

1.	IRS: S-24	Electric Point Machine
2.	RDSO SPN 153	Subsidiary LED Signal
3.	RDSO SPN 199	Main LED Signal
4.	RDSO SPN 146	Audio Frequency Track Circuit
5.	IRS S 105	Block Proving by Axle Counter using UFSBI
6.	RDSO SPN 175	Solid State Block Proving by Digital Axle Counter
7.	RDSO SPN 176	Multi-section Digital Axle Counter
8.	RDSO/SPN/177	Single section Digital Axle Counter
9.	IRS: S-99	Data Logger System.
10.	RDSO/SPN/165/2023	Integrated Power Supply
11.	IRS: :S 63	PVC insulated underground unscreened cable for Railway signaling
12.	IRS: S 76	PVC insulated indoor cables for Railway signaling
13.	RDSO/SPN/144	Safety and reliability requirement of electronic signalling equipment.
14.	RDSO/SPN/197	Code of Practice for Earthing and Bonding System for Signalling Equipments.
15.	IRS:S 89/2013	Track Feed battery charger
16.	BRS:938-A for QT2 & BRS:939-A for QTA2	Track Relay
17.	IRS:S 88 2004	Lead Acid Battery
18.	RDSO/SPN/208/2012	Electric Lifting Barrier(ELB)
19.	RDSO/SPN/TL/23/99, SMPS Power Plant	Telecom Charger of OFC
20.	EN 50121*	Railway Applications - Characteristics of Railway Systems that affect EMC behaviour
21.	ISO 9001	Quality Systems- model for quality assurance in design, development, production, installation and serving.
22.	IEC 60688	Electrical measuring transducers for converting a.c. and d.c. electrical quantities to analogue or digital signals.
23.	IEC61326	Electrical equipment for measurement, control and laboratory use –EMC requirements
24.	TEC 31318:2025 or latest	Telecommunication Engineering Center Code of practice for securing Consumer Internet of Things (IoT)
25.	IEC 60603	Connectors for electronic equipment - Part 7: Detail Specification for 8-way, unshielded, free and fixed connectors.
26.	IEC 61754-20	Fiber optic interconnecting devices and passive components - Fiber optic connector interfaces - Part 20: Type LC connector family.
27.	ISO/IEC JTC 1/SC 42	AI/ML references

Note:

1. * Equivalent Recognized International standards may also be agreed subject to their acceptability in a foreign Railway for same application. The supplier shall submit a copy of the same for verification.
2. Whenever, reference to any Specification appears in this document, it shall be taken as a reference to the latest version of that Specification.

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ABBREVIATIONS :

SNO.	Abbreviation	Expanded Form
1.	IoT	Internet of Things
2.	AI	Artificial Intelligence
3.	ML	Machine Learning
4.	ABS	Automatic Block Signalling
5.	IBS	Intermediate Block Signalling
6.	I/O	Input/output
7.	CCIP	Control Cum Indication Panel
8.	MTBF	Mean Time Between Failure
9.	MTBWSF	Mean Time Between Wrong Side Failure
10.	MTTR	Mean Time to Repair
11.	PCB	Printed Circuit Board
12.	RDSO	Research Designs and Standards Organisation
13.	OFC	Optical Fiber Cable
14.	IPS	Integrated Power Supply
15.	AFTC	Audio Frequency Track Circuit
16.	DAC	Digital Axle Counter
17.	SEM	Signal Engineering Manual
18.	MSDAC	Muti Section Digital Axle Counter
19.	UFSBI	Universal Failsafe Block Interface
20.	LED	Light Emitting Diode
21.	SM	Station Master
22.	RTC	Real Time Clock
23.	BOM	Bill of Material
24.	FOM	Fiber Optic Modem
25.	OS	Operating System
26.	ROM	Read Only Memory
27.	STR	Schedule of Technical Requirements
28.	LCD	Liquid Crystal Display
29.	RE	Railway Electrification
30.	QA	Quality Assurance
31.	QAP	Quality Assurance Program
32.	MTTR	Mean Time To Repair
33.	PF contact	Potential Free contact
34.	OPC-UA	Open Platform Communications – Unified Architecture
35.	LoRa	Long Range communication (a wireless Technology)
36.	IRNSS	Indian Regional Navigation Satellite System
37.	CCSP Layer	C-DOT Common Service Platform
38.	OneM2M	Machine to Machine IoT Standards
39.	MQTT	Message Queuing Telemetry Transport
40.	AE	Application Entity
41.	SMMS	Signalling Maintenance Management System
41.	ISP	It is intermediate service platform which handles to and fro transmission of data packets between RDPMS Application & Station Gateway
42.	Cloud RDPMS Application	RDPMS Application software hosted as decided by Zonal Railway

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Annexure A

Standard Nomenclature used in data packets between Station gateway/SMMS and RDPMS Application

Station Gateway is nodal device which collects data from various means as following

1. Sensors/PF contacts through IoT.
2. Datalogger Port
3. IPS Port
4. Diagnostic Port of various Equipment (as applicable)

Gateway then sends these data to Application software in standard data format using standard nomenclatures of different parameters as defined below. Apart from parameter value, few other important data is also sent/received like configuration, diagnostic, confirmation and time stamp etc. These Nomenclatures are also used in Logics, APIs and RDPMS Application. The details are given below.

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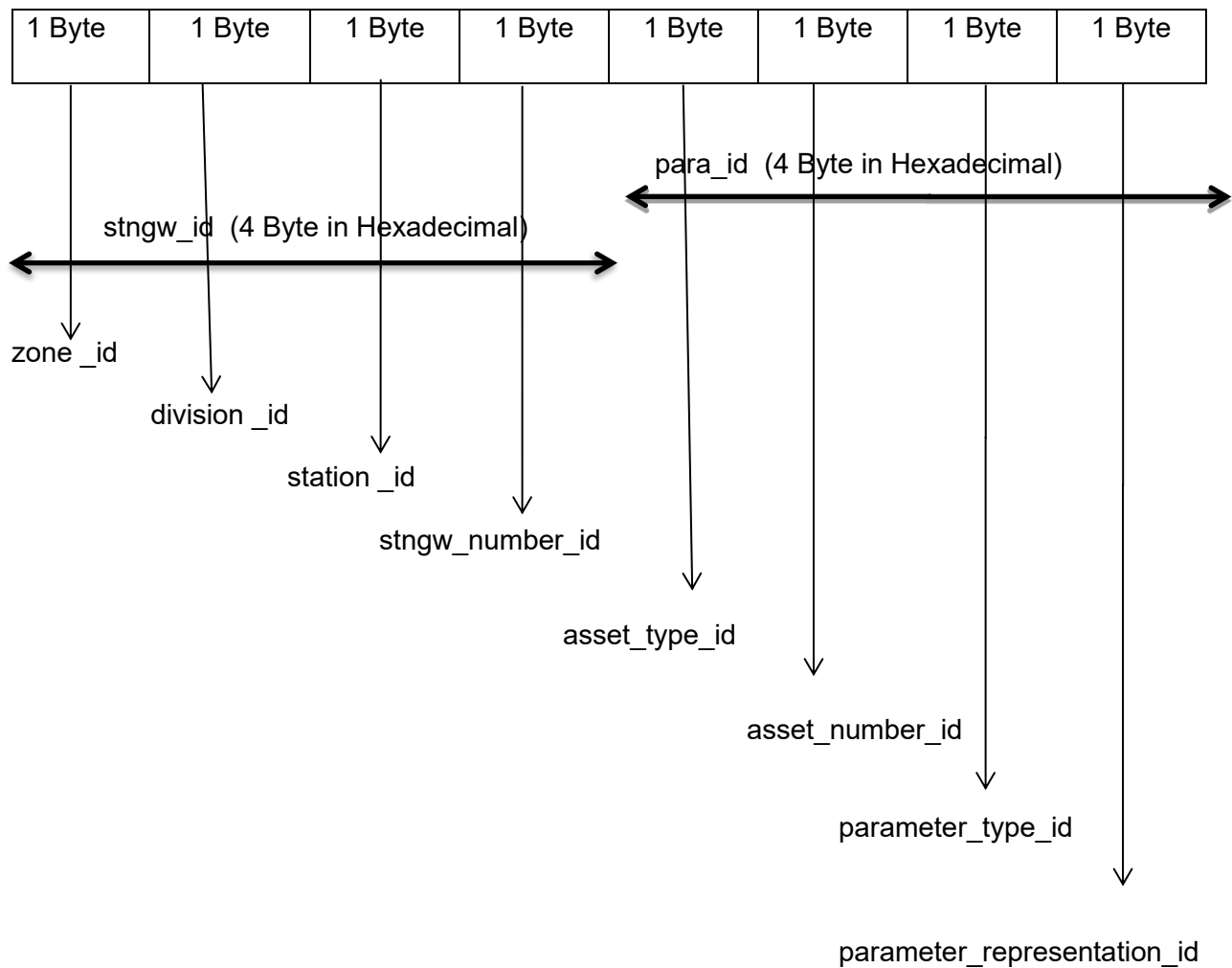
1. Station gateway id:

It is 4 Byte (in Hexadecimal) id to uniquely define each gateway in Indian Railways. It is being used in standard data format. It is represented as “stngw_id” in data packet.

2. Parameter id :

It is 4 Byte (in Hexadecimal) id to uniquely define each parameter within each station gateway. It may either be sensor input or digital status of relays from Data logger port. It is represented as “para_id” in data packet. para_id and stngw_id, when used together, will uniquely define each parameter in Indian Railways. It is being used in standard data format.

Details of para_id and stngw_id has been shown below:



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3. Details of each byte of station gateway id and parameter id :

a. Zone details: zone_id is part of stngw_id denoted in one byte in Hexadecimal. Zone information can be taken from this by Application software. Details given below. Short name of Zone code is represented as “zc” in data packet.

b. Division details: division_id is part of stngw_id denoted in one byte in Hexadecimal. Division information can be taken from this by Application software. Details given below. Short name of Division Code is represented as “dc” in data packet.

S. No.	Zone name (string) zone_name	Zone code (string) zc	Zone id (One Byte-shown in Hexadecimal) zone_id	Division name (string) division_name	Division code (string) dc	Division id (One Byte-shown in Hexadecimal) division_id
1	CENTRAL RAILWAY	CR	00	BHUSAVAL	BSL	00
				MUMBAI	CSTM	01
				NAGPUR	NGP	02
				PUNE	PUNE	03
				SOLAPUR	SUR	04
2	EAST CENTRAL RAILWAY	ECR	01	DANAPUR	DNR	00
				DHANBAD	DHN	01
				PT.DEEN DAYAL UPADHYAYA	DDU	02
				SAMASTIPUR	SPJ	03
				SONPUR	SEE	04
3	EAST COST RAILWAY	ECoR	02	KHURDA ROAD	KUR	00
				SAMBALPUR	SBP	01
				WALTAIR	WAT	02
4	EASTERN RAILWAY	ER	03	ASANSOL	ASN	00
				HOWRAH	HWH	01
				MALDA	MLDT	02
				SEALDAH	SDAH	03
5	NORTH CENTRAL RAILWAY	NCR	04	AGRA	AGRA	00
				JHANSI	JHS	01
				PRAYAGRAJ	PYRJ	02
6	NORTH EASTERN RAILWAY	NER	05	IZZATNAGAR	IZN	00
				LUCKNOW	LJN	01
				VARANASI	BSB	02
7	NORTH FRONTIOR RAILWAY	NFR	06	ALIPURDUAR	APD	00
				KATIHAR	KIR	01
				LUMDING	LMG	02
				RANGIYA	RNY	03
				TINSUKIA	TSK	04

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8	NORTHERN RAILWAY	NR	07	AMBALA	UMB	00
				DELHI	DLI	01
				FEROZPUR	FZP	02
				LUCKNOW	LKO	03
				MORADABAD	MB	04
9	NORTH WESTERN RAILWAY	NWR	08	AJMER	AIJ	00
				BIKANER	BKN	01
				JAIPUR	JP	02
				JODHPUR	JU	03
10	SOUTH CENTRAL RAILWAY	SCR	09	GUNTAKAL	GTL	00
				GUNTUR	JNT	01
				HYDERABAD	HYB	02
				NANDED	NED	03
				SECUNDERA-BAD	SC	04
				VIJAYAWADA	BZA	05
11	SOUTH EAST CENTRAL RAILWAY	SECR	0A	BILASPUR	BSP	00
				NAGPUR	NGP	01
				RAIPUR	R	02
12	SOUTH EASTERN RAILWAY	SER	0B	ADRA	ADRA	00
				CHAKARDHARPUR	CKP	01
				KHARAGPUR	KGP	02
				RANCHI	RNC	03
13	SOUTHERN RAILWAY	SR	0C	CHENNAI	MAS	00
				MADURAI	MDU	01
				PALAKKAD	PGT	02
				SALEM	SA	03
				THIRUVANANTHURAM	TVC	04
				TIRUCHCHIRAPPALLI	TPJ	05
14	SOUTH WESTERN RAILWAY	SWR	0D	BENGALURU	SBC	00
				HUBBALLI	UBL	01
				MYSURU	MYS	02
15	WEST CENTRAL RAILWAY	WCR	0E	BHOPAL	BPL	00
				JABALPUR	JBP	01
				KOTA	KOTA	02
16	WESTERN RAILWAY	WR	0F	AHEMDABAD	ADI	00
				BHAVNAGAR	BVC	01
				MUMBAI CENTRAL	BCT	02
				RAJKOT	RJT	03
				RATLAM	RTM	04
				VADODARA	BRC	05

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c. Station details: station_id is part of stngw_id denoted in one byte in Hexadecimal. Station information can be taken from this by Application software. It is being used in standard data format. Details given below. Short name of station code is represented as “sc” in data packet.

S No	Station name (station name used in data format in string)	Station code (station code used in data format in string)	Station id (One Byte – shown in Hexadecimal)
	station_name	sc	station_id
1.	Lucknow	LJN	12

Note: station_id should be allotted by Division from 00 onwards in Lexicographical order of Station code within each Division.

- d. Station Gateway details:** It is a number given to station gateway from 00 onwards at each station if multiple gateways are installed. It is part of stngw_id. It has been denoted as stngw_number_id (one byte in Hexadecimal).
- e. Asset type:** asset_type_id is part of para_id (parameter id) denoted in one byte in Hexadecimal. It is being used in standard data format. asset_type_code should be used in UIs, APIs & reports where Asst type is mentioned. It is represented as asset_type_code in data packet. These types are same as being used in SMMS. Details given below.

S No	asset_type_name (String)	as- set_type_code (String)	asset_type_id allotted by RDPMS (One Byte – shown in Hexa- decimal)	Remarks
1.	Point Machine	EOP	00	Code as per SMMS
2.	Main Signal	LED	10	
3.	Shunt Signal	LES	11	
4.	Calling On Signal	LEC	12	
5.	Route Signal	LER	13	No code defined in SMMS
6.	DC Track Circuit	DCT	20	Code as per SMMS
7.	Single Section Axle Counter - Eldyne	SSE	21	
8.	Single Section Axle Counter - CEL	SSC	22	

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9.	Single Section Axle Counter – G.G. Tronics	SSG	23	No code defined in SMMS
10.	Multi Section Axle Counter - Eldyne	MSE	24	
11.	Multi Section Axle Counter - CEL	MSC	25	
12.	Multi Section Axle Counter - Siemens	MSS	26	
13.	Multi Section Axle Counter – Siemens ACM200	ACM	27	
14.	Multi Section Axle Counter - Frauscher	MSF	28	
15.	Multi Section Axle Counter - Medha	MSM	29	
16.	Multi Section Axle Counter – G.G. Tronics	MSG	2A	
17.	Multi Section Axle Counter – Sigma Altpro	MSA	2B	
18.	High Availability Single Section Digital Axle Counter - GGtronics	HAG	2C	
19.	Audio Frequency Track Circuit-Ansaldo/Alstom	AFA	2D	
20.	Audio Frequency Track Circuit- Siemens	AFS	2E	
21.	Audio Frequency Track Circuit- Bombardier	AFB	2F	
22.	Block Proving Axle Counter with UAC	BUA	30	
23.	Block Proving Axle Counter with UFSBI	BUF	31	
24.	SGE Double line Block Instrument	SGE	32	
25.	DIODO Type Single Line Block Instrument	DDT	33	
26.	Block Instrument – Push Button	PBT	34	
27.	Block Instrument – Push Button – Neal's Token	NLT	35	
28.	Mechanical Level Crossing Gate	MLC	40	
29.	Electrical Level Crossing Gate	ELC	41	
30.	Integrated Power Supply	IPS	50	
31.	SPD	SPD	51	

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32.	Earth Leakage Detector	ELD	60	Code as per SMMS
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- f. Asset number id (hexadecimal) :** It is part of para_id (parameter id). It is a numerical number allotted to asset to uniquely define parameter id. It has been denoted as asset_number_id (one byte in Hexadecimal). It may not be same as actual asset number at station. This number will start from 00 to FF for each asset type pertaining to one gateway. It is being represented as “asset_number_id” in standard data format.
- g. SMMS asset code (string):** It is unique id assigned to each asset in SMMS (Signalling Maintenance Management System) system. SMMS Asset Code is system generated code which is created when the asset is defined (created) in the SMMS system. This SMMS code will be received from SMMS using API provided by SMMS system. This API interface is described in the annexure later in this document. SMMS will provide API based interface through which RDPMS will get details and meta data for each asset. This is denoted as smms_asset_code. This is similar to asset_number_id of RDPMS as mentioned above in (f).
- h. Asset number code (string):** It is used for asset number like PT-01, TC-101 which is actual asset number at station. It has been denoted as asset_number_code (in string). It should be used in standard data format, UIs, APIs and reports where asset number is mentioned. It is represented as “asset_number_code” in standard data packet.
- i. SMMS asset name (string):** It is used by SMMS for asset name like PT-01, TC-101 which is actual asset name at station. This will be received from SMMS using API provided by SMMS system. This API interface is described in the annexure later in this document. This is represented as smms_asset_name. This is actually same as asset_number_code (asset_number_code) of RDPMS system as defined in (h) above. There shall be a table in application software to map all 4 referred in (f), (g), (h) & (i) as given below.

smms_asset_code (String) (SMMS)	smms_asset_name (string) (SMMS)	asset_number_id (one byte in Hexadecimal) (RDPMS)	asset_number_code (string) (RDPMS)
EOPMIU00013	PT-101	00	PT-101
EOPMIU00014	PT-102	01	PT-102
DCTMIU00047	TC-100	10	TC-100
DCTMIU00048	TC-101	11	TC-101

- j. Equipment Room Type id:** Equipment room parameters Temperature, Humidity, Relay room opening/Closing have been defined ahead for different type of Rooms such as Relay

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room, IPS room etc. Here equipment room type is represented as eqpmntroom_type_id. This id is same as asset_type_id for signalling asset. This shall be used in place of asset_type_id as part of para_id (parameter id) denoted in one byte in Hexadecimal to define equipment room parameters. It is being used in standard data format.

eqpmntroom_type_code should be used in UIs, APIs & reports where Equipment room is mentioned. Details given below.

S No	eqpmntroom_name (String)	eqpmntroom_type_code (String)	eqpmntroom_type_id (One Byte – in Hexadecimal)	Re- marks
1	Relay Room	RR	F0	
2	IPS Room	IPS	F1	
3	Battery Room	BATT	F2	
4	Maintainer Room	MAIN	F3	
5	Generator Room	GEN	F4	
6	Outdoor	OUTDOOR	F5	
7	Location Box	LOC	F6	

k. Equipment Room Number id (hexadecimal) : It is same as asset_number_id used in para_id (parameter id). It is a numerical number allotted to one type of equipment room such as Relay room 1, 2, to uniquely define parameter id. It has been denoted as eqpmntroom_number_id (one byte in Hexadecimal). It may not be same as actual room number at station. This number will start from 00 to FF for each type of room pertaining to one gateway. However, outdoor and location box shall have only 00 as only one outdoor and one location shall be used. It is being represented as eqpmntroom_number_id in standard data format.

l. Parameter type and unit details: It is part of para_id (parameter id). It defines type and unit of parameter. It has been denoted as parameter_type_id (one byte in Hexadecimal). Details given below.

Parameter type	parameter_type_id (one byte in Hexadecimal)	Remarks
Current DC (unit –A)	00	Other units of current DC can be allotted bytes from 02 to 0F
Current DC (unit - mA)	01	
Current AC (unit –A)	10	Similar analogy as above
Current AC (unit - mA)	11	
Voltage DC (unit – V)	20	Similar analogy as above
Voltage DC (Unit-mV)	21	
Voltage AC (unit – V)	30	Similar analogy as above

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Voltage AC (Unit-mV)	31	
Digital	40	Similar analogy as above
Temperature (Unit – C (centigrade))	50	Similar analogy as above
Humidity (Relative humidity in %)	51	Similar analogy as above
Vibration	60	Similar analogy as above
Frequency (Unit - Khz)	70	Similar analogy as above
Frequency (Unit – Hz)	71	
Resistance (Unit-Ohm)	80	Similar analogy as above
Resistance (Unit-KOhm)	81	
Time (Unit – seconds)	90	Similar analogy as above
Time (Unit – mili seconds)	91	
String	A0	

m. Parameter representation details:

parameter_representation_id is part of para_id (parameter id). It is represented in one byte in hexadecimal. parameter_representation_code is short nomenclature to define parameter of asset like current of point machine, voltage of point machine etc. in string. It is being used in logics in Annexure C and can be used in Application Software for logic processing and in UI for displaying parameters.

parameter_representation_name (string)	parameter_representation_code (string)	parameter_representation_id (one byte)
110 DC at Loc box for Normal.	V _{PT} 110 DC Loc N	0A

n. Nomenclature of Parameters for various assets:

- Parameters of IPS, Point machine, all types of Signal, DC track circuit, ELD, ELB, Temperature, humidity of equipment room, AXLE Counter, UFSBI etc have been shown below. There can be four types of Parameters ie through Sensor, through Digital (Datalogger /IoT), diagnostic port and derived. All out efforts have been made to cover all Parameters. However, there are still spaces left in parameter_representation_id (one byte hexadecimal). Vendors can define new parameters with suitable parameter_representation_id (one byte hexadecimal) and parameter_representation_code (string) in consultation with RDSO so that suitable amendments in this Annexure may be issued after review.

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- para_id as defined for each parameter in beginning of this Annexure A contains 4 sub ids, ie. asset_type_id (eqpmntroom_type_id for equipment room), asset_number_id (eqpmntroom_number_id for equipment room), parameter_type_id, parameter_representation_id. First two sub ids have already been defined earlier. Now parameters have been defined for different type of assets as following with other two sub ids.
- These parameters have been used in standard logics in Annexure C. There are various types of values defined for each parameter (except digital) as given below.
- Avg value of parameter is defined as average operating parameter taking Avg value of 15 days (not counting failure data). Number of days ie 15 for taking average should be configurable in Application. In case of replacement of asset, this value should be counted afresh. Avg and vale and value at any instance are different , for example, average value of operating voltage of point machine for 15 days may be 105 V whereas it may have different current value at any instance.
- Min safe value of parameter shall be the min boundary value for safe operation.
- Min fail value of parameter shall be the min boundary value below which the equipment will fail.
- All Boundary values shall be decided by Zonal Railway as per field condition. Though some of the values have been given for reference based on operating range of parameters as per Specifications of asset. “__” shown in value column represents value not required. “*” shown in the column requires value to be decided by Zonal Railways only as per site requirement.
- Digital status sent through packet should be a stable digital status and shall exclude chattering of Relays to avoid confusion.

Nomenclature of Parameters of Integrated Power Supply

Note: Nomenclatures for additional parameters received from diagnostic ports can be further decided.

Sr no.	Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id (one byte)	parameter_type_id (one byte)	Value in V/A		Remarks
						Min safe	Min fail	
1	IPS Diagnostic	IPS 110 DC O/P Voltage	VIPS 110 DC	00	20	*	*	* Zonal Railway shall decide the values

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	Port as per latest Specs							as per site requirement.
2	OR Using Sensors	IPS 110 AC Sig-1 O/P Voltage	VIPS SIG-1 110 AC	10	30	*	*	
3		IPS 110 AC Sig-2 O/P Voltage	VIPS SIG-2 110 AC	11	30	*	*	
4		IPS 110 AC Sig-3 O/P Voltage	VIPS SIG-3 110 AC	12	30	*	*	
5		IPS 110 AC Sig-4 O/P Voltage	VIPS SIG-4 110 AC	13	30	*	*	
6		IPS 110 AC TR-1 O/P Voltage	VIPS TR-1 110 AC	20	30	*	*	
7		IPS 110 AC TR-2 O/P Voltage	VIPS TR-2 110 AC	21	30	*	*	
8		IPS 110 AC TR-3 O/P Voltage	VIPS TR-3 110 AC	22	30	*	*	
9		IPS 110 AC TR-4 O/P Voltage	VIPS TR-4 110 AC	23	30	*	*	
10		IPS SMR-1 Voltage	VIPS SMR-1 110 DC	30	20	*	*	
11		IPS SMR-2 Voltage	VIPS SMR-2 110 DC	31	20	*	*	
12		IPS SMR-3 Voltage	VIPS SMR-3 110 DC	32	20	*	*	
13		IPS SMR-4 Voltage	VIPS SMR-4 110 DC	34	20	*	*	
14		IPS SMR-5 Voltage	VIPS SMR-5 110 DC	35	20	*	*	
15		IPS DC-DC Relay Internal Voltage	VIPS DC R INT	40	20	*	*	
16		IPS DC-DC Relay External Voltage	VIPS DC R EXT	48	20	*	*	
17		IPS DC-DC AXLE COUNTER Voltage	VIPS DC AXLE C	50	20	*	*	
18		IPS DC-DC PANEL INDICATION Voltage	VIPS DC PAN IND	58	20	*	*	
19		IPS DC-DC BLOCK LOCAL Voltage	VIPS DC BLOCK LOCAL	5C	20	*	*	
20		IPS DC-DC HKT MAGNETO Voltage	VIPS DC HKT MAG	60	20	*	*	
21		IPS DC-DC BLOCK LINE UP Voltage	VIPS DC BLOCK LINE UP	62	20	*	*	
22		IPS DC-DC BLOCK LINE DN Voltage	VIPS DC BLOCK LINE DN	63	20	*	*	
23		IPS DC-DC BLOCK TELE UP Voltage	VIPS DC BLOCK TEL UP	65	20	*	*	

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24		IPS DC-DC BLOCK TELE DN Voltage	V _{IPS DC BLOCK TEL DN}	66	20	*	*	
25		IPS DC-DC DAT-ALOGGER Voltage	V _{IPS DC DATALOG}	68	20	*	*	
26		IPS DC-DC EI Voltage	V _{IPS DC EI}	70	20	*	*	
27		IPS Battery charging Current	I _{IPS BATT CHAR 110 DC}	78	00	*	*	It shall be possible to measure current in both direction ie during charging and during discharging.
28		IPS input Voltage	V _{IPS I/P}	79	30	*	*	
29		Individual battery voltage (optional)	V _{IPS BATT-1} V _{IPS BATT-2.....}	80-BF	20	*	*	parameter_representation_id can be used from 80 upto BF.

Nomenclature of Parameters of SPD

Sr	Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id (one byte)	parameter_type_id (one byte)	Remarks
1	Data-log-ger/IoT	SPD-1 To SPD-n	SPD ₁ to SPD _n	00-FF	40	

Nomenclature of Parameter of Point Machine

(Total number of Sensors = 8 for one point machine (Minimum), if NWKR and operation is independent for each machine)

- Parameters are to be defined for individual point (One Point machine) treating as one asset.
- For common entity to both the ends of points like NWKR etc, parameters for both the ends of points should be defined, however Station gateway shall send same value/status of both parameters. While displaying such parameter in Application software, it may be displayed as one entity pertaining to crossover pair, as per field requirement.
- NWKR and RWKR are normal and reverse indication relays respectively. NWCR and RWCR are Contactor Relay through which 110 V feed is passed for operation of individual or both points.

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Sr	Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id (one byte)	parameter_type_id (one byte)	Values in V/A/Sec			Remarks
						Max safe	Min safe	Min fail	
		Relay Room							
1	Sensor	24 VDC at RR from Loc Normal	V _{PT} NWKR	00	20	—	21	18	NWKR of individual point OR One NWKR for both Points.
2	Sensor	24 VDC at RR from Loc Reverse	V _{PT} RWKR	01	20	—	21	18	RWKR of individual point OR One RWKR for both Points
3	Digital input from Data-logger	Digital status of NWKR	NWKR	10	40	—	—	—	NWKR of individual point OR One NWKR for both Points
4	Digital input from Data-logger	Digital status of RWKR	RWKR	11	40	—	—	—	RWKR of individual point OR One RWKR for both Points
5	Digital input from Data-logger	Digital status of NWCR	NWCR	12	40	—	—	—	NWCR of individual or both points.
6	Digital input from Data-logger	Digital status of RWCR	RWCR	13	40	—	—	—	RWCR of individual or both points
		Location Box							

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7	Sensor	110 DC at Loc box for Normal.	V _{PT 110 DC LOC N}	20	20	—	90	82	For individual Point
8	Sensor	110 DC at Loc box for Reverse.	V _{PT110 DC LOC R}	21	20	—	90	82	For individual Point
9	Sensor	Point Machine Current Normal	I _{PT N}	30	00	—	*	*	For individual Point
10	Sensor	Point Machine Current Reverse	I _{PT R}	31	00	—	*	*	For individual Point
11	Sensor	24 V DC going to Relay Room after detection at LOC for Normal	V _{PT 24 DC LOC N}	40	20	—	—	—	For individual Point
12	Sensor	24 V DC going to Relay Room after detection at LOC for Reverse	V _{PT 24 DC LOC R}	41	20	—	—	—	For individual Point
13	Sensor	Vibration	X	50	60	—	—	—	For individual Point (Optional)
14	Derived	Normal Operation time	T _{PT N}	60	90	8	—	—	T_{PT N} Time should be calculated from current signature for individual point. Max safe time should be about 1.5 sec less than WJR timer time so that obstruction can be declared when when T_{PT N} > max safe time during obstruction.
15	Derived	Reverse Operation time	T _{PT R}	61	90	8	—	—	T_{PT R} Time should be calculated from current signature for individual point. Max safe time should be about 1.5 sec less than WJR timer time so that obstruction can be

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									declared when when $T_{PTN} > \text{max}$ safe time during obstruction.
--	--	--	--	--	--	--	--	--	--

Nomenclature of Parameters of DC Track Circuit

(Total number of Sensors = 9 for One DC track circuit)

TR Relay is track relay which picks up at site by receiving end feed when track is not occupied. TPR is repeater Relay in Relay room repeating the status of Track relay at site. TPR normally picks up through front contact of TR Relay by 24 V DC feed coming from site.

Sr no.	Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id (one byte)	parameter_type_id (one byte)	Value in V/mA			Remarks
		Relay Room				Max safe	Max safe	Max safe	
1	Sensor	24 V DC TPR I/P	$V_{TC\ 24\ DC\ TPR\ I/P}$	00	20	—	21	18	
2	Digital input from Datalogger	Digital status of TPR Relay	TPR	10	40	—	—	—	Repeater Relay in Relay room repeating the status of Track relay
		Feed End							
3	Sensor	Track Feed Charger I/P Voltage	$V_{TC\ TFC\ I/P}$	20	30	—	*	—	* Should be defined by Zonal railways depending on input tapping used for different

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									track circuits.
4	Sensor	Track Feed Charger O/P Voltage (on load)	$V_{TC\ TFC\ O/P}$	21	20	—	*	*	* Should be defined by Zonal railways. for 3 or 2 battery used.
5	Sensor	Track Feed Charger O/P Current	$I_{TC\ TFC\ O/P}$	22	01	—	—	—	
6	Sensor	Voltage drop at feed end choke	$V_{TC\ CH\ FEED\ END}$	23	20	—	—	—	
7	Sensor	Track Feed end voltage (going to Rails)	$V_{TC\ FEED\ END}$	24	20	—	—	—	
8	Sensor	Track Feed Current	$I_{TC\ FEED\ END}$	25	01	—	—	—	
9	Derived	Bat Charging Current	$I_{TC\ BATT\ CHARG}$	26	01	*	*	—	* Should be defined by Zonal railway for each Track ckt. $I_{TC\ BATT\ CHARG} = I_{TC\ TFC\ O/P} - I_{TC\ FEED\ END}$
10	Derived	Voltage at Variable Track Resistance	$V_{TC\ VAR\ RES}$	27	20	—	—	—	$V_{TC\ VAR\ RES} = V_{TC\ TFC\ O/P} - V_{TC\ FEED\ END} - V_{TC\ CH\ FEED\ END}$
11	Derived	Feed end choke resistance	$R_{TC\ CH\ FEED\ END}$	28	80	—	—	—	$R_{TC\ CH\ FEED\ END} = V_{TC\ CH\ FEED\ END} / I_{TC\ FEED\ END}$
12	Derived	Variable resistance	$R_{TC\ VAR\ RES}$	29	80	—	—	—	$R_{TC\ VAR\ RES} = V_{TC\ VAR\ RES} / I_{TC\ FEED\ END}$
		Relay End							
13	Sensor	Track Relay end Current	$I_{TC\ RELAY\ END}$	41	01	—	210	100	
14	Derived	Track Relay Voltage QTA2 (1.4 V)	$V_{TC\ TR}$	42	20	4.2	2.1	—	May be different for other Relays.

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									$V_{TC\ TR} = I_{TC\ RELAY\ END}$ multiplied by Track Relay resistance. (By physical measurement at the time of installation or as per specification)
15	Sensor	24 V DC going to TPR after TR pick up contact	$V_{TC\ 24\ DC\ LOC}$	44	20	—	—	—	
16	Derived	Ballast/Sleeper Current	I_{BALST}	60	01	—	—	—	$I_{TC\ FEED\ END} - I_{TC\ RELAY\ END}$
17	Derived	Rail Resistance	R_{RAIL}	61	80	—	—	—	$2*(V_{TC\ FEED\ END} - V_{TC\ TR}) / (I_{TC\ FEED\ END} + I_{TC\ RELAY\ END})$

Nomenclature of Parameters of Main Signal

(Total number of Sensors = 3 for One OFF aspect and 2 for One ON aspect)

Number of sensors given in table below is for four aspect signal having RED, YELLOW, DOUBLE YELLOW and GREEN to give clarity in logics for one full Signal.

RECR/HECR/HHECR/DECR are relays showing Signal lit status for Red/Yellow/Double yellow/Green. HR/HHR/DR are relays whose pick up status shows that 110 V AC feed for Signal is extended to Signal for Yellow/Double yellow/Green.

Sr no.	Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id	parameter_type_id (one byte)	Value in V/mA

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				(one byte)				
		Relay Room				Max safe	Min Safe	Min fail
1	Digital input from Datalogger	Digital status of HECR	HECR	00	40	—	—	—
2	Digital input from Datalogger	Digital status of RECR	RECR	01	40	—	—	—
3	Digital input from Datalogger	Digital status of DECR	DECR	02	40	—	—	—
4	Digital input from Datalogger	Digital status of HHECR	HHECR	03	40	—	—	—
5	Digital input from Datalogger	Digital status of DR	DR	10	40	—	—	—
6	Digital input from Datalogger	Digital status of HR	HR	11	40	—	—	—
7	Digital input from Datalogger	Digital status of HHR	HHR	12	40	—	—	—
		Location Box						
8	Sensor	DPR Voltage	V _{SIG DPR}	20	20	—	21	18
9	Sensor	HPR Voltage	V _{SIG HPR}	21	20	—	21	18
10	Sensor	HHPR Voltage	V _{SIG HHPR}	22	20	—	21	18
11	Sensor	Green Aspect Voltage	V _{SIG DG}	30	30	—	90	82
12	Sensor	Yellow Aspect Voltage	V _{SIG HG}	31	30	—	90	82
13	Sensor	Double Yellow Aspect Voltage	V _{SIG HHG}	32	30	—	90	82
14	Sensor	Red Aspect Voltage	V _{SIG RG}	33	30	—	90	82
15	Sensor	Green Aspect current	I _{SIG DG}	40	11	150	110	90
16	Sensor	Yellow Aspect current	I _{SIG HG}	41	11	150	110	90
17	Sensor	Double yellow Aspect current	I _{SIG HHG}	42	11	150	110	90
18	Sensor	Red Aspect current	I _{SIG RG}	43	11	150	110	90

Nomenclature of Parameters Calling ON Signal

(Total number of Sensors = 3 per Calling On)

CO-HECR is relay showing calling ON Signal lit status. CO-HR is relay whose pick up status shows that 110 V AC feed is extended to Signal.

Sr no.	Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id	parameter_type_id	Value in V/mA
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				(one byte)	(one byte)			
		Relay Room				Max safe	Min Safe	Min fail
1	Digital input from Data-logger	Digital status of Calling ON HECR	CO-HECR	00	40	—	—	—
2	Digital input from Data-logger	Digital status of Calling ON HR	CO-HR	10	40	—	—	—
3		Location Box						
4	Sensor	Calling ON HPR Voltage	V _{COSIG HPR}	20	20	—	21	18
5	Sensor	Calling ON Aspect Voltage	V _{COSIG}	30	30	165	120	90
6	Sensor	Calling ON Aspect current	I _{COSIG}	40	11	—	93	87

Nomenclature of Parameters Route Signal

(Total number of Sensors = 3 per Route Signal)

UECR is relay showing Route Signal lit status. UHR is relay whose pick up status shows that 110 V AC feed is extended to Signal.

Sr no.	Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id (one byte)	parameter_type_id (one byte)	Value in V/mA		
		Relay Room				Max safe	Min Safe	Min fail
1	Digital input from Datalogger	Digital status of UCER	UECR	00	40	—	—	—
2	Digital input from Datalogger	Digital status of UHR	UHR	10	40	—	—	—
		Location Box						
3	Sensor	Route HPR Voltage	V _{ROSIG HPR}	20	20	—	21	18
4	Sensor	Route Aspect Voltage	V _{ROSIG}	30	30	131	119	90
5	Sensor	Route Aspect current	I _{ROSIG}	40	11	—	93	87

Nomenclature of Parameters Shunt Signal

(Total number of Sensors = 6 for one independent Shunt Signal)

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SH-HECROFF is relay showing Shunt Signal lit status in OFF condition. SH-HECROn is relay showing Shunt Signal lit status in ON condition. SH-HR is relay whose pick up status shows that 110 V AC feed is extended to Signal.

Sr no.	Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id (one byte)	parameter_type_id (one byte)	Value in V/mA		
						Max safe	Min Safe	Min fail
		Relay Room						
1	Digital input from Data-logger	Digital status of SH-ECROFF	SH-ECROFF	00	40	—	—	—
2	Digital input from Data-logger	Digital status of SH-ECROn	SH-ECROn	01	40	—	—	—
3	Digital input from Data-logger	Digital status of SH-HR	SH-HR	10	40	—	—	—
		Location Box						
4	Sensor	Shunt HPR Voltage	VSHSIG HPR	20	20	—	21	18
5	Sensor	Shunt ON Aspect Voltage	VSHSIG ON	30	30	58	52	90
6	Sensor	Shunt OFF Aspect Voltage	VSHSIG OFF	31	30	—	93	87
7	Sensor	Shunt ON Aspect current	ISHSIG ON	40	11	58	52	90
8	Sensor	Shunt OFF Aspect current	ISHSIG OFF	41	11	58	52	—
9	Sensor	Shunt PILOT Aspect current	ISHSIG PILOT	42	11	—	93	87

	Nomenclature of Parameter of Electric Lifting Barrier				
	Note: These are indicative parameter list. Actual shall be prepared as per site and OEM of ELB under intimation to RDSO.				
Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id (one byte)	parameter_type_id (one byte)	Remarks
	Location Box/Gate hut				
	Voltage of Motor (AC ELB)				

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Sensor	V AC at ELB -A Feed going to motor for closing/opening	V _{ELB-A AC}	00	30	
Sensor	V AC at ELB -B Feed going to motor for closing/opening	V _{ELB-B AC}	01	30	
	Current of Motor (AC ELB)				
Sensor	ELB-A Closing current	I _{ELB-A AC CLOSE}	10	10	
Sensor	ELB-A Opening current	I _{ELB-A AC OPEN}	11	10	
Sensor	ELB-B Closing current	I _{ELB-B AC CLOSE}	12	10	
Sensor	ELB-B Opening current	I _{ELB-B AC OPEN}	13	10	
	Voltage of Motor (DC ELB)				
Sensor	V AC at ELB -A Feed going to motor for closing/opening	V _{ELB-A DC}	20	20	
Sensor	V AC at ELB -B Feed going to motor for closing/opening	V _{ELB-B DC}	21	20	
	Current of Motor (DC ELB)				
Sensor	ELB-A Closing current	I _{ELB-A DC CLOSE}	30	00	
Sensor	ELB-A Opening current	I _{ELB-A DC OPEN}	31	00	
Sensor	ELB-B Closing current	I _{ELB-B DC CLOSE}	32	00	
Sensor	ELB-B Opening current	I _{ELB-B DC OPEN}	33	00	
	Other Inputs				
Sensor	ELB-A close Relay voltage (suitable Relay which indicates that button for operation has been initiated)	V _{ELB-A CLOSE BUTTON}	40	20	Tentative and cab be decided as per site and shared to RDSO.
Sensor	ELB -A closed & Locked indication Relay voltage. (suitable final Relay which extends the 24 V to RR for closed and locked status)	V _{ELB-A CLOSED & LOCKED IND}	41	20	
Sensor	ELB-B close Relay (suitable Relay which indicates that button for operation has been initiated)	V _{ELB-B CLOSE BUTTON}	50	20	
Sensor	ELB -B closed & Locked indication Relay. (suitable final Relay which extends the 24 V to RR for closed and locked status)	V _{ELB-B CLOSED & LOCKED IND}	51	20	
Sensor	Final Relay for gate closed and locked slot transfer	V _{ELB FINAL SLOT}	60	20	
Datalogger	Status of other suitable Relays can be taken from Datalogger		70 onwards	40	

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Nomenclature of Parameter of AXLE COUNTER					
Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id (one byte)	parameter_type_id (one byte)	Remarks
	Location Box/Gate hut				
Sensor	Power supply of Equipment	V _{AXCO} EQ	00-0F	20	
Sensor	Current of Equipment	I _{AXCO} EQ	10-1F	00	
Sensor	24 V DC going to VPR (or similar) after VR (or similar) pick up contact	V _{AXCO} 24 DC LOC	20	20	
Sensor	24 V DC of VPR (or similar) Relay	V _{AXCO} 24 DC VPR I/P	30	20	
Diagnostic port	Other parameters	Nomenclatures ids to be decided based on protocol shared by OEM and to be shared to RDSO			

Nomenclature of Parameter of UFSBI					
Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id (one byte)	parameter_type_id (one byte)	Remarks
Sensor	Power supply of Equipment	V _{UFSBI} EQ	00-0F	20	
Sensor	Current of Equipment	I _{UFSBI} EQ	10-1F	00	
Data logger/IoT	Status of UFSBI Relays	Nomenclatures and ids to be decided and to be shared to RDSO		40	
Data logger/IoT	PF contacts	Nomenclatures and ids to be decided based on information shared by OEM and to be shared to RDSO		40	

Nomenclature of Parameter of Earth Leakage Detector

(ELD can be numbered as 00 to FF in asset_number_id if more than 1 ELD being used)

Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id (one byte)	parameter_type_id (one byte)
Digital input from Datalogger (IoT if not provided by Datalogger)	Channel 1	ELD _{CH1}	00	40
Digital input from Datalogger (IoT if not provided by Datalogger)	Channel 2	ELD _{CH2}	01	40

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Digital input from Datalogger (IoT if not provided by Datalogger)	Channel 3	ELD _{CH3}	02	40
Digital input from Datalogger (IoT if not provided by Datalogger)	Channel 4	ELD _{CH4}	03	40
Digital input from Datalogger (IoT if not provided by Datalogger)	Channel 5	ELD _{CH5}	04	40
Digital input from Datalogger (IoT if not provided by Datalogger)	Channel 6	ELD _{CH6}	05	40
Digital input from Datalogger (IoT if not provided by Datalogger)	Channel 7	ELD _{CH7}	06	40
Digital input from Datalogger (IoT if not provided by Datalogger)	Channel 8	ELD _{CH8}	07	40
Digital input from Datalogger (IoT if not provided by Datalogger)	Channel 9	ELD _{CH9}	08	40
Digital input from Datalogger (IoT if not provided by Datalogger)	Channel 10	ELD _{CH10}	09	40
Digital input from Datalogger (IoT if not provided by Datalogger)	Channel 11	ELD _{CH11}	0A	40
Digital input from Datalogger (IoT if not provided by Datalogger)	Channel 12	ELD _{CH12}	0B	40
Digital input from Datalogger (IoT if not provided by Datalogger)	Channel 13	ELD _{CH13}	0C	40
Digital input from Datalogger (IoT if not provided by Datalogger)	Channel 14	ELD _{CH14}	0D	40
Digital input from Datalogger (IoT if not provided by Datalogger)	Channel 15	ELD _{CH15}	0E	40
Digital input from Datalogger (IoT if not provided by Datalogger)	Channel 16	ELD _{CH16}	0F	40

Nomenclature of Parameter of Relay Room

Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id (one byte)	parameter_type_id (one byte)
Datalogger/IoT	Door Open/Close of Relay Room	DOOR _{RR}	00	40
Sensor	Temperature of Relay Room	TEMP _{RR}	01	50
Sensor	Humidity of Relay Room	HUMD _{RR}	02	51

Nomenclature of Parameter of IPS Room

Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id (one byte)	parameter_type_id (one byte)
Sensor	Temperature of IPS Room	TEMP _{IPS}	01	50
Sensor	Humidity of IPS Room	HUMD _{IPS}	02	51

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Nomenclature of Parameter of Battery Room

Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id (one byte)	parameter_type_id (one byte)
Sensor	Temperature of Battery Room	TEMP _{BATT}	01	50
Sensor	Humidity of Battery Room	HUMD _{BATT}	02	51

Nomenclature of Parameter of Maintainer Room

Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id (one byte)	parameter_type_id (one byte)
Sensor	Temperature of Maintainer Room	TEMP _{MAIN}	01	50
Sensor	Humidity of Maintainer Room	HUMD _{MAIN}	02	51

Nomenclature of Parameter of Generator Room

Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id (one byte)	parameter_type_id (one byte)
Sensor	Temperature of Generator Room	TEMP _{GEN}	01	50
Sensor	Humidity of Generator Room	HUMD _{GEN}	02	51

Nomenclature of Parameter of Outdoor

Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id (one byte)	parameter_type_id (one byte)
Sensor	Temperature of Outdoor	TEMP _{OUTDOOR}	01	50
Sensor	Humidity of Outdoor	HUMD _{OUTDOOR}	02	51

Nomenclature of Parameter of Location Box

Type of Input	parameter_representation_name	parameter_representation_code	parameter_representation_id (one byte)	parameter_type_id (one byte)
Sensor	Temperature of Location Box	TEMP _{LOC}	01	50

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Sensor	Humidity of Location Box	HUMD _{LOC}	02	51
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n. Sensor Range:

Approximate Voltage and Current ranges of different types of parameters of Point Machine, DC track Circuits and LED Signal lamps are given below. 8 clusters for sensors have been suggested from A to H by grouping parameters having close values. Indicative range of measurement by sensor has been given against each cluster for guidance purpose. Vendor shall select sensor with appropriate range for measurement keeping in view the overall requirement of Specification. Even clusters can be further narrowed down for better results.

Sensors for temperature and humidity shall be selected keeping in view the temperature and humidity range as per Indian climate.

	Signaling Gear parameters		Approximate Voltage/Current range of parameter	Indicative Range of measurement by Sensor
	DC Voltage			
A	All Voltage of track circuit (except TFC input Voltage)			0 V DC – 10 V DC
	1	TFC charger Output Voltage	0 – 8 V DC	
	2	Voltage across Choke at Feed end	0 – 2 V DC	
	3	Voltage across Rail at Feed end	0 – 6 V DC	
	4	Voltage across Rail at Relay end	0 – 5 V DC	
	5	Voltage across Choke at Relay end	0 – 2 V DC	
	6	Voltage across Track Relay.	0 – 5 V DC	
B	1	All voltages driving 24 V DC Relays	0– 32 V DC	0 V DC – 40 V DC
	2	IPS DC-DC Converter Voltage of 24 V DC module/AXLE COUNTER/UFSBI equipment Voltage/ELB (DC)	0– 32 V DC	
C	1	Point Machine Operating voltage	0– 140 V DC	0 – 150 V DC
	2	IPS Battery/SMR Voltage	0– 140 V DC	
	3	IPS DC-DC Converter Voltage upto 66 V	0– 66 V DC	
	4	All voltages driving 50 V DC Relays	0– 50 V DC	
	AC Voltage			
D	1	Signal Voltage	0– 140 V AC	0 V AC – 150 V AC
	2	IPS-All Inverter output of 110 V AC	0– 140 V AC	
	3	TFC input Voltage	0– 140 V AC	
	4	ELB (AC)	0– 140 V AC	
	5	220 V AC input to IPS	0– 300 V AC	0 V AC – 300 V AC
	DC Current			
E	All Current of Track circuit			0 A DC – 1 A DC
	1	Feed End current at Rail	0– 1 A DC	

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	2	Relay End current from Rail	0– 400 mA DC	
F	1	Point Machine Current	0– 6 A DC	0 A DC – 10 A DC
	2	TFC output Current	0– 5 A DC	
G	1	IPS Battery charging Current (It should be possible to measure current in both direction)	0– 30 A DC	0 A DC – 50 A DC
	AC Current			
H		Signal Current	0– 150 mA AC	0 A AC – 1 A AC

4. Parameter value details: parameter value is being used in Standard data format to send the value of given parameter.

It is represented as “prv” (in float for analogue values) (00 for DN and 01 for UP for digital value) (string in case diagnostic port data of any equipment requires) in data packet.

5. Parameter time details: parameter time is being used in Standard data format to associate time of sampling of each parameter. It is represented in string. Details given below:

It is represented as “prt” in data packet.

(DD-MM-YYYY HH:mm:ss.SSS) example: 04-11-2025 16:27:45.123 IST

DD → Day (2 digits, 01–31)

MM → Month (2 digits, 01–12)

YYYY → Year (4 digits)

HH → Hour (00–23, 24-hour format)

mm → Minutes (00–59)

ss → Seconds (00–59)

SSS → Milliseconds (000–999)

IST → Indian Standard Time label

6. Configuration details: It is being used in Standard data format. It is used to configure station gateway as per configuration id (represented as config_id in data packet) and configuration value (represented as config_val in data packet) given by RDPMS Application. Indicative values for configuration have been given below, however, these values can be decided by Zonal Railways. Details given below :

sr	Configuration details	config_id (one byte hexadecimal)	config_val (float)
1	parameter data packet sending frequency	01	5 (second)

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2	RDPMS health data packet sending frequency	02	30 (min)
3	% variation in value for sending analog parameters	03	2 (%)
4	Parameter data collection frequency for Point Motor/ELB etc	04	20 (ms)

7. Sensor health details: Station gateway gives health details of Sensor in form of sensor health id to RDPMS Application. It is represented as sh_id (in one byte hexadecimal) in data-format. Sensor health code (in string as shown) is represented as sh_code and should be referred in UI, API. sh_t is time stamp with either healthy or faulty. Details given below :

	Sensor health details	sh_code (sensor health code) (string)	sh_id (sensor health id) (one byte in hexadecimal)	sh_t time stamp with either healthy or faulty.	Remarks
	Faulty Sensor	"Sensor Faulty"	01		To be sent along with "para_id" to correlate.
	Healthy Sensor	"Sensor Healthy"	00		To be sent along with "para_id" to correlate.

8. IoT health details: Station gateway gives health details of IoT in form of IoT health id to RDPMS Application. It is represented as ioth_id (in one byte hexadecimal) in dataformat. IoT health code (in string as shown) is represented as ioth_code and should be referred in UI, API. ioth_t is time stamp with either healthy or faulty. Details given below :

	IoT health details	ioth_code (IoT health code) (string)	ioth_id (IoT health id) (one byte in hexadecimal)	ioth_t is time stamp with either healthy or faulty.	Remarks
	Faulty IoT	"IoT Faulty"	01		To be sent along with all parameters of faulty IoT to correlate it

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	Healthy IoT	"IoT Healthy"	00		To be sent along with all parameters of faulty IoT to correlate it
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9. Station gateway health details: Station gateway gives health details of station gateway in form of stngwh health id to RDPMS Application. It is represented as stngwh_id (in one byte hexadecimal) in dataformat. stngwh health code (in string as shown) is represented as stngwh_code and should be referred in UI, API. stngwh_t is time stamp with either healthy or faulty. Details given below :

	Station gateway health details	stngwh_code (station gateway health code) (string)	stngwh_id (station gateway health id) (one byte in hexadecimal)	stngwh_t is time stamp with either healthy or faulty.	Remarks
	Faulty Station gateway	"Station gateway Faulty"	01		
	Healthy Station gateway	"Station gateway Healthy"	00		

10. Network health details: Station gateway gives health details of network in the form of network health id to RDPMS Application. It is represented as neth_id (in one byte hexadecimal) in dataformat. Network health code (in string as shown) is represented as neth_code and should be referred in UI, API. neth_t is time stamp with either healthy or faulty. Details given below :

	Network health details	neth_code (Network health code) (string)	neth_id (Network health id) (one byte in hexadecimal)	net_des (Network description to identify which network has failed) (In string)	neth_t is time stamp with either healthy or faulty.	Remarks
	Faulty Network	"Network Faulty"	01	Short description for		To be sent along with all parameters

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				faulty network		of faulty IoT to correlate it
	Healthy Network	"Network Healthy"	00	Short description for faulty network		To be sent along with all parameters of faulty IoT to correlate it

11. RDPMS Application Time stamp details: RDPMS Application informs Station gateway about its time with date in Indian Standard Time (IST) to make Station gateway sync with Application. It is represented in string. (DD-MM-YYYY HH:mm:ss.SSS ~~IST~~) example: 04-11-2025 16:27:45.123 ~~IST~~. It is represented as "clt" in data packet.

DD → Day (2 digits, 01–31)
MM → Month (2 digits, 01–12)
YYYY → Year (4 digits)
HH → Hour (00–23, 24-hour format)
mm → Minutes (00–59)
ss → Seconds (00–59)
SSS → Milliseconds (000–999)
IST → Indian Standard Time label

12. Asset make details: Asset make is being used in Standard data format, Standard APIs, UI & reports. Denoted in string format. It is represented as "make" in data packet.

13. Asset Model number: Asset model No is being used in Standard data format, UI & reports. Denoted in in string format. Details given below. (May be used, if required)
It is represented as "model" in data packet.

14. Sub Asset Type details: Sub Asset type is being used in Standard data format, UI & reports. Denoted in string format. Details given below. (May be used, if required)
It is represented as "attr1" and "attr2" in data packet.

15. Parameter location details: Parameter location is being used in Standard data format, UI & reports. Denoted in string format. Details given below.
It is represented as "prloc".

16. Vendor details: It is part of Standard data format, UI to uniquely identify Vendor of Field devices (station gateway and IOT/Sensor) and vendor of RDPMS Application in Indian Railways. Details given below with their short names used in data packet.

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Vendor code of RDPMS Application Software (represented as vcc) (Vendor code of RDPMS Application used in string)	Vendor name of RDPMS Application Software (represented as vcn) (Vendor name of RDPMS Application used in string)	Vendor code of station field devices (vgc) (code of Vendor of Station field devices in string)	Vendor name of station field devices (vgn) (Name of Vendor of Station Field devices in string)
xyz	Xyz technologies	abc	Abc technologies

17. Version details: It is part of Standard data format to uniquely identify Version of Application Software of Station Gateway to keep track of Software update in Indian Railways. Details given below.

S No	stngw_ver (version code used in standard data format in string)
1.	"1.0"

18. Command: "cmd" is used in data packets to define the packet as "IMAGE", "INFO", "TIME_SYNC" or "CONFIG". (ref-Annexure B).

19. Base address: it is used in Standard API for query address of RDPMS Application vendors. It should be possible to raise a query to RDPMS Application to take reports as described in Annexure F. This is to be used for APIs. Denoted as: base_address.

20. Sampling details of parameters:

Asset	Parameter	Standard operating value	Sampling
Point Machine	Operating Voltage	110 V DC	Absolute value every 20 ms OR sampling frequency as decided
	Operating Current	1.5 A	Absolute value every 20 ms OR sampling frequency as decided
	NWKR/RWKR Voltage	24 VDC	± 2% of Standard operating value OR % as decided
	Operation Time	4 seconds	± 2% of Standard operating value OR % as decided
Signal	Operating Voltage	110 V AC	± 2% of Standard operating value OR % as decided

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DC Track Circuit	Operating Current	130 mA	± 2% of Standard operating value OR % as decided
	HPR/HHPR/DPR	24 V DC	± 2% of Standard operating value OR % as decided
	TFC Charger Input	110 V AC	± 2% of Standard operating value OR % as decided
	TFC charger Output	10 V DC	± 2% of Standard operating value OR % as decided
	TFC Charger Current	2 A	± 2% of Standard operating value OR % as decided
	Voltage across Choke at Feed end	1 V DC	± 2% of Standard operating value OR % as decided
	Voltage across Rail at Feed end	5 V DC	± 2% of Standard operating value OR % as decided
	Feed End current at Rail	500 mA	± 2% of Standard operating value OR % as decided
	Relay End current from Rail	350 mA	± 2% of Standard operating value OR % as decided
	Voltage across Rail at Relay end	4 V DC	± 2% of Standard operating value OR % as decided
	Voltage across Choke at Relay end	1 V DC	± 2% of Standard operating value OR % as decided
	Voltage across Track Relay.	3.5 V DC	± 2% of Standard operating value OR % as decided
	Voltage at Loc after TR relay going to TPR	24 V DC	± 2% of Standard operating value OR % as decided
IPS	SMR/Battery output Voltage	110 V DC	± 2% of Standard operating value OR % as decided
	All 110 V AC	110 V AC	± 2% of Standard operating value OR % as decided
	All 24 V DC	24 V DC	± 2% of Standard operating value OR % as decided
	Others	Mid value of given range as per IPS Specification	± 2% of Standard operating value OR % as decided
Other Equipments		Mid value of given range/nominal operating value as per Specification	± 2% of Standard operating value OR % as decided

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Summary of nomenclatures

Sr	Description	Short name used	Remarks	Use case
1	Station gateway id	stngw_id	4 byte hexadecimal	Data format, APPLICATION
2	Gateway number id	stngw_number_id	One byte hexadecimal	Part of station_gw_id
3	Request id	rqi	Unique string to identify requests uniquely	Data format, APPLICATION
4	Response id	resi	Response ID for a request, this should be same as rqi	Data format, APPLICATION
5	Vendor code of RDPMS Application Software	vcc	string	Data format, APPLICATION , API
6	Vendor code of station field devices	vgc	string	Data format, APPLICATION, API
7	Version of Application Software of Station gateway	stngw_ver	string	Data format, APPLICATION
8	Station id	station_id	One byte Hexadecimal	Part of parameter_id
9	Station code	sc	string	Data format, APPLICATION, API
10	Station name	station_name	string	Data format, APPLICATION
11	Division id	division_id	One byte hexadecimal	Part of parameter_id
12	Division code	dc	string	APPLICATION , API
13	Division name	division_name	string	APPLICATION
14	Zone id	zone_id	One byte hexadecimal	Part of parameter_id
15	Zone code	zc	string	APPLICATION , API
16	Zone name	zone_name	string	APPLICATION
17	Asset number code	asset_number_code	string	Data format, API
18	Asset number id	asset_number_id	One byte hexadecimal	Data format, API
19	SMMS asset name	smms_asset_name	string	Data format, API
20	SMMS asset code	smms_asset_code	string	Data format, API
21	Equipment room number id	eqpmntroom_number_id	One byte hexadecimal	
22	Asset type id	asset_type_id	One byte hexadecimal	Part of parameter_id
23	Asset type code	asset_type_code	string	Data format, API
24	Equipment room type id	eqpmntroom_type_id	One byte hexadecimal	
25	Equipment room type code	eqpmntroom_type_code	string	
26	Asset make	make	string	Data format, API
27	Asset model No	model	string	Data format,

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28	Sub Asset type1	attr1	string	Data format,
29	Sub Asset type2	attr2	string	Data format,
30	Parameter id	para_id	4 byte hexadecimal	Data format,
31	Parameter type id	parameter_type_id	One byte hexadecimal	Part of parameter_id
32	Parameter representation id	parameter_representation_id	One byte hexadecimal	Part of parameter_id
33	Parameter representation code	parameter_representation_code	string	APPLICATION
34	Parameter representation name	parameter_representation_name	string	APPLICATION (for reference)
35	Parameter loc	prloc	string	Data format,
36	Parameter time	prt	hexadecimal	Data format,
37	Parameter value	prv	hexadecimal	Data format,
38	Sensor health code	sh_code	string	APPLICATION
39	Sensor health id	sh_id	One byte hexadecimal	Data format,
40	Sensor health time	sh_t	hexadecimal	
41	IoT health code	ioth_code	string	APPLICATION
42	IoT health id	ioth_id	One byte hexadecimal	Data format,
43	IoT health time	ioth_t	hexadecimal	
44	Station gateway health code	stngwh_code	string	APPLICATION
45	Station gateway health id	stngwh_id	One byte hexadecimal	Data format,
46	Station gateway health time	stngwh_t	hexadecimal	
47	Network health code	neth_code	string	
48	Network health id	neth_id	One byte hexadecimal	
49	Network description	net_des	string	
50	Network health time	neth_t	hexadecimal	
51	RDPMS Application time	clt	hexadecimal	Data format,
52	Configuration id	config_id	One byte hexadecimal	Data format,
53	Configuration value	config_val	hexadecimal	Data format,
54	command	cmd	Its value shall be one of these. "IMAGE", "INFO", "CONFIG", "TIME_SYNC"	Data format,

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Annexure-B

Standard Data Format between Station gateway/SMMS and RDPMS Application

1.1 Types of Data Packet in Standard Data format:

1.1.1 Asset details from SMMS (Signalling Maintenance Management System):

RDPMS Application shall take asset details from SMMS as per request and response format described in Annexure A & B.

1.1.2 Discovery: This data packet shall be based on publish and subscribe model. Station gateway sends a data packet to Application software for connection at the time of new installation Or at addition/deletion of sensors/IOT Or at reboot or power failure. (Annexure B) (Annexure A also to be referred for Standard Nomenclature).

1.1.3 Time Sync: This data packet shall be based on publish and subscribe model. Application software sends this data packet with Time sync information to station gateway in response to discovery data packet (para 10.2.2) to set the time of station gateway with application time.

In addition to sending this time sync packet in response to discovery packet, Application Software may also initiate a time sync data packet on its own and send it to station gateway to set the time at a definite interval of 7 days or as and when required or as decided by Zonal Railway. This duration should be programable. (Annexure B)(Annexure A also to be referred for Standard Nomenclature).

1.1.4 Time Sync Confirmation: This data packet shall be based on publish and subscribe model. This data packet pertains to confirmation of Time sync process done by Station gateway. Station gateway sends a data packet to Application software in response to Time sync packet received from application software after completing Time sync process. (Annexure B) (Annexure A also to be referred for Standard Nomenclature).

1.1.5 Information: This data packet shall be based on publish and subscribe model. This data packet pertains to Asset and other basic information. Application software initiates a request packet (with asset information taken from SMMS) in response to discovery packet from station gateway. In response to request packet, Station gateway sends information data packet (After modifying/adding information taken from SMMS) to Application software.

In addition to sending the request packet in response to discovery packet, Application software may also initiate a request packet (with asset information

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taken from SMMS) on its own to Station Gateway at a definite interval of one month or as and when required or as decided by Zonal Railway. This duration should be programable. In response to request packet, Station gateway sends information data packet (After modifying/adding information taken from SMMS) to Application software. (Annexure B)(Annexure A also to be referred for Standard Nomenclature).

1.1.6 Image: This data packet shall be based on publish and subscribe model. This data packet pertains to value of all parameters at approximate one instance with time stamp. Application software initiates a request packet in response to discovery packet from station gateway. In response to request packet, Station gateway sends image data packet to Application software.

In addition to sending the request packet in response to discovery packet, Application software may also initiate a request packet on its own to Station Gateway at a definite interval of 7 days or as and when required or as decided by Zonal Railway. This duration should be programmable. In response to request packet, Station gateway sends image data packet to Application software. (Annexure B)(Annexure A also to be referred for Standard Nomenclature).

1.1.7 Parameter:

1.1.7.1 Parameter (for fix time internal): This data packet shall be based on publish and subscribe model. This data packet pertains to
(a) Value of all those analog parameters (except parameter at 1.1.7.2) whose value changes with $\pm 2\%$ (or as defined) with respect to standard value (Refer Annexure A) with time stamp.

(b) Value of all Digital status changed with time stamp.

Station gateway sends this data packet to Application software at every 5 second (configurable) or as decided by Zonal Railway. So this data packet shall contain value of all parameters as per (a) & (b) during last 5 seconds (configurable).

1.1.7.2 Parameter (for event-based parameters): This data packet shall be based on publish and subscribe model. This data packet pertains to value of those analog parameters whose value is sampled every 20 ms (Voltage and current of Point Machine and Electric Lifting Barrier during operation) with time stamp of first sample. Sampling time 20 ms is configurable. Station gateway sends this data packet to Application software after completion of operation of point machine/ELB. (Annexure A & B to be referred for Standard Nomenclature)

1.1.8 RDPMS health: This data packet shall be based on publish and subscribe model. This data packet pertains to health status of sensor/IoT/Station gateway whether they are healthy or faulty with time stamp parameter. Station gateway sends a data

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packet to Application software at every 30 min Duration is configurable (Annexure B) (Annexure A also to be referred for Standard Nomenclature).

1.1.9 Configuration: This data packet shall be based on publish and subscribe model. This data packet pertains to configuration command details for configuring station gateway. Application software sends data packet to station gateway. (Annexure B) (Annexure A also to be referred for Standard Nomenclature).

1.1.10 Configuration Confirmation: This data packet shall be based on publish and subscribe model. This data packet pertains to confirmation of Configuration process done by Station gateway. Station gateway sends a data packet to Application software in response to configuration packet received from application software after completing configuration process. (Annexure B) (Annexure A also to be referred for Standard Nomenclature).

1.1.11 Telemetry data from RDPMS Application to SMMS (Signalling Maintenance Management System): SMMS shall take Telemetry data (Live parameter values of asset) from RDPMS application as per request and response data format described in Annexure A & B.

1.1.12 RDPMS Vendor shall send image data packet and parameter data packets to Railway Cloud (CRIS or as decided by Railway management) from Station gateway or Intermediate service platform (as decided by Railway management based on technical feasibility) for further analysis or use as Railway being the custodian of Data. Necessary APIs/data format should be designed in RDPMS system in consultation with Railway Cloud (CRIS or as decided by Railway management) to ensure smooth flow of desired packets. Details of packets have been given ahead in this Annexure B.

1.2 Intermediate Service Platform (ISP): The Intermediate Service Platform (ISP) is a secure and intelligent middleware to facilitate seamless communication between Station Gateways and the RDPMS Application. It manages bidirectional command and data exchanges, including time synchronization, system discovery, configuration commands, continuous parameter monitoring, and sensor health updates. ISP should be compliant to MQTT Specifications.

The ISP ensures reliable and secure data transmission through built-in acknowledgment mechanisms, retry logic, and encryption. It supports logging and auditing of all transactions, handles large data payloads, and is designed to scale across distributed infrastructure with fault tolerance for continuous uptime.

Furthermore, the ISP provides an option for **authorized third-party systems** to access real-time data streams via a **subscription model**.

ISP shall operate on a publish-subscribe communication model as follows:

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- **Publishers** (message senders) may publish messages to designated topics.
- **Subscribers** (message receivers) can subscribe to these topics to receive the corresponding messages.
- A single topic can support multiple concurrent subscribers, all of whom will receive the messages published to that topic.

Technical and Security Considerations:

Certificate-Based Authentication: All publishers and subscribers must be authenticated using digital certificates issued by a trusted Certificate Authority (CA). Only clients with valid certificates will be allowed to establish a connection to the ISP.

Certificate Blacklisting and Revocation: The ISP shall maintain a dynamic Certificate Revocation List (CRL) or implement Online Certificate Status Protocol (OCSP) to ensure that compromised, expired, or unauthorized certificates are denied access. Clients with blacklisted certificates will be automatically disconnected and denied any further interaction with the platform.

Encrypted Communication: All communication between publishers, subscribers, and the ISP shall be secured using Transport Layer Security (TLS) to ensure data confidentiality and integrity.

Access Control and Topic Authorization: Publishers and subscribers shall be allowed to publish or subscribe only to authorized topics, as defined in the access control policies. Unauthorized attempts will be logged and rejected.

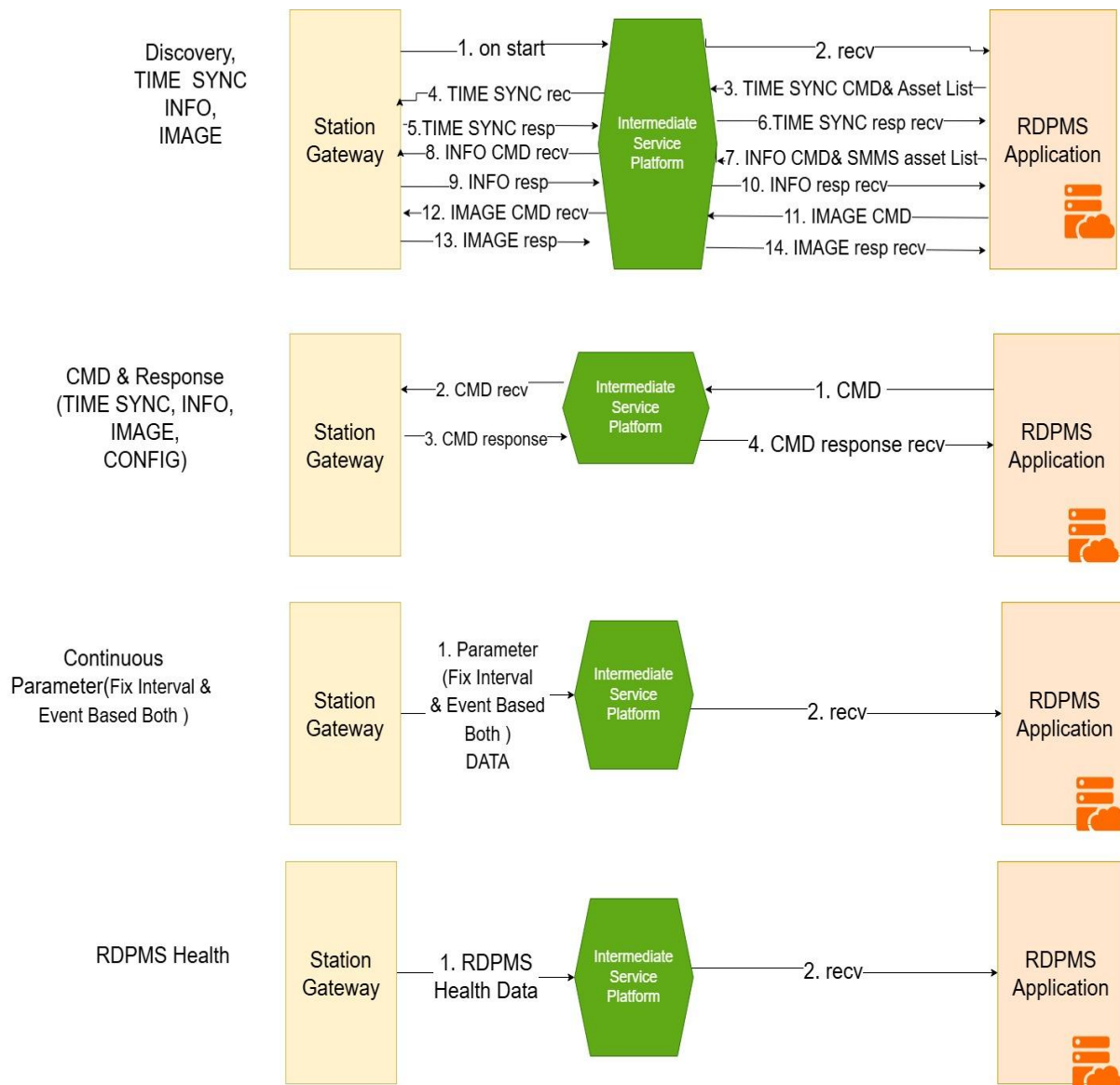
Logging and Monitoring: All publish-subscribe transactions, authentication events, and security violations shall be logged for auditing and monitoring purposes.

- 1.3 In future, If it is planned to use CCSP LAYER (C-DOT COMMON SERVICE PLATFORM) based on OneM2M standards using MQTT protocol in place of ISP, Vendors shall comply it. Vendor shall acquire basic understanding of oneM2M standards. (Refer <https://coi.cdota.in/docs/oneM2MGuideline.html>).

1.4 High level flow

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2 Long Name-Short Name Mapping for payloads: Pl refer Annexure A.

3. STATION GATEWAY

3.1 MQTT Packet and MQTT Topics for station gateway and RDPMS Application

MQTT Packet: <Client-Id> (as a simple string)

→ **ClientID will be set to station gateway id**

Publish topic:

discovery/{stngw_id}/{vcc}

parameter_f/{stngw_id}/{vcc}

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parameter_e/{stngw_id}/{vcc}
rdpms_health/{stngw_id}/{vcc}
cmd/{stngw_id}/{vcc}

Topic to be subscribed:

cmd/{vcc}/{ stngw_id }

4. RDPMS APPLICATION

4.1 MQTT Packet: <Client-Id> (as a simple string) → ClientID will be set to vcc

Publish topic:

cmd/{vcc}/{stngw_id }

Topic to be Subscribed topic:

discovery/{stngw_id }/{vcc}
parameter_f/{ stngw_id }/{vcc}
parameter_e/{stngw_id}/{vcc}
rdpms_health/{ stngw_id }/{vcc}
cmd/{ stngw_id }/{vcc}

Note : MQTT topics are designed with following naming convention
=> {role}/{sender_id}/{receiver_id}

5. MESSAGE EXCHANGE FORMAT BETWEEN SMMS, STATION GATEWAY AND RDPMS APPLICATION

5.1 RDPMS Application to SMMS request for Asset Data information

API URL : /get_asset_list/{zc}/{dc}/{sc}

The API will be called by RDPMS Application by providing zone code, division code and station code (as provided in this document earlier).

Example: /get_asset_list/NCR/PYRJ/MIU

SMMS host will exchange API Security Key to enable the communication.

Request packet from RDPMS Application to SMMS for Asset Data information	
Structure	Sample

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<pre>{ "rqi": "string-unique", "vcc": "string", "zc": "string", // zone code "dc": "string", // division code "sc": "string", // station code }</pre>	<pre>{ "rqi": "01JM50ETKYVV1WHC9KVFVPAEP", "vcc": "XYZ", "zc": "NCR", "dc": "PYRJ", "sc": "MIU", }</pre>
Response packet from SMMS to RDPMS Application for Asset Data information	
Structure	Sample
<pre>{ "resi": "string-unique", "vcc": "string", "zc": "string", // zone code "dc": "string", // division code "sc": "string", // Station code "info": [{ "smms_asset_name": "string", // readable name of asset that will be used "smms_asset_code": "string", "additional_info": [{ "key": "string", "value": "string", }] }] }</pre>	<pre>{ "resi": "01JM50ETKYVV1WHC9KVFVPAEP ", "vcc": "XYZ", "zc": "NCR", "dc": "PYRJ", "sc": "MIU", "info": [{ "smms_asset_name": "PT-01", "smms_asset_code": "EOPMIU00021", "additional_info": [{ "key": "make", "value": "ABC" }, { "key": "modelabc1", "value": "UVZ" }] }, { "smms_asset_name": "PT-01", "smms_asset_code": "EOPMIU00023", "additional_info": [{ "key": "make", "value": "ABC" }, { "key": "model", "value": "UVZ" }] }] }</pre>

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5.2 Station Gateway to RDPMS Application Discovery data packet (This will be sent on start of station gateway as described)

Topic : *discovery/{ stngw_id }/{vcc}*

Structure	Sample
<pre>{ "rqi" : "string-unique", "stngw_id" : "string (4 Byte hexadecimal)", "vcc" : "string", "vgc" : "string", "stngw_ver" : "string" }</pre>	<pre>{ "rqi" : "01JM4ZZ9AQ384MW63JT6EBVVC7", "stngw_id" : "456523AB ", "vcc" : "XYZ", "vgc" : "ABC", "stngw_ver" : "1.0" }</pre>

5.3 RDPMS Application to Station gateway time sync data packet (in response to discovery packet as well as and independent request)

Topic : *cmd/{vcc}/{stngw_id}*

Structure	Sample
<pre>{ "rqi" : "string-unique", "stngw_id" : "string (4 Byte hexadecimal)", "vcc" : "string", "vgc" : "string", "stngw_ver" : "string ", "cmd" : "TIME_SYNC", "clt" : "DD-MM-YYYY HH:mm:ss.SSS" // in string }</pre>	<pre>{ "rqi" : "01JMCGE4EK9N1481N8KQQHH3SJ", "stngw_id" : "456523AB", "vcc" : "XYZ", "vgc" : "ABC", "stngw_ver" : "1.0", "cmd" : "TIME_SYNC", "clt" : "04-11-2025 16:27:45.123" }</pre>

5.4 Station gateway to RDPMS Application Time Sync Confirmation data packet

Topic : *cmd/{ stngw_id }/{vcc}*

Structure	Sample
<pre>{ "resi" : "string-unique", "stngw_id" : "string (4 Byte hexadecimal)", "vcc" : "string", "vgc" : "string", "stngw_ver" : "string ", "cmd" : "TIME_SYNC", "clt" : "DD-MM-YYYY HH:mm:ss.SSS" // in string }</pre>	<pre>{ "resi" : "01JMCGE4EK9N1481N8KQQHH3SJ", "stngw_id" : "456523AB", "vcc" : "XYZ", "vgc" : "ABC", "stngw_ver" : "1.0", "cmd" : "TIME_SYNC", "clt" : "04-11-2025 16:27:45.123" }</pre>

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5.5 RDPMS Application to Station gateway Information request data packet (in response to discovery packet as well as and independent request)
(Note: Asset Information received from SMMS is also sent to station gateway through this)

Topic :cmd/{vcc}/{stngw_id}

Structure	Sample
<pre> { "rqi": "string-unique", "stngw_id": "string (4 Byte hexadecimal)", "vcc": "string", "vgc": "string", "stngw_ver": "string ", "cmd": "INFO" "info": [{ "smms_asset_name ": "string", // readable name of asset that will be used "smms_asset_code": "string", "additional_info": [{ "key": "string", "value": "string", }] }] }</pre>	<pre> { "rqi": "01JM50ETKYV1WHC9KVFEVPAEP", "stngw_id": "456523AB", "vcc": "XYZ", "vgc": "ABC", "stngw_ver": "1.0", "cmd": "INFO" "info": [{ "smms_asset_name ": "PT-01", "smms_asset_code": "EOPMIU00021", "additional_info": [{ "key": "make", "value": "ABC" }, { "key": "model", "value": "UVZ" }] }, { "smms_asset_name ": "PT-01", "smms_asset_code": "EOPMIU00023", "additional_info": [{ "key": "make", "value": "ABC" }, { "key": "model", "value": "UVZ" }] }] }</pre>

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5.6 Station gateway to RDPMS Application Information data packet

Topic :cmd/{ stngw_id }/{vcc}

Structure	Sample
<pre> { "resi": "string-unique", "stngw_id": "string (4 Byte hexadecimal)", "vcc": "string", "vgc": "XYZ", "stngw_ver": "string", "sc": "string", // Station code "station_name": "String value", // station name "cmd": "INFO", "info": [{ "asset_number_code": "string", // readable name of asset that will be used "asset_number_id": "string (1 Byte hexadecimal)", "asset_type_code": "string", "smms_asset_code": "string", // readable name of asset that will be used "additional_info": { "make": "string", // make of asset "model": "string", // model of asset "attr1": "string", // more attribute info (dynamic verndor can add more as they see fit "attr2": "string" } "para_id": [], // list of "string (4 Byte hexadecimal)", "prloc": [], // list of string that represent monitoring locations for each parameterseg LB-1, RR etc }] }</pre>	<pre> { "resi": "01JM50ETKYVV1WHC9KVFVPAEP ", "stngw_id": "456523AB", "vcc": "XYZ", "vgc": "ABC", "stngw_ver": "1.0", "sc": "LKO", "station_name": "Lucknow", "cmd": "INFO", "info": [{ "asset_number_code": "PT-01", "asset_number_id": "01", "asset_type_code": "EOPMIU00023", "smms_asset_code": "EOPMIU00023", "additional_info": { "make": "ABC", "model": "UVZ", "attr1": "TWS" }, "para_id": ["0001000C", "0001000D", "0001120A"], "prloc": ["LB-1", "RR", "LB-2"] }, { "asset_number_code": "PT-02", "asset_number_id": "02", "asset_type_code": "EOPMIU00023", "smms_asset_code": "EOPMIU00024", "additional_info": { "make": "DEF", "model": "KLM", "attr1": "IRS" }, "para_id": ["0002000A", "0002000B"], "prloc": ["LB-3", "RR"] }] }</pre>

Note : Parameter ids have been defined as per Annexure A.

0001000C	Normal current of PT-01
0001000D	Reverse current of PT-01
0001120A	Normal operation voltage of PT-01
0001120B	Reverse operation Voltage of PT-01
00012604	NWKR Digital

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00012605 RWKR Digital

5.7 RDPMS Application to Station gateway Image request data packet (in response to discovery packet as well as and independent request)

Topic :cmd/{vcc}/{stngw_id }

Structure	Sample
<pre>{ "rqi": "string-unique", "stngw_id": "string (4 Byte hexadecimal)", "cmd": "IMAGE" }</pre>	<pre>{ "rqi": "01JMCG6R4MC37N3XVC2G2GT16W", "stngw_id": "456523AB", "cmd": "IMAGE" }</pre>

5.8 Station gateway to RDPMS Application Image data packet

Topic :cmd/{stngw_id }/{vcc}

Structure	Sample
<pre>{ "resi": "string-unique", "stngw_id": "string (4 Byte hexadecimal)", "cmd": "IMAGE", "image": [{ "para_id": "parameter id string (4 Byte hexadecimal)", "prv": [], //last float value for the para_id measured. (This shall exclude event type parameter like current of point machine.) "prt": [] // last time stamp of the measurement value in format DD-MM-YYYY HH:mm:ss.SSS // string }] }</pre>	<pre>{ "resi": "01JMCG6R4MC37N3XVC2G2GT16W", "stngw_id": "456523AB", "cmd": "IMAGE", "image": [{ "para_id": "0001000C", "prv": [12.34], "prt": ["04-11-2025 16:27:45.123"] }, { "para_id": "0001000D", "prv": [98], "prt": ["04-11-2025 16:27:45.123 "] }] }</pre>

5.9 Station gateway to RDPMS Application Parameter data packet on fixed intervals (excluding event based parameter)

Topic :parameter_f/{stngw_id }/{vcc}

Structure	Sample
<pre>{ "rqi": "string-unique", "stngw_id": "string (4 Byte hexadecimal)", "parameters": [{</pre>	<pre>{ "rqi": "01JMCG7JS749C1NEGP9BWXE1XP", "stngw_id": "456523AB", "parameters": [{</pre>

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<pre> "para_id" : "parameter id string (4 Byte hexadecimal)", "prv" : [], //array of all value changes in float happened within (x seconds as mentioned in the Specification. "prt" : [] // array of station gateway Time stamp –in DD-MM-YYYY HH:mm:ss.SSS of all corresponding values in string },] }</pre>	<pre> "para_id" : "0001000C", "prv" : [1.34, 1.35, 1.45, 1.46], "prt" :["04-11-2025 16:27:45.123" , "04-11-2025 16:27:45.125" , "04-11-2025 16:27:45.127" , " 04-11-2025 16:27:45.130"] }, { "para_id": "0001000B", "prv" :[1.33, 1.36, 1.44, 1.45], "prt" :["04-11-2025 16:27:45.123" , "04-11-2025 16:27:45.125" , "04-11-2025 16:27:45.127" , " 04-11-2025 16:27:45.130"] }]</pre>
---	--

5.10 Station gateway to RDPMS Application Parameter data packet after completing the event for each event based parameter as described in 1.1.7 of this Annexure

Topic : *parameter_e/{stngw_id}/{vcc}*

Structure	Sample
<pre> { "rqi": "string-unique", "stngw_id": "string (4 Byte hexadecimal)", "parameters": [{ "para_id" : "parameter id string (4 Byte hexadecimal)", // (for event based parameter like Point machine & ELB current) "prv" : [], //array of all values in float happened within event period "prt" : [] // time stamp of first value in DD-MM-YYYY HH:mm:ss.SSS (string). Time stamp of further values can be calculated by adding 20 ms or as configured at application level },] }</pre>	<pre> { "rqi": "01JMCG7JS749C1NEGP9BWXE1XP", "stngw_id": "456523AB", "parameters": [{ "para_id" : "0001000C", "prv" : [1.34, 1.35, 1.45, 1.46], "prt" : "04-11-2025 16:27:45.123" },] }</pre>

5.11 Station gateway to RDPMS Application RDPMS health data packet

Topic : *rdpms_health/{stngw_id}/{vcc}*

Structure	Sample
Packet to be sent every 30 min OR as decided by Zonal Railway	
{	{

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<pre> "rqi": "string-unique", "stngw_id": "string (4 Byte hexadecimal)", "rdpms_health": [{ "para_id": "parameter id string (4 Byte hexadecimal)", "sh_id": "string (1 Byte hexadecimal)", // 01 means faulty "sh_t": "time stamp" }] { "para_id": "all parameter ids pertaining to faulty IoT, string (4 Byte hexadecimal)", "ioth_id": "string (1 Byte hexadecimal)", // 01 means faulty "ioth_t": "time stamp" } } { "stngwh_id": "string (1 Byte hexadecimal)", // 01 means faulty" "stngwh_t": "time stamp" } } { "para_id": "all parameter ids pertaining to faulty network, string (4 Byte hexadecimal)", "neth_id": "string (1 Byte hexadecimal)", // 01 means faulty" "neth_t": "time stamp" "net_des": "description of network" } }] } </pre>	<pre> "rqi": "01JMC GA8Z2JJ31NPG06NPBNG42", "stngw_id": "456523AB", "rdpms_health": [{ "para_id": "0001000D", "sh_id": "01" }, { "para_id": "0001000E", "sh_id": "01" "sh_t": "04-11-2025 16:27:45.123" }] { "para_id": ["0001000E", "0001000F", "0001000D"] "ioth_id": "01" "ioth_t": "04-11-2025 16:27:45.123" } } { "stngwh_id": "00", "stngwh_t": "04-11-2025 16:27:45.123" } } { "para_id": ["0001000E", "0001000F", "0001000D"] "neth_id": "00", "neth_t": "04-11-2025 16:27:45.123" "net_des": "xyz" }] } </pre>
--	--

5.12 RDPMS Application to Station gateway Configuration data packet

Topic :cmd/{vcc}/{stngw_id }

Structure	Sample
<pre> { "rqi": "string-unique", "stngw_id": "string (4 Byte)", "cmd": "CONFIG", "config": [{ "config_id": "string", // string (one byte hexadecimal) "config_val": float, // representing config value }] } </pre>	<pre> { "rqi": "01JMC GE4EK9N1481N8KQ QHH3SJ", "stngw_id": "456523AB", "cmd": "CONFIG", "config": [{ "config_id": "01", "config_val": 01 }, { "config_id": "02", "config_val": 01 }] } </pre>

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}	]
	}

5.13 Station gateway to RDPMS Application Configuration Confirmation data packet

Topic :cmd/{stngw_id}/{vcc}

Structure	Sample
<pre>{ "resi": "string-unique", "stngw_id": "string (4 Byte hexadecimal)", "cmd": "CONFIG", "config": [{ "config_id": "string", // string (one byte hexadecimal) to confirm that command has been executed "config_val" : float, // representing config value },] }</pre>	<pre>{ "resi": "01JMCGE4EK9N1481N8KQQHH3SJ", "stngw_id": "456523AB", "cmd": "CONFIG", "config": [{ "config_id": "01", "config_val": 01 }, { "config_id": "02", "config_val": 01 }] }</pre>

5.14 SMMS to RDPMS Application for Telemetry Data

API URL : /get_asset_telemetry/{zc}/{dc}/{sc}/{smms_asset_code}/{para_id}

The API will be called by SMMS by providing zone code, division code, station code , smms asset code and parameter id (as provided in this document earlier).

If smms_asset_code is not provided, then the response will contain all asset telemetry data of that station. Also, if para_id is null then data will be sent for all para_id, otherwise only for specified para_id.

Request from SMMS to RDPMS Application for Telemetry Data	
Structure	Sample
<pre>{ "rqi": "string-unique", "vcc": "string", "zc": "string", // zone code "dc": "string", // division code "sc": "string", // Station code "smms_asset_code": "string", "para_id": [], // list of "string (4 Byte hexadecimal)" }</pre>	<pre>{ "rqi": "01JMCGE4EK9N1481N8KQQHH3SJ", "vcc": "XYZ", "zc": "NCR", "dc": "PYRJ", "sc": "MIU", "smms_asset_code": "EOPMIU00021", "para_id": ["0001000C", "0001000D", "0001120A"] }</pre>
Response from RDPMS Application to SMMS for Telemetry Data	

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Structure	Sample
<pre> { "resi": "string-unique", "vcc": "string", "zc": "string", // zone code "dc": "string", // division code "sc": "string", // Station code "telemetry_data": [{ "smms_asset_code": "string" }], "parameters": [{ "para_id" : "parameter id string (4 Byte hexadecimal)", "prv" : float, // last available value "prt" : Time stamp –in DD-MM-YYYY HH:mm:ss.SSS of corresponding values in string. Time stamp should be approximately same as of requested packet to ensure that parameter value is current. }], } </pre>	<pre> { "resi": "01JM50ETKYVV1WHC9KVFEVPAEP ", "vcc": "XYZ", "zc": "NCR", "dc": "PYRJ", "sc": "MIU", "telemetry_data": [{ "smms_asset_code": "EOPMIU00021", "parameters": [{ "para_id" : "0001000C", "prv" : 1.34, "prt" : "04-11-2025 16:27:45.123" }, { "para_id": "0001000B", "prv" : 1.33, "prt" : "04-11-2025 16:27:45.123" }] }], } </pre>

6. SECURITY

Security will be achieved by mTLS

mTLS Implementation Guide for Three-Entity Architecture

What is a Certificate Authority (CA)?

A Certificate Authority (CA) is like a trusted third party that issues digital ID cards (certificates) to people or systems like government office issues Aadhaar cards or passports:

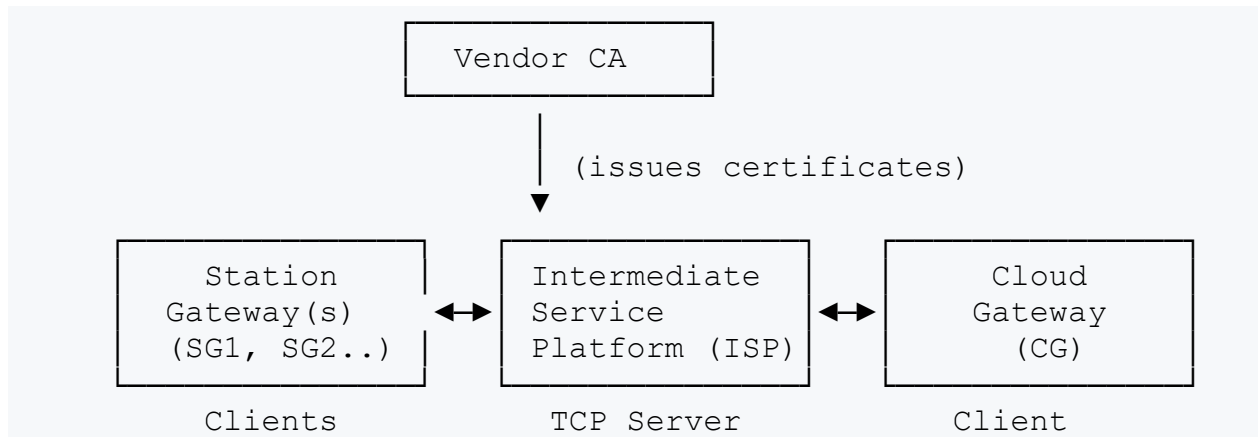
1. The CA verifies your identity before issuing you a certificate
2. The certificate contains your public information and is signed by the CA
3. When you present this certificate to others, they can trust you're legitimate because they trust the CA that issued your certificate
4. The CA's signature on your certificate is like an official stamp that can't be forged

In this system, each vendor acts as their own CA, issuing certificates to all their entities. This allows all entities within a vendor's ecosystem to trust each other's identities.

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Architecture Overview



Part 1: Understanding mTLS in this architecture of ISP, RDPMS Application and station gateway

mTLS Authentication Process

1. Initialization:

- Vendor acts as the Certificate Authority (CA)
- Each entity receives its own private key and certificate
- All certificates are signed by the Vendor CA

2. Connection Process:

- When an SG or CG (For RDPMS Application) connects to the ISP:
 - TCP connection established
 - TLS handshake initiated
 - Client sends "Client Hello" with supported cipher suites
 - Server responds with "Server Hello" and chosen cipher suite
 - Server sends its certificate
 - Client verifies server certificate using CA certificate
 - Client sends its certificate
 - Server verifies client certificate using CA certificate
 - Key exchange occurs and session keys established
 - Encrypted application data flows after successful TLS handshake

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3. Certificate Verification:

- Each entity validates the other's certificate by:
 - Verifying the digital signature using the CA's public key
 - Checking certificate expiration dates
 - Confirming the certificate isn't revoked (via CRL (certificate revocation list) or OCSP (online certificate status protocol))
 - Verifying the certificate's purpose is appropriate

4. Encryption/Decryption Process:

- After successful mTLS handshake, all communication is encrypted
- Data is encrypted using negotiated session keys (not the certificate keys)
- Common cipher suites shall be chosen as prescribed in Chapter 2 (List of Cryptographic Controls), Table-2 of ITSAR No: ITSAR001962411, issued by National Centre for Communication Security (NCCS), Department of Telecommunications, Ministry of Communications, Government of India.

Part 2: Step-by-Step Certificate Creation Process

Files Security Classification

Private (Never Share or Expose):

- All private keys (*.key files)
- CA private key (ca.key) is especially sensitive and should have the highest protection

Restricted (Internal Use Only):

- CA certificate signing request (ca.csr)
- Entity certificate signing requests (*.csr files)
- CA database files (index.txt, serial, etc.)

Public (Can Be Shared):

- CA certificate (ca.crt) - must be distributed to all entities
- Entity certificates (isp.crt, cg.crt, sg1.crt, etc.)

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- Certificate Revocation Lists (CRLs)

1. Create Certificate Authority (CA)

```
# Create directory structure
mkdir -p ca/certs ca/private ca/newcerts ca/crl
touch ca/index.txt
echo 1000 > ca/serial

# Create CA configuration file
cat > ca/openssl.cnf << EOF
[ ca ]
default_ca = CA_default

[ CA_default ]
dir                = ./ca
certs              = $dir/certs
crl_dir            = $dir/crl
new_certs_dir      = $dir/newcerts
database           = $dir/index.txt
serial             = $dir/serial
RANDFILE           = $dir/private/.rand

private_key        = $dir/private/ca.key
certificate         = $dir/certs/ca.crt

crlnumber          = $dir/crlnumber
crl                = $dir/crl/ca.crl
crl_extensions     = crl_ext
default_crl_days   = 30

default_md         = sha256
name_opt           = ca_default
cert_opt           = ca_default
default_days       = 3650
preserve          = no
policy             = policy_strict

[ policy_strict ]
countryName        = match
stateOrProvinceName = match
organizationName   = match
organizationalUnitName = optional
commonName         = supplied
emailAddress       = optional

[ req ]
default_bits       = 4096
distinguished_name = req_distinguished_name
```

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```

string_mask      = utf8only
default_md       = sha256
x509_extensions  = v3_ca

[ req_distinguished_name ]
countryName      = Country Name (2 letter code)
stateOrProvinceName = State or Province Name
localityName     = Locality Name
0.organizationName = Organization Name
organizationalUnitName = Organizational Unit Name
commonName       = Common Name
emailAddress      = Email Address

[ v3_ca ]
subjectKeyIdentifier = hash
authorityKeyIdentifier = keyid:always,issuer
basicConstraints = critical, CA:true
keyUsage = critical, digitalSignature, cRLSign, keyCertSign

[ server_cert ]
basicConstraints = CA:FALSE
nsCertType = server
nsComment = "OpenSSL Generated Server Certificate"
subjectKeyIdentifier = hash
authorityKeyIdentifier = keyid,issuer:always
keyUsage = critical, digitalSignature, keyEncipherment
extendedKeyUsage = serverAuth

[ client_cert ]
basicConstraints = CA:FALSE
nsCertType = client
nsComment = "OpenSSL Generated Client Certificate"
subjectKeyIdentifier = hash
authorityKeyIdentifier = keyid,issuer:always
keyUsage = critical, digitalSignature, keyEncipherment
extendedKeyUsage = clientAuth
EOF

# Generate CA private key (KEEP THIS SECURE!)
openssl genrsa -out ca/private/ca.key 4096

# Generate CA certificate (This can be shared publicly)
openssl req -config ca/openssl.cnf -key ca/private/ca.key -new -x509 -
days 36500 -sha256 -extensions v3_ca -out ca/certs/ca.crt -subj
"/C=IN/ST=UttarPradesh/L=Noida/O=Bitcomm/OU=IoT/CN=VendorCA"

# View CA certificate
openssl x509 -noout -text -in ca/certs/ca.crt

```

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2. Create Certificate for the Intermediate Service Platform (ISP)

```
# Generate ISP private key (KEEP THIS SECURE!)
openssl genrsa -out isp.key 2048

# Create Certificate Signing Request (CSR) - temporary file, not sensitive once used
openssl req -new -key isp.key -out isp.csr -subj "/C=IN/ST=UttarPradesh/L=Noida/O=Bitcomm/OU=IoT/CN=isp"

# Sign the certificate with CA
openssl ca -config ca/openssl.cnf -extensions server_cert -days 36500 -notext -md sha256 -in isp.csr -out isp.crt

# Verify the certificate
openssl verify -CAfile ca/certs/ca.crt isp.crt

# The isp.crt file can be shared publicly
# The isp.key file must be kept private and secure
```

3. Create Certificate for Cloud Gateway (CG)

```
# Generate CG private key (KEEP THIS SECURE!)
openssl genrsa -out cg.key 2048

# Create Certificate Signing Request (CSR) - temporary file, not sensitive once used
openssl req -new -key cg.key -out cg.csr -subj "/C=IN/ST=UttarPradesh/L=Noida/O=Bitcomm/OU=IoT/CN=cloud-gateway"

# Sign the certificate with CA
openssl ca -config ca/openssl.cnf -extensions client_cert -days 36500 -notext -md sha256 -in cg.csr -out cg.crt

# Verify the certificate
openssl verify -CAfile ca/certs/ca.crt cg.crt

# The cg.crt file can be shared publicly
# The cg.key file must be kept private and secure
```

4. Create Certificates for Station Gateways (SGs)

```
# This example creates certificates for SG1 - repeat for each SG with a unique common name
# Generate SG1 private key (KEEP THIS SECURE!)
openssl genrsa -out sg1.key 2048
```

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```
# Create Certificate Signing Request (CSR) - temporary file, not sensitive once used
openssl req -new -key sg1.key -out sg1.csr -subj "/C=IN/ST=UttarPradesh/L=Noida/O=Bitcomm/OU=IoT/CN=station-gateway-1"

# Sign the certificate with CA
openssl ca -config ca/openssl.cnf -extensions client_cert -days 36500 -notext -md sha256 -in sg1.csr -out sg1.crt

# Verify the certificate
openssl verify -CAfile ca/certs/ca.crt sg1.crt

# The sg1.crt file can be shared publicly
# The sg1.key file must be kept private and secure

# Repeat for additional SGs (sg2, sg3, etc.)
```

5. Distributing Certificates

1. For each entity, distribute:

- Its own private key (keep secure!)
- Its own certificate (can be public)
- The CA certificate (can be public)

2. Distribution Matrix:

Entity	Files Needed	Security Classification
ISP	isp.key	PRIVATE - Keep secure
	isp.crt	PUBLIC - Can be shared
	ca.crt	PUBLIC - Can be shared
CG	cg.key	PRIVATE - Keep secure
	cg.crt	PUBLIC - Can be shared
	ca.crt	PUBLIC - Can be shared

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Entity	Files Needed	Security Classification
SG1	sg1.key	PRIVATE - Keep secure
	sg1.crt	PUBLIC - Can be shared
	ca.crt	PUBLIC - Can be shared
SG2	sg2.key	PRIVATE - Keep secure
	sg2.crt	PUBLIC - Can be shared
	ca.crt	PUBLIC - Can be shared
...	...	...

Part 3: TCP Server and Client Configuration

Configuring the ISP (TCP Server)

Mqtt servers support mTLS. Here's a typical configuration structure:

```
# Example server configuration
port = 8883
ca_certificate = /path/to/ca.crt
server_certificate = /path/to/isp.crt
server_key = /path/to/isp.key
require_client_certificate = true
verify_client = true
nection
```

Part 4: Certificate Revocation and Management

Certificate Revocation

```
# Create a Certificate Revocation List (CRL)
openssl ca -config ca/openssl.cnf -gencrl -out ca/crl/ca.crl

# Revoke a compromised certificate
```

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```
openssl ca -config ca/openssl.cnf -revoke ca/newcerts/1001.pem -
crl_reason keyCompromise

# Regenerate the CRL after revocation
openssl ca -config ca/openssl.cnf -gencrl -out ca/crl/ca.crl

# The CRL (ca.crl) can be publicly shared
```

Best Practices

1. **Key Length:** Use at least 2048-bit RSA keys (4096 for CA)
2. **Certificate Validity:**
 - CA certificates: 5-10 years
 - Entity certificates: 1-3 years
3. **Password Protection:**
 - Protect private keys with strong passwords
 - Use secure key storage where available
4. **Certificate Naming:**
 - Use meaningful Common Names (CN) to identify entities
 - Consider including entity IDs in certificate Subject Alternative Names (SAN)
5. **Automation:**
 - Script certificate creation for multiple SGs
 - Consider using automated certificate management tools for large deployments

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Annexure – C

Standard Failure/Prediction Logics and AI/Model guidelines for RDPMS Application

1. PREDICTION LOGICS AND FAILURE LOGICS OF ASSETS: Static Logics of failure and prediction for all types of Signals, Point machine, DC Track circuit have been given. However, logics for rest of the assets, Earthing, SPDs, temperature and humidity and data taken from diagnostic ports of various types of assets like IPS, Axle counter etc. should be prepared by vendors and approved by Zonal Railways under intimation to RDSO so that this Annexure C can be suitably amended based on experience after review.

2. AI/ML Model guideline: AI/ML models for predictive Alerts should be developed taking various inputs from asset parameter. Tentative inputs are being given as following. Vendors shall develop models based on experience of failures/prediction, Railway inputs, Alerts feedback by Railways, data collected by RDPMS, temperature, humidity and weather conditions.

Tentative inputs for models:

IPS Battery:

1. Battery charging current pattern (during partially/fully discharged to charged condition)
2. Battery discharging current pattern (during charged to partially/fully discharged condition).
3. Battery Voltage pattern during charging and discharging
4. Battery Voltage pattern when it is fully charged.
5. Alert feedback.
6. Weather condition including temperature and humidity
7. Any other

Point Machine:

1. Motor Current and Voltage pattern during entire operation from normal to reverse and vice-versa.
2. Normal and Reverse operation time.
3. Voltage and current pattern during slipping.
4. Alert feedback.
5. Weather condition including temperature and humidity
6. Any other

Electric Lifting Barrier:

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1. Motor Current and Voltage pattern during entire operation from Close to open and vice-versa.
2. Opening and Closing operation time.
3. Alert feedback.
4. Weather condition including temperature and humidity
5. Any other

Track Circuit:

Track circuit Battery:

1. Battery charging current (during partially/fully discharged to charged condition) pattern.
2. Battery discharging current (during charged to partially/fully discharged condition) pattern.
3. Battery Voltage pattern during charging and discharging
4. Battery Voltage pattern when it is fully charged.
5. Alert feedback.
6. Weather condition including temperature and humidity
7. Any other

Track circuit other issues:

1. Pattern of Feed End current, Receive end Current and current through Bal-last/Sleeper
2. Pattern of various other voltages
3. Alert feedback.
4. Weather condition including temperature and humidity.
5. Any other

Signal:

1. Voltage and current pattern of LED Signal.
2. Alert feedback.
3. Weather condition including temperature and humidity
4. Any other

Other Signalling Assets:

1. Voltage and current pattern of Signalling Asset.
2. Alert feedback.
3. Weather condition including temperature and humidity
4. Any other

Note: Logics based on threshold values with associated cause have been given below for predictive and failure alerts. However, vendors shall develop AI/ML models to predict alerts over and above hard logics based on threshold values. Additional predictive Alerts with cause should be prepared by vendors and approved by Zonal Railways under intimation to RDSO so that Annexure C of this Specification can be suitably amended based on experience after review.

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2.1 (a) Prediction Logics of Integrated Power Supply

Note:

1. These logics are primitive logics for Prediction of failures. However, in addition to these primitive logics, these primitive logics, various AI/ML Techniques and learning models need to be developed and applied by Vendors for prediction of failures. The models used by them shall be shared with Zonal Railways and RDSO.
2. Comparisons in logics have been made with deviation in parameter value in terms of certain % of avg value OR deviation from Min/Max safe. Although % deviation has been given for predictive logics for reference, however, % may be modified as decided by Zonal Railway. **This % is being denoted by "LD" for lower side deviation. For IPS LD = 90.**
3. For Predictive logic, the analog continuous deviation (say < 80% of avg value) should occur for at least certain duration to avoid predictive alerts for short transients. Indicative timing has been mentioned against logics. However, time duration can be decided by Zonal Railways as per site requirement.
4. The deviation should occur for entire duration of at least 1 operation in discrete cases like voltage and current of point machine.
5. Prediction Logics have been given based on defined parameters. However, Vendors may add new logics and modify given logics based on field experience OR otherwise also under approval of Railways and under intimation to RDSO so that suitable amendment may be issued in Specification after review.

	cause_detail	Logic	cause_code	Remarks on logics
1	IPS 220 AC Voltage is low.	$V_{IPS\ I/P} < \text{LD \% of avg value OR Min safe value}$	IPS I/P VOLT LOW	Logic to be checked for 15 sec
2	IPS 110 DC O/P Voltage low.	$V_{IPS\ 110\ DC} < \text{LD \% of avg value OR Min safe value}$ &&	IPS 110 DC BATT VOLT LOW	Avg value of $V_{IPS\ 110\ DC}$ should be calculated when IPS Battery charging Current ie $I_{IPS\ Batt}$

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		$I_{IPS \text{ BATT CHAR}} > 0^+$		Char 110 DC is more than 0^+. 0^+ can be taken reasonable positive value to confirm that current is not zero. Logic to be checked for 15 sec when $I_{IPS \text{ Batt Char 110 DC}}$ is more than 0^+.
3	IPS 110 DC O/P Voltage low.	$V_{IPS \text{ 110 DC}} < \text{LD \% of avg value OR Min safe value}$ && $I_{IPS \text{ BATT CHAR}} < 0$	IPS 110 DC BATT VOLT LOW	Avg value of $V_{IPS \text{ 110 DC}}$ should be calculated when IPS Battery charging Current is negative ie Battery is on load. Logic to be checked for 15 sec when $I_{IPS \text{ Batt Char 110 DC}}$ negative ie Battery is on load.
4	IPS 110 AC Sig-1 O/P Voltage low.	$V_{IPS \text{ SIG-1 110 AC}} < \text{LD \% of avg value OR Min safe value}$	IPS 110 AC Sig-1 VOLT LOW	Logic to be checked for 15 sec. To be done for all similar modules, if required.
5	IPS 110 AC TR-1 O/P Voltage low.	$V_{IPS \text{ TR-1 110 AC}} < \text{LD \% of avg value OR Min safe value}$	IPS 110 AC TR-1 VOLT LOW	Logic to be checked for 15 sec. To be done for all similar modules, if required.

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6	IPS SMR-1 Voltage low.	$V_{IPS\ SMR-1\ 110\ DC} < LD\ \% \text{ of avg value OR Min safe value}$	IPS SMR-1 VOLT LOW	Logic to be checked for 15 sec. To be done for all similar modules, if required.
7	IPS DC-DC Relay Internal Voltage low.	$V_{IPS\ DC\ R\ INT} < LD\ \% \text{ of avg value OR Min safe value}$	IPS DC-DC R INT VOLT LOW	Logic to be checked for 15 sec.
8	IPS DC-DC Relay External Voltage Low.	$V_{IPS\ DC\ R\ EXT} < LD\ \% \text{ of avg value OR Min safe value}$	IPS DC-DC R EXT VOLT LOW	Logic to be checked for 15 sec.
9	IPS DC-DC AXLE COUNTER Voltage Low.	$V_{IPS\ DC\ AXLE\ C} < LD\ \% \text{ of avg value OR Min safe value}$	IPS DC-DC AXLE C VOLT LOW	Logic to be checked for 15 sec.
10	IPS DC-DC PANEL INDICATION Voltage Low.	$V_{IPS\ DC\ PAN\ IND} < LD\ \% \text{ of avg value OR Min safe value}$	IPS DC-DC PAN IND VOLT LOW	Logic to be checked for 15 sec.
11	IPS DC-DC BLOCK LOCAL Voltage Low.	$V_{IPS\ DC\ BLOCK\ LOCAL} < LD\ \% \text{ of avg value OR Min safe value}$	IPS DC-DC BLOCK LOCAL VOLT LOW	Logic to be checked for 15 sec.
12	IPS DC-DC HKT MAGNETO voltage Low.	$V_{IPS\ DC\ HKT\ MAG} < LD\ \% \text{ of avg value OR Min safe value}$	IPS DC-DC HKT MAG VOLT LOW	Logic to be checked for 15 sec.
13	IPS DC-DC BLOCK LINE UP Voltage Low.	$V_{IPS\ DC\ BLOCK\ LINE\ UP} < LD\ \% \text{ of avg value OR Min safe value}$	IPS DC-DC BLOCK LINE UP VOLT LOW	Logic to be checked for 15 sec.

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14	IPS DC-DC BLOCK LINE DN Voltage Low.	$V_{IPS\ DC\ BLOCK\ LINE\ DN} < LD\ \% \text{ of avg value OR Min safe value}$	IPS DC_DC BLOCK LINE DN VOLT LOW	Logic to be checked for 15 sec.
15	IPS DC-DC BLOCK TELE UP Voltage Low.	$V_{IPS\ DC\ BLOCK\ TEL\ UP} < LD\ \% \text{ of avg value OR Min safe value}$	IPS DC-DC BLOCK TEL UP VOLT LOW	Logic to be checked for 15 sec.
16	IPS DC-DC BLOCK TELE DN Voltage Low.	$V_{IPS\ DC\ BLOCK\ TEL\ DN} < LD\ \% \text{ of avg value OR Min safe value}$	IPS DC-DC BLOCK TEL DN VOLT LOW	Logic to be checked for 15 sec.
17	IPS DC-DC DATALOGGER Voltage Low.	$V_{IPS\ DC\ DATALOG} < LD\ \% \text{ of avg value OR Min safe value}$	IPS DC-DC DATA-LOG VOLT LOW	Logic to be checked for 15 sec.
18	IPS DC-DC EI Voltage Low.	$V_{IPS\ DC\ EI} < LD\ \% \text{ of avg value OR Min safe value}$	IPS DC-DC EI VOLT LOW	Logic to be checked for 15 sec.
19	IPS Battery Charging current low	$I_{IPS\ BATT\ CHAR\ 110\ DC} < LD\ \% \text{ of avg value OR Min safe value}$	IPS BATT CHAR CURR LOW	Logic to be checked for 15 sec.

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2.1 (b) Failure Logics of Integrated Power Supply				
Note: Failure Logics have been given based on defined parameters. However, Vendors may add new logics and modify given logics based on field experience OR otherwise also under approval of Railways and under intimation to RDSO so that suitable amendment may be issued in Specification after review.				
	cause_detail	Logic	cause_code	Remarks on logics
1	IPS 220 AC Voltage failed.	$V_{IPS\ I/P} < \text{Min Fail value}$	IPS I/P VOLT FAIL	Logic to be checked for 10 sec.
2	IPS 110 DC O/P Voltage failed	$V_{IPS\ 110\ DC} < \text{Min Fail value}$	IPS 110 DC VOLT FAIL	Logic to be checked for 10 sec.
3	IPS 110 AC Sig-1 O/P Voltage failed.	$V_{IPS\ SIG-1\ 110\ AC} < \text{Min Fail value}$	IPS 110 AC Sig-1 VOLT FAIL	Logic to be checked for 10 sec.
4	IPS 110 AC TR-1 O/P Voltage failed.	$V_{IPS\ TR-1\ 110\ AC} < \text{Min Fail value}$	IPS 110 AC TR-1 VOLT FAIL	Logic to be checked for 10 sec.
5	IPS SMR-1 Voltage failed.	$V_{IPS\ SMR-1\ 110\ DC} < \text{Min Fail value}$	IPS SMR-1 VOLT FAIL	Logic to be checked for 10 sec.
6	IPS DC-DC Relay Internal Voltage failed.	$V_{IPS\ DC\ R\ INT} < \text{Min Fail value}$	IPS DC-DC R INT VOLT FAIL	Logic to be checked for 10 sec.
7	IPS DC-DC Relay External Voltage failed.	$V_{IPS\ DC\ R\ EXT} < \text{Min Fail value}$	IPS DC-DC R EXT VOLT FAIL	Logic to be checked for 10 sec.
8	IPS DC-DC AXLE COUNTER Voltage failed.	$V_{IPS\ DC\ AXLE\ C} < \text{Min Fail value}$	IPS DC-DC AXLE C VOLT FAIL	Logic to be checked for 10 sec.
9	IPS DC-DC PANEL INDICATION Voltage failed.	$V_{IPS\ DC\ PAN\ IND} < \text{Min Fail value}$	IPS DC-DC PAN IND VOLT FAIL	Logic to be checked for 10 sec.
10	IPS DC-DC BLOCK LOCAL Voltage failed.	$V_{IPS\ DC\ BLOCK\ LOCAL} < \text{Min Fail value}$	IPS DC-DC BLOCK LOCAL VOLT FAIL	Logic to be checked for 10 sec.

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11	IPS DC-DC HKT MAGNETO failed.	V _{IPS DC HKT MAG} < Min Fail value	IPS DC-DC HKT MAG VOLT FAIL	Logic to be checked for 10 sec.
12	IPS DC-DC BLOCK LINE UP Voltage failed.	V _{IPS DC BLOCK LINE UP} < Min Fail value	IPS DC-DC BLOCK LINE UP VOLT FAIL	Logic to be checked for 10 sec.
13	IPS DC-DC BLOCK LINE DN Voltage failed.	V _{IPS DC BLOCK LINE DN} < Min Fail value	IPS DC-DC BLOCK LINE DN VOLT FAIL	Logic to be checked for 10 sec.
14	IPS DC-DC BLOCK TELE UP Voltage failed.	V _{IPS DC BLOCK TEL UP} < Min Fail value	IPS DC-DC BLOCK TEL UP VOLT FAIL	Logic to be checked for 10 sec.
15	IPS DC-DC BLOCK TELE DN Voltage failed.	V _{IPS DC BLOCK TEL DN} < Min Fail value	IPS DC-DC BLOCK TEL DN VOLT FAIL	Logic to be checked for 10 sec.
16	IPS DC-DC DATA-LOGGER Voltage failed.	V _{IPS DC DATALOG} < Min Fail value	IPS DC-DC DAT-ALOG VOLT FAIL	Logic to be checked for 10 sec.
17	IPS DC-DC EI Voltage failed.	V _{IPS DC EI} < Min Fail value	IPS DC-DC EI VOLT FAIL	Logic to be checked for 10 sec.
18	IPS Battery Charging current failed.	I _{IPS BATT CHAR 110 DC} < Min Fail value	IPS DC-DC BATT CHAR CURR FAIL	Logic to be checked for 10 sec.

2.2 (a) Prediction Logics of Points

Note:

1. These logics are primitive logics for Prediction of failures. However, in addition to these primitive logics, these primitive logics, various AI/ML Techniques and learning models need to be developed and applied by Vendors for prediction of failures. The models used by them shall be shared with Zonal Railways/RDSO.
2. Comparisons in logics have been made with deviation in parameter value in terms of certain % of avg value OR deviation from Min/Max safe. Although % deviation has been given for predictive logics for reference, however, % may be modified as decided by Zonal Railway. **This % is being denoted as LD1 , LD2 for lower side deviation and “HD” for higher side deviation. For Points, LD1 = 80, LD2 = 90 and HD = 150.**

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3. For Predictive logic, the analog continuous deviation (say < than 80% of avg value) should occur for at least certain duration to avoid predictive alerts for short transients. Indicative timing has been mentioned against logics. However, time duration can be decided by Zonal Railways as per site requirement.
4. The deviation should occur for entire duration of at least 1 operation in discrete cases like voltage and current of point machine.
5. Prediction Logics have been given based on defined parameters. However, Vendors may add new logics and modify given logics based on field experience OR otherwise also under approval of Railways and under intimation to RDSO so that suitable amendment may be issued in Specification after review.
6. Point predictive logics have been shown for one individual Point (not cross over) assuming that NWKR is separate for each point (not cross over). However, if one NWKR is used for cross over, remarks have been given against each logic.

	cause_detail	Logic	cause_code	Remarks on logics
1	Predictive Alert for PT---. Voltage OR Current for Normal operation Low at Loc.	[NWCR $\uparrow$ (entire UP duration or 12 sec whichever is less) && $I_{IPS \text{ BATT CHAR } 110 \text{ DC}} > 0^+$ && $V_{IPS \text{ 110 DC}} \geq \text{Min safe value}$ && ($V_{PT \text{ 110 DC LOC N}} < \text{LD1 \% of avg value OR Min safe value}$ $I_{PT \text{ N}} < \text{LD1\% of avg value OR Min safe value}$)]	PT N VOLT/CURR LOW	(1) Avg value of $V_{PT \text{ 110 DC Loc N}}$ should be calculated when IPS Battery charging Current ie $I_{IPS \text{ Batt Char 110 DC}}$ is more than 0^+. A small positive value can be used in place of 0^+. (2) [] Logic is checked for entire period of NWCR UP (or 12 sec whichever is less) and $I_{IPS \text{ Batt Char 110 DC}}$ is more than 0^+. (3) Value of voltage and current should be taken after transients settles.

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		NWKR ↑ (within 2 sec after above logic)		
2	Predictive Alert for PT---. Voltage OR current for Reverse operation Low at Loc.	[RWCR ↑ (entire UP duration or 12 sec whichever is less) && $I_{IPS \text{ BATT CHAR } 110 \text{ DC}} > 0^+$ && $V_{IPS \text{ 110 DC}} \geq \text{Min safe value}$ && ($V_{PT \text{ 110 DC LOC R}} < \text{LD1 \% of avg value OR Min safe value}$ $I_{PT \text{ R}} < \text{LD1 \% of avg value OR min safe value}$)] RWKR ↑ (within 2 sec after above logic)	PT R VOLT/CURR LOW	(1) Avg value of $V_{PT \text{ 110 DC LOC R}}$ should be calculated when IPS Battery charging Current ie $I_{IPS \text{ Batt Char 110 DC}}$ is more than 0^+. A small positive value can be used in place of 0^+. (2) [] Logic is checked for entire period of RWCR UP (or 12 sec whichever is less) and $I_{IPS \text{ Batt Char 110 DC}}$ is more than 0. (3) Value of voltage and current should be taken after transients settles.
3	Predictive Alert for PT---. Normal Indication Voltage Low at Loc after detection.	[NWKR ↑ && $V_{IPS \text{ DC R EXT}} \geq \text{Min safe value}$ && $V_{PT \text{ 24 DC LOC N}} < \text{LD1 \% of avg value}$]	PT N IND VOLT LOW AT LOC	(1) [] Logic is checked for 15 sec duration. (2) If one NWKR is being used after taking indication of both points in series, then logic may be modified to consider individual indication voltage of each point at

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				location ie $V_{PT\ 24\ DC\ LOC\ N}$ in such a way to arrive at conclusion whether which point has low voltage.
4	Predictive Alert for PT---. Reverse Indication Voltage Low at Loc after detection.	[RWKR ↑ && $V_{IPS\ DC\ R\ EXT} \geq \text{Min safe value}$ && $V_{PT\ 24\ DC\ LOC\ R} < \text{LD1 \% of avg value}$]	PT R IND VOLT LOW AT LOC	(1) [] Logic is checked for 15 sec duration. (2) If one RWKR is being used after taking indication of both points in series, then logic may be modified to consider individual indication voltage of each point at location ie $V_{PT\ 24\ DC\ LOC\ R}$ in such a way to arrive at conclusion whether which point has low voltage.
5	Predictive Alert for PT---. Normal indication voltage is ok at loc but low at Indoor NWKR. Cable may be defective.	[NWKR ↑ && $V_{PT\ 24\ DC\ LOC\ N} \geq \text{LD2 \% of avg value}$ && $V_{PT\ NWKR} < \text{LD1 \% of avg value OR Min safe value}$]	PT VOLT LOW AT NWKR	(1) [] Logic is checked for 15 sec duration. (2) If one NWKR is being used after taking indication of both points in series, then $V_{PT\ 24\ DC\ LOC\ N}$ of point which is

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				last in series shall be taken for comparison. In this case Alert msg can be given for both points.
6	Predictive Alert for PT---. Reverse indication voltage ok at Loc but low at Indoor RWKR. Cable may be defective.	[RWKR↑ && $V_{PT\ 24\ DC\ LOC\ R} \geq LD2\ \% \text{ of avg value}$ && $V_{PT\ RWKR} < LD1\ \% \text{ of avg value OR Min safe value}$]	PT VOLT LOW AT RWKR	(1) [] Logic is checked for 15 sec duration. (2) If one RWKR is being used after taking indication of both points in series, then $V_{PT\ 24\ DC\ LOC\ R}$ of point which is last in series shall be taken for comparison. In this case Alert msg can be given for both points.
7	Predictive Alertg for PT---. Normal Operation Time high	[NWCR ↑ (entire UP duration or 12 sec whichever is less) NWKR ↑ (within 2 sec after above logic) $T_{PT\ N} > HD\ \% \text{ of avg value}$]	PT N TIME HIGH	(1) Normal operation time is calculated as per parameter details in this Annexure A.
8	Predictive Alert for PT---. Reverse Operation Time high	[RWCR ↑ (entire UP duration or 12 sec whichever is less) RWKR ↑ (within 2 sec after above logic)	PT R TIME HIGH	(2) Reverse operation time is measured as per parameter details in Annexure A.

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		$T_{PTR} > \text{HD \% of avg value}$ J		
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2.2 (b) Failure Logics of Points

Note:

1.Failure Logics have been given based on defined parameters. However, Vendors may add new logics and modify given logics based on field experience OR otherwise also under approval of Railways and under intimation to RDSO so that suitable amendment may be issued in Specification after review. **The % deviation for Failure of Points is being denoted as LD1, LD2 for lower side deviation. LD1=70 & LD2=90.**

2. Point failure logics have been shown for one individual Point (not cross over) assuming that NWKR is separate for each point (not cross over). However, if one NWKR is used for cross over, remarks have been given against each logic.

	cause_detail	Logic	cause_code	Remarks on Logics
1	Point No. PT---- failed in Normal. Normal Indication Voltage at Loc is low/failed/detection break.	[NWCR ↓ && RWCR ↓ && RWKR ↓ && NWKR ↑ to ↓ && $V_{IPS\ DC\ R\ EXT} \geq \text{Min fail value}$ && $V_{PT\ 24\ DC\ LOC\ N} < \text{LD1 \% of avg value}$]	PT N IND VOLT FAIL AT LOC	(1) [] Logic is checked for 2 sec duration. after NWKR goes from UP to DN. (2) If one NWKR is being used after taking indication of both points in series, then logic may be modified to consider individual indication voltage of each point at location ie $V_{PT\ 24\ DC\ LOC\ N}$ in such a way to arrive at conclusion whether which point has low voltage.

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2	Point No. PT---- failed in Reverse. Reverse indication Voltage at Loc is low/failed/detection break.	[NWCR ↓ && RWCR ↓ && NWKR ↓ && RWKR ↑ to ↓ && $V_{IPS\ DC\ R\ EXT} \geq \text{Min fail value}$ && $V_{PT\ 24\ DC\ LOC\ R} < \text{LD1 \% of avg value}$]	PT R IND VOLT FAIL AT LOC	(1) [] Logic is checked for 2 sec duration. after RWKR goes from UP to DN. (2) If one RWKR is be- ing used after taking indica- tion of both points in se- ries, then logic may be modified to consider indi- vidual indica- tion voltage of each point at location ie $V_{PT\ 24\ DC\ LOC\ R}$ in such a way to arrive at con- clusion whether which point has low volt- age.
3	Point No. PT---- failed in Normal. Voltage at Loc is ok but fail at Indoor NWKR.	[NWCR ↓ && RWCR ↓ && RWKR ↓ && NWKR ↑ to ↓ && $V_{PT\ 24\ DC\ LOC\ N} \geq \text{LD2 \% of avg value}$ && $V_{PT\ NWKR} < \text{Min Fail value}$]	PT VOLT FAIL AT NWKR	(1) [] Logic is checked for 2 sec duration. after NWKR goes from UP to DN. (2) If one NWKR is be- ing used after taking indica- tion of both points in se- ries, then $V_{PT\ 24\ DC\ LOC\ N}$ of point which is last in series shall be taken

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		]		for comparison. In this case Alert msg can be given for both points.
4	Point No. PT---- failed in Reverse. Voltage at Loc is ok but fail at Indoor RWKR.	[NWCR ↓ && RWCR ↓ && NWKR ↓ && RWKR ↑ to ↓ && $V_{PT\ 24\ DC\ LOC\ R} \geq LD2\ \% \text{ of avg value}$ && $V_{PT\ RWKR} < \text{Min Fail value}$]	PT VOLT FAIL AT RWKR	(1) [] Logic is checked for 2 sec duration. after RWKR goes from UP to DN. (2) If one RWKR is being used after taking indication of both points in series, then $V_{PT\ 24\ DC\ LOC\ R}$ of point which is last in series shall be taken for comparison. In this case Alert msg can be given for both points.
5	Point No. PT---- failed in Normal. Voltage at NWKR is Ok. NWKR Relay may be defective.	[NWCR ↓ && RWCR ↓ && RWKR ↓ && NWKR ↑ to ↓ && $V_{PT\ NWKR} \geq \text{Min Safe value}$]	PT NWKR RELAY DEFECT	(1) [] Logic is checked for 2 sec duration. after NWKR goes from UP to DN. (2) If one NWKR is being used after taking indication of both points in series, then Alert msg can be given for both points.

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6	Point No. PT---- failed in Reverse. Voltage at RWKR is ok. RWKR Relay may be defective.	[NWCR ↓ && RWCR ↓ && NWKR ↓ && RWKR ↑ to ↓ && $V_{PT\ RWKR} \geq \text{Min Safe value}$]	PT RWKR RELAY DEFECT	(1) [] Logic is checked for 2 sec duration. after RWKR goes from UP to DN. (2) If one RWKR is being used after taking indication of both points in series, then Alert msg can be given for both points.
7	Point No. PT---- in Normal. Voltage Or Current for normal operation in Loc failed.	[NWCR ↑ (Entire UP duration or 12 sec whichever is less) && $V_{IPS\ 110\ DC} \geq \text{Min fail value}$ && ($V_{PT\ 110\ DC\ LOC\ N} < \text{Min Fail value}$ OR $I_{PT\ N} < \text{Min Fail value}$)] NWKR ↓ (remaining DN up to 2 sec after completion of above logic)	PT N VOLT/CURR FAIL	(1) [] Logic is checked for entire period of NWCR UP (or 12 sec). It is followed by NWKR remaining DN upto 2 sec after completion of [] logic. (2) Value of voltage and current should be taken after transients settles.
8	Point No. PT---- failed in Reverse.	[RWCR ↑ (Entire UP duration or 12 sec whichever is less)	PT R VOLT/CURR FAIL	(1) [] Logic is checked for entire period of RWCR UP (or 12 sec). It

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	Voltage Or Current for reverse operation in Loc failed.	&& $V_{IPS\ 110\ DC} \geq \text{Min fail value}$ && ($V_{PT\ 110\ DC\ LOC\ R} < \text{Min Fail value}$ OR $I_{PT\ R} < \text{Min Fail value}$)] RWKR ↓ (remaining DN up to 2 sec after completion of above logic)		is followed by RWKR re-remaining DN upto 2 sec after completion of [] logic. (2) Value of voltage and current should be taken after transients settles.
9	Point No. PT---- failed in Normal. Normal indication Voltage at Loc failed OR Point in Obstruction.	[NWCR ↑ (Entire UP duration or 12 sec whichever is less) && $V_{PT\ 110\ DC\ LOC\ N} \geq \text{Min safe value}$ && $I_{PT\ N} \geq \text{Min safe value}$] NWKR ↓ (remaining DN upto 2 sec after completion of above logic) $T_{PT\ N} > \text{Max Safe Value}$	PT N OBS	(1) [] Logic is checked for entire period of NWCR UP (or 12 sec). (2) Value of operation voltage and current should be taken after transients settles.
10	Point No. PT---- failed in Reverse. Reverse indication Voltage at Loc failed OR Point in Obstruction.	[RWCR ↑ (Entire UP duration or 12 sec whichever is less) && $V_{PT110\ DC\ LOC\ R} \geq \text{Min safe value}$ &&	PT R OBS	(1) [] Logic is checked for entire period of NWCR UP (or 12 sec). (2) Value of operation voltage and current should be

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		$I_{PT R} \geq \text{Min safe value}$] RWKR ↓ (remaining DN up to 2 sec after completion of above logic) $T_{PT R} > \text{Max Safe Value}$		taken after transients settles.
11	Point No. PT---- failed in Normal. Normal indication Voltage at LOC is ok but failed at NWKR.	[NWCR ↑ (Entire UP duration or 12 sec whichever is less) && $V_{PT 110 DC LOC N} \geq \text{Min safe value}$ && $I_{PT N} \geq \text{Min safe value}$] [NWKR ↓ (remaining DN upto 2 sec after completion of above logic) && $V_{PT 24 DC LOC N} \geq \text{LD2 \% of avg value}$ && $V_{PT NWKR} < \text{Min Fail value}$]	PT VOLT FAIL AT NWKR	(1) First [] Logic is checked for entire period of NWCR UP (or 12 sec). (2) Value of operation voltage and current should be taken after transients settles. (3) In Second [] logic, Normal indication voltage at Loc and at NWKR are compared for 2 sec after completion of just previous logic. (4) If one NWKR is being used after taking indication of both points in series, then $V_{PT 24 DC LOC N}$ of point which is last in series shall be taken

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				for comparison. In this case Alert msg can be given for both points.
12	Point No. PT---- failed in Reverse. Reverse indication Voltage at LOC is ok but failed at RWKR.	[RWCR ↑ (Entire UP duration or 12 sec whichever is less) && $V_{PT110\ DC\ LOC\ R} \geq \text{Min safe value}$ && $I_{PT\ R} \geq \text{Min safe value}$] [RWKR ↓ (remaining DN upto 2 sec after completion of above logic) && $V_{PT\ 24\ DC\ LOC\ R} \geq \text{LD2 \% of avg value}$ && $V_{PT\ RWKR} < \text{Min Fail value}$]	PT VOLT FAIL AT RWKR	(1) First [] Logic is checked for entire period of RWCR UP (or 12 sec). (2) Value of operation voltage and current should be taken after transients settles. (3) In Second [] logic, Reverse indication voltage at Loc and at NWKR are compared for 2 sec after completion of just previous logic. (4) If one RWKR is being used after taking indication of both points in series, then $V_{PT\ 24\ DC\ LOC\ R}$ of point which is last in series shall be taken for comparison. In this case Alert

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				msg can be given for both points.
13	Point No. PT---- failed in Normal. Normal indication voltage at NWKR is ok. NWKR Relay may be defective.	[NWCR ↑ (Entire UP duration or 12 sec whichever is less) && $V_{PT\ 110\ DC\ LOC\ N} \geq \text{Min safe value}$ && $I_{PT\ N} \geq \text{Min safe value}$] [NWKR ↓ (remaining DN upto 2 sec after completion of above logic) && $V_{PT\ NWKR} \geq \text{Min safe value}$]	PT NWKR RELAY DEFECT	(1) First [] Logic is checked for entire period of NWCR UP (or 12 sec). (2) Value of operation voltage and current should be taken after transients settles. (3) In Second [] logic, Normal indication voltage at NWKR is compared for 2 sec after completion of just previous logic. (4) If one NWKR is being used after taking indication of both points in series, then Alert msg can be given for both points.
14	Point No. PT---- failed in Reverse. Reverse indication voltage at RWKR is ok. RWKR Relay may be defective.	[RWCR ↑ (Entire UP duration or 12 sec whichever is less) &&	PT RWKR RELAY DEFECT	(1) [] Logic is checked for entire period of RWCR UP (or 12 sec). It is followed by RWKR

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		$V_{PT\ 110\ DC\ LOC\ R} \geq \text{Min safe value}$ && $I_{PT\ R} \geq \text{Min safe value}$] [RWKR ↓ (remaining DN upto 2 sec after completion of above logic) && $V_{PT\ RWKR} \geq \text{Min safe value}$]		remaining DN upto 2 sec after completion of [] logic. (2) Value of operation voltage and current should be taken after transients settles. (3) In Second [] logic, Reverse indication voltage at NWKR is compared for 2 sec after completion of just previous logic. (4) If one RWKR is being used after taking indication of both points in series, then Alert msg can be given for both points.
15	Point No. PT---- failed in Normal but reasons not diagnosed.	[NWCR ↓ && RWCR ↓ && RWKR ↓ &&	PT N FAIL UNKNOWN	

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		NWKR ↑ to ↓ If any of remaining logic at Sr 1, 3 & 5 do not qualify]		
16	Point No. PT---- failed in Reverse but reasons not diagnosed.	[NWCR ↓ && RWCR ↓ && NWKR ↓ && RWKR ↑ to ↓ If any of remaining logic at Sr 2, 4 & 6 do not qualify]	PT R FAIL UNKNOWN	
17	Point No. PT---- failed in Normal but reasons not diagnosed.	[NWCR ↑ (Entire UP duration or 12 sec whichever is less) && If any of remaining logic at Sr 7, 9, 11 & 13 do not qualify NWKR ↓ (remaining DN upto 2 sec after completion of above logic)]	PT N FAIL UNKNOWN	
18	Point No. PT---- failed in Reverse but reasons not diagnosed.	[RWCR ↑ (Entire UP duration or 12 sec whichever is less) && If any of remaining logics from sr 8, 10, 12 & 14 do not qualify	PT R FAIL UNKNOWN	

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		RWKR ↓ (remaining DN upto 2 sec after completion of above logic)]		
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	2.3 (a) Prediction Logics of DC Track Circuit			
Note:				
1. These logics are primitive logics for Prediction of failures. However, in addition to these primitive logics, these primitive logics, various AI/ML Techniques and learning models need to be developed and applied by Vendors for prediction of failures. The models used by them shall be shared with Zonal Railways/RDSO. 2. Comparisons in logics have been made with deviation in parameter value in terms of certain % of avg value OR deviation from Min/Max safe. Although % deviation has been given for predictive logics for reference, however, % may be modified as decided by Zonal Railway. This % is being denoted as LD1, LD2, LD3 for lower side deviation and as HD1, HD2 for higher side deviation. For DC Track ckts, LD1=80, LD3=90, LD2=50, HD1=120, HD2=150. 3. For Predictive logic, the analog continuous deviation (say < than 80% of avg value) should occur for at least certain duration to avoid predictive alerts for short transients. Indicative timing has been mentioned against logics. However, time duration can be decided by Zonal Railways as per site requirement. 4. The deviation should occur for entire duration of at least 1 operation in discrete cases like voltage and current of point machine. 5. Prediction Logics have been given based on defined parameters. However, Vendors may add new logics and modify given logics based on field experience OR otherwise also under approval of Railways and under intimation to RDSO so that suitable amendment may be issued in Specification after review.				
	cause_detail	Logic	cause_code	Remarks on Logics
1	Track Ckt predictive Alert for T----: TFC input voltage Low/failed in Loc.	[TPR ↑ && $V_{IPS\ TR-1\ 110\ AC} \geq \text{Min safe value}$ && $V_{TC\ TFC\ I/P} < \text{LD1\% of avg value OR Min safe value}$]	TC TFC I/P VOLT LOW	[] Logic is checked for 15 min.

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2	Track Ckt predictive Alert for T----: Battery charging but TFC output voltage Low.	[TPR ↑ && $I_{TC \text{ BATT CHARG}} > 0^+$ && $V_{IPS \text{ TR-1 110 AC}} \geq \text{Min safe value}$ && $V_{TC \text{ TFC O/P}} < \text{LD1\% of avg value OR Min safe value}$]	TC TFC O/P VOLT LOW	[] Logic is checked for 15 sec when Battery is being charged. Avg value of $V_{TC \text{ TFC O/P}}$ should be calculated when Track Battery charging Current ie $I_{TC \text{ BATT CHARG}}$ is more than 0. A small positive value can be used in place of 0^+ to account for errors.
3	Track Ckt predictive Alert for T----: Battery not charging and TFC output voltage Low/fail.	[TPR ↑ && $I_{TC \text{ BATT CHARG}} < 0$ && $V_{IPS \text{ TR-1 110 AC}} \geq \text{Min safe value}$ && $V_{TC \text{ TFC O/P}} < \text{LD1\% of avg value OR Min safe value}$]	TC TFC O/P VOLT LOW	[] Logic is checked for 15 sec when Battery is being discharged. Avg value of $V_{TC \text{ TFC O/P}}$ should be calculated when Track Battery is discharging ie $I_{TC \text{ BATT CHARG}}$ is negative.
4	Track Ckt predictive Alert for T----: Batt charging current high.	[TPR ↑ && $I_{TC \text{ BATT CHARG}} > \text{Max safe value}$]	TC BT CHG CURR HIGH	[] Logic is checked for 15 sec.
5	Track Ckt predictive Alert for T----: Batt not charging	[TPR ↑ && $I_{TC \text{ BATT CHARG}} < \text{Min safe value}$]	TC BT CHG CURR LOW	[] Logic is checked for 15 sec.

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6	Track Ckt predictive Alert for T----: Feed End Choke Resistance Low or short.	[TPR ↑ && $R_{TC\ CH\ FEED\ END} < LD2\ \% \text{ of avg value}$]	TC CH RES LOW	[] Logic is checked for 15 sec. Voltage and Current (for R) shall be taken after transient settles.
7	Track Ckt predictive Alert for T----: Feed End Choke Resistance High.	[TPR ↑ && $R_{TC\ CH\ FEED\ END} > HD2\ \% \text{ of avg value}$]	TC CH RES HIGH	[] Logic is checked for 15 sec. Voltage and Current (for R) shall be taken after transient settles.
8	Track Ckt predictive Alert for T----: Feed End Variable Resistance Low or short.	[TPR ↑ && $R_{TC\ VAR\ RES} < LD2\ \% \text{ of avg value}$]	TC VAR RES LOW	[] Logic is checked for 15 sec. Voltage and Current (for R) shall be taken after transient settles.
9	Track Ckt predictive Alert for T----: Feed End Variable Resistance high.	[TPR ↑ && $R_{TC\ VAR\ RES} > HD2\ \% \text{ of avg value}$]	TC VAR RES HIGH	[] Logic is checked for 15 sec. Voltage and Current (for R) shall be taken after transient settles.
10	Track Ckt predictive Alert for T----: Track Relay voltage/Current is low due to Bal-last/sleeper low resistance.	[TPR ↑ && $I_{TC\ RELAY\ END} < LD1\ \% \text{ of avg value OR Min safe value}$ && $I_{BALST} > HD1\ \% \text{ of avg value}$ &&	TC BALST/SLPR RES LOW	[] Logic is checked for 15 sec. Current shall be taken after transient settles. Δ indicates difference (only magnitude without sign) between present parameter and avg parameter.

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		$\Delta I_{BALST} > \Delta I_{TC \text{ RELAY END}}$]		
11	Track Ckt predictive Alert for T----: Track Relay voltage is low due to high series Rail resistance.	[TPR $\uparrow$ && $R_{RAIL} > \text{HD2 \% of avg value}$]	TC RAIL RES HIGH	[] Logic is checked for 15 sec. Voltage and Current (for R) shall be taken after transient settles.
12	Track Ckt predictive Alert for T----: Track Relay Voltage Low/ Under energization.	[TPR $\uparrow$ && $I_{TC \text{ RELAY END}} < \text{LD1 \% of avg value OR Min safe value}$ (If logic of sr 9 & 10 above do not qualify)]	TC TR VOLT LOW	[] Logic is checked for 15 sec. Voltage shall be taken after transient settles.
13	Track Ckt predictive Alert for T----: Track Relay Voltage high/Over energization.	[TPR $\uparrow$ && $I_{TC \text{ RELAY END}} > \text{HD1 \% of avg value OR Max}$]	TC TR OVER ENERIZATION	[] Logic is checked for 15 sec. Voltage shall be taken after transient settles.
14	Track Ckt predictive Alert for T----: Track Relay contact resistance high	[TPR $\uparrow$ && $V_{IPS \text{ DC R EXT}} \geq \text{Min safe value}$ && $V_{TC \text{ 24 DC LOC}} < \text{LD1\% of avg value}$]	TC TR CONTACT RES HIGH	[] Logic is checked for 15 sec. Voltage shall be taken after transient settles.

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15	Track Ckt predictive Alert for T----: TPR input voltage Low in relay room. Cable may be defective.	[TPR ↑ && $V_{TC\ 24\ DC\ LOC} \geq LD3\% \text{ of avg value}$ $V_{TC\ 24\ DC\ TPR\ I/P} < LD1\ \% \text{ of avg value OR Min safe value}$]	TC TPR I/P VOLT LOW	[] Logic is checked for 15 sec. Voltage shall be taken after transient settles.
16	Track Ckt predictive Alert for T2/T1: TR current low in adjacent track. Glued joint between T1 and T2 may be defective.	//Logics in square bracket below to be checked for T1 [TPR ↓ && $I_{TC\ RELAY\ END} < \text{Min Fail value}$ && $I_{TC\ RELAY\ END} > 0^+$ && $I_{BALST} > HD1\ \% \text{ of avg value}$ $\Delta I_{BALST} > \Delta I_{TC\ RELAY\ END}$] && //Logics in square bracket below to be checked for T2 [TPR ↑ && $\Delta I_{TC\ RELAY\ END} > 10\% \text{ of } I_{TC\ RELAY\ END}$ END	TC GJ SHORT	T1 and T2 are adjacent tracks. When due to short or train movement, any track ckt (T1) is shunted and simultaneously in adjacent track ckt (T2), track relay current lowers by more than 10% of original value (although adjacent track ckt is UP). It may indicate possibility of Glued joint defect/short. Δ indicates difference (only magnitude without sign) between parameter before TPR (T1) UP and after TPR (T1) DN for T1 parameter. In $\Delta I_{TC\ RELAY\ END}$ for T2, Δ indicates difference (only magnitude without sign) between parameter before TPR (T1) UP and after TPR (T1) DN when $I_{TC\ RELAY\ END}$ of T2 goes down.

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		// (I _{TC RELAY END} is value of T2 when TPR (T1) was UP)]		
--	--	--	--	--

	2.3 (b) Failure Logics of DC Track Circuit			
	Note: 1. Failure Logics have been given based on defined parameters. However, Vendors may add new logics and modify given logics based on field experience OR otherwise also under approval of Railways and under intimation to RDSO so that suitable amendment may be issued in Specification after review. The % deviation for Failure of Track is being denoted as LD1, LD2 for lower side deviation and as HD1 for higher side deviation. LD1= 70, LD2 = 90 and HD1=120. 2. Logic for declaring track circuit failure: Suppose T1, T2 & T3 are track circuits in sequence. Then Logic to ascertain failure of track circuit T2 is (to exclude it from occupancy): T1 ↑ && T3 ↑ At least delay of 5 sec T2 ↓ (remain DN for 10 sec) Logics for ascertaining failure of First and last track circuits at station shall be decided by Zonal Railways. 3. Time for Alert logic as given against each logic below shall be counted after confirmation of track circuit failure as per above logic.			
	cause_detail	Logic	cause_code	Remarks on Logic
1	Track Ckt No. T---- failed. TR Down. Possible shorting in track.	[TPR ↓ && I _{TC RELAY END} < Min Fail value	TC SHORT	[] Logic is checked for 10 sec. Current shall be taken after transient settles.

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		&& $I_{TC \text{ RELAY END}} > 0^+$ && $I_{BALST} > \text{HD1 \% of avg value}$ $\Delta I_{BALST} > \Delta I_{TC \text{ RELAY END}}$]		A small positive value can be used in place of 0^+ to confirm that current is not zero. Δ indicates difference (only magnitude without sign) between parameter before TPR UP and after TPR DN.
2	Track Ckt No. T---- failed. TR Down. Feed End choke resistance high Or open	[TPR ↓ && $I_{TC \text{ RELAY END}} < \text{Min Fail value}$ && $I_{TC \text{ FEED END}} > 0^+$ && $R_{TC \text{ CH FEED END}} > 2 * R_{TC \text{ CH FEED END (avg)}}$]	TC CH RES OPEN	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient settles. A small positive value can be used in place of 0^+ to confirm that current is not zero.
3	Track Ckt No. T---- failed. TR Down. Feed End variable resistance high Or open.	[TPR ↓ && $I_{TC \text{ RELAY END}} < \text{Min Fail value}$ && $I_{TC \text{ FEED END}} > 0^+$ && $R_{TC \text{ VAR RES}} > 2 * R_{TC \text{ VAR RES (avg)}}$	TC VAR RES OPEN	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient settles. A small positive value can be used in place of

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		]		0 ⁺ to confirm that current is not zero.
4	Track Ckt No. T---- failed. TR Down. Rail resistance high OR Open/TLJB cable open	[TPR ↓ && $I_{TC \text{ RELAY END}} < \text{Min Fail value}$ && $I_{TC \text{ RELAY END}} > 0^+$ && $I_{TC \text{ FEED END}} > 0^+$ && $R_{RAIL} > 2 * R_{RAIL} (\text{avg})$]	TC RAIL RES OPEN	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient settles. A small positive value can be used in place of 0 ⁺ to confirm that current is not zero.
5	Track Ckt No. T---- failed. TR Down. TFC Output Voltage Fail	[TPR ↓ && $I_{TC \text{ RELAY END}} < \text{Min Fail value}$ && $I_{TC \text{ FEED END}} > 0^+$ && $V_{TC \text{ TFC O/P}} < \text{Min Fail value}$]	TC TFC O/P VOLT FAIL	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient settles. A small positive value can be used in place of 0 ⁺ to confirm that current is not zero.
6	Track Ckt No. T---- failed. TR Down. Circuit open.	[TPR ↓ && ($I_{TC \text{ RELAY END}} < 0^+$ $I_{TC \text{ FEED END}} < 0^+$]	TC CKT OPEN	A small positive value can be used in place of 0 ⁺ to confirm that voltage is zero.

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		)]		
7	Track Ckt No. T---- failed. TR Current is Ok but TR Relay may be defective.	[TPR ↓ && $I_{TC\ RELAY\ END} \geq \text{Min safe value}$ && $V_{IPS\ DC\ R\ EXT} \geq \text{Min fail value}$ && $V_{TC\ 24\ DC\ LOC} < \text{LD1 \% of avg value}$]	TC TR RE- LAY DE- FECT	[] Logic is checked for 10 sec. Voltage and Current shall be taken af- ter transient settles.
8	Track Ckt No. T---- failed. Track relay UP. TPR in- put voltage in Relay Room low/zero. Possi- ble cable fault.	[TPR ↓ && $I_{TC\ RELAY\ END} \geq \text{Min safe value}$ && $V_{TC\ 24\ DC\ LOC} \geq \text{LD2 \% of avg value}$ && $V_{TC\ 24\ DC\ TPR\ I/P} < \text{Min Fail value}$]	TC TR UP TPR DN	[] Logic is checked for 10 sec. Voltage and Current shall be taken af- ter transient settles.
9	Track Ckt No. T---- failed. Track relay UP. TPR in- put voltage in Relay Room is OK. TPR Re- lay may be defective	[TPR ↓ && $I_{TC\ RELAY\ END} \geq \text{Min safe value}$ && $V_{TC\ 24\ DC\ TPR\ I/P} \geq \text{Min safe value}$]	TC TPR RE- LAY DE- FECT	[] Logic is checked for 10 sec. Voltage and Current shall be taken af- ter transient settles.
10	Track Ckt No. T---- failed. Reason not diagnosed.	[TPR ↓ &&	REASON UNKNOWN	

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		If any of above Logics do not qualify]		
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	2.4 (a) Prediction Logic for Main Signal			
Note: - These logics are primitive logics for Prediction of failures. However, in addition to these primitive logics, these primitive logics, various AI/ML Techniques and learning models need to be developed and applied by Vendors for prediction of failures. The models used by them shall be shared with Zonal Railways/RDSO. - Comparisons in logics have been made with deviation in parameter value in terms of certain % of avg value OR deviation from Min/Max safe. Although % deviation has been given for predictive logics for reference, however, % may be modified as decided by Zonal Railway. This % is being denoted by “LD” for lower side deviation and “HD” for higher side deviation. For Main Signal, LD=80 and HD=120. - For Predictive logic, the analog continuous deviation (say < than 80% of avg value) should occur for at least certain duration to avoid predictive alerts for short transients. Indicative timing has been mentioned against logics. However, time duration can be decided by Zonal Railways as per site requirement. - The deviation should occur for entire duration of at least 1 operation in discrete cases like voltage and current of point machine. - Prediction Logics have been given based on defined parameters. However, Vendors may add new logics and modify given logics based on field experience OR otherwise also under approval of Railways and under intimation to RDSO so that suitable amendment may be issued in Specification after review.				
	cause_detail	Logic	cause_code	Remarks on Logics
1	Sig predictive Alert for S---: Voltage or Current of RG Aspect Low.	[RECR ↑ && V _{IPS SIG-1 110 AC} ≥ Min safe value &&	SIG RG VOLT/CURR LOW	[] Logic is checked for 15 sec. Voltage and Current shall be

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		(V _{SIG RG} < LD % of avg value OR Min safe value I _{SIG RG} < LD % of avg value OR Min safe value)]		taken after transient settles.
2	Sig predictive Alert for S---: Voltage or Current of HG Aspect Low.	[HECR ↑ && V _{IPS SIG-1 110 AC} ≥ Min safe value && (V _{SIG HG} < LD % of avg value OR Min safe value I _{SIG HG} < LD % of avg value OR Min safe value)]	SIG HG VOLT/CURR LOW	[] Logic is checked for 15 sec. Voltage and Current shall be taken after transient settles.
3	Sig predictive Alert for S---: Voltage or Current of HHG Aspect Low.	[HHECR ↑ && V _{IPS SIG-1 110 AC} ≥ Min safe value && (V _{SIG HHG} < LD % of avg value OR Min safe value I _{SIG HHG} < LD % of avg value OR Min safe value)]	SIG HHG VOLT/CURR LOW	[] Logic is checked for 15 sec. Voltage and Current shall be taken after transient settles.

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4	Sig predictive Alert for S---: Voltage or Current of DG Aspect Low.	[DECR ↑ && $V_{IPS\ SIG-1\ 110\ AC} \geq \text{Min safe value}$ && ($V_{SIG\ DG} < \text{LD \% of avg value}$ OR Min safe value $I_{SIG\ DG} < \text{LD \% of avg value}$ OR Min safe value)]	SIG DG VOLT/CURR LOW	[] Logic is checked for 15 sec. Voltage and Current shall be taken after transient settles.
5	Sig predictive Alert for S---: Current of RG Aspect high.	[RECR ↑ && $I_{SIG\ RG} > \text{HD \% of avg value}$ OR Max safe value]	SIG RG CURR HIGH	[] Logic is checked for 15 sec. Current shall be taken after transient settles.
6	Sig predictive Alert for S---: Current of HG Aspect high.	[HECR ↑ && $I_{SIG\ HG} > \text{HD \% of avg value}$ OR Max safe value]	SIG HG CURR HIGH	[] Logic is checked for 15 sec. Current shall be taken after transient settles.
7	Sig predictive Alert for S---: Current of HHG Aspect high.	[HHECR ↑ && $I_{SIG\ HHG} > \text{HD \% of avg value}$ OR Max safe value]	SIG HHG CURR HIGH	[] Logic is checked for 15 sec. Current shall be taken after transient settles.
8	Sig predictive Alert for S---:	[DECR ↑	SIG DG CURR HIGH	[] Logic is checked for 15 sec.

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	Current of DG Aspect high.	&& $I_{SIG\ DG} > \begin{matrix} HD \% \text{ of avg value} \\ OR \text{ Max safe value} \end{matrix}$]		Current shall be taken after transient settles.
9	Sig predictive Alert for S---: Voltage of HPR is Low in Loc Box	[HCCR ↑ && $V_{IPS\ DC\ R\ EXT} \geq \text{Min safe value}$ && $V_{SIG\ HPR} < \begin{matrix} LD \% \text{ of avg value} \\ OR \text{ Min safe value} \end{matrix}$]	SIG HPR VOLT LOW	[] Logic is checked for 15 sec. Voltage shall be taken after transient settles.
10	Sig predictive Alert for S---: Voltage of HHPR is Low in Loc Box	[HCCR ↑ && $V_{IPS\ DC\ R\ EXT} \geq \text{Min safe value}$ && $V_{SIG\ HHPR} < \begin{matrix} LD \% \text{ of avg value} \\ OR \text{ Min safe value} \end{matrix}$]	SIG HHPR VOLT LOW	[] Logic is checked for 15 sec. Voltage shall be taken after transient settles.
11	Sig predictive Alert for S---: Voltage of DPR is Low in Loc Box	[DECR ↑ && $V_{IPS\ DC\ R\ EXT} \geq \text{Min safe value}$ && $V_{SIG\ DPR} < \begin{matrix} LD \% \text{ of avg value} \\ OR \text{ Min safe value} \end{matrix}$]	SIG DPR VOLT LOW	[] Logic is checked for 15 sec. Voltage shall be taken after transient settles.

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2.4 (b) Failure Logic for Main Signal				
Note: Failure Logics have been given based on defined parameters. However, Vendors may add new logics and modify given logics based on new parameters added OR otherwise also under approval of Railways and under intimation to RDSO so that suitable amendment may be issued in Specification after review.				
	cause_detail	Logic	cause_code	Remarks on logics
1	Sig No --- RG Aspect failed: HR DN. Signal blank in ON position. Voltage or current of RG Aspect failed.	[HR ↓ && RECR ↓ && $V_{IPS\ SIG-1\ 110\ AC} \geq \text{Min fail value}$ && ($V_{SIG\ RG} < \text{Min Fail value}$ $I_{SIG\ RG} < \text{Min Fail value}$)]	SIG RG VOLT/CURR FAIL	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient set-tles.
2	Sig No --- RG Aspect failed: HR DN. Signal blank in ON position. RECR Relay may be defective.	[HR ↓ && RECR ↓ && ($V_{SIG\ RG} \geq \text{Min safe value}$ && $I_{SIG\ RG} \geq \text{Min safe value}$)]	SIG RG RECR RE- LAY DEFECT	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient set-tles.
3	Sig No --- RG Aspect failed:	[HR ↓	SIG RG UN- KOWN	

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	HR DN. Signal blank in ON position. Reasons not diagnosed.	&& RECR ↓ If any of the remaining logic at sr 1 & 2 does not qualify]		
4	Sig No --- HG Aspect failed: Voltage of HPR OK at Loc. But Voltage or Current of HG aspect failed.	[HR ↑ && HHR ↓ && DR ↓ && HECR ↓ && RECR ↑ && $V_{SIG\ HPR} \geq \text{Min safe value}$ && $V_{IPS\ SIG-1\ 110\ AC} \geq \text{Min fail value}$ && ($V_{SIG\ HG} < \text{Min Fail value}$ $I_{SIG\ HG} < \text{Min Fail value}$)]	SIG HG VOLT/CURR FAIL	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient settles.
5	Sig No. --- HG Aspect failed. Voltage of HPR failed.	[HR ↑ && HHR ↓ && DR ↓	SIG HG HPR VOLT FAIL	[] Logic is checked for 10 sec. Voltage shall be taken after transient settles.

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		&& HECR ↓ && RECR ↑ && $V_{IPS\ DC\ R\ EXT} \geq \text{Min fail value}$ && $V_{SIG\ HPR} < \text{Min Fail value}$]		
6	Sig No. --- HG Aspect failed. HECR Relay may be defective.	[HR ↑ && HHR ↓ && DR ↓ && HECR ↓ && RECR ↑ && $V_{SIG\ HPR} \geq \text{Min safe value}$ && $V_{SIG\ HG} \geq \text{Min safe value}$ && $I_{SIG\ HG} \geq \text{Min safe value}$]	SIG HG HECR RE- LAY DEFECT	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient settles.
7	Sig No. --- HG Aspect failed. Reasons not diagnosed.	[HR ↑ && HHR ↓ && DR ↓	SIG HG UN- KOWN	

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		&& HECR ↓ && RECR ↑ If any of the remaining logic at sr 4, 5 & 6 does not qualify.]		
8	Sig No. --- HHG Aspect failed. Voltage of HPR & HHPR OK at loc but Voltage or Current of HHG aspect failed.	[HR ↑ && HHR ↑ && DR ↓ && (HECR ↓ HHECR ↓) && (RECR ↑ HECR ↑ HHECR ↑) && $V_{SIG\ HPR} \geq \text{Min safe value}$ && $V_{SIG\ HHPR} \geq \text{Min safe value}$ && $V_{IPS\ SIG-1\ 110\ AC} \geq \text{Min fail value}$ && ($V_{SIG\ HG} < \text{Min Fail value}$ $V_{SIG\ HHG} < \text{Min Fail value}$ $I_{SIG\ HG} < \text{Min Fail value}$	SIG HHG VOLT/CURR FAIL	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient set- tles.

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		$\begin{aligned} & \parallel \\ & I_{SIG\ HHG} < \text{Min Fail value} \\ &) \\ &] \end{aligned}$		
9	Sig No. --- HHG Aspect failed. Voltage of HPR OR HHPR failed.	$\begin{aligned} & [\\ & HR \uparrow \\ & \& \\ & HHR \uparrow \\ & \& \\ & DR \downarrow \\ & \& \\ & (HECR \downarrow \parallel HHECR \downarrow) \\ & \& \\ & (RECR \uparrow \parallel HECR \uparrow \parallel HHECR \\ & \uparrow) \\ & \& \\ & V_{IPS\ DC\ R\ EXT} \geq \text{Min fail value} \\ & \& \\ & (\\ & V_{SIG\ HPR} < \text{Min Fail value} \\ & \parallel \\ & V_{SIG\ HHPR} < \text{Min Fail value} \\ &) \\ &] \end{aligned}$	SIG HHG HHPR VOLT FAIL	[] Logic is checked for 10 sec. Voltage shall be taken after transient settles.
10	Sig No. --- HHG Aspect failed. HECR OR HHECR Relay may be defective.	$\begin{aligned} & [\\ & HR \uparrow \\ & \& \\ & HHR \uparrow \\ & \& \\ & DR \downarrow \\ & \& \\ & (HECR \downarrow \parallel HHECR \downarrow) \end{aligned}$	SIG HHG HHECR RE- LAY DEFECT	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient settles.

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		&& (RECR ↑ HECR ↑ HHECR ↑) && $V_{SIG\ HPR} \geq \text{Min safe value}$ && $V_{SIG\ HHPR} \geq \text{Min safe valu}$ && $V_{SIG\ HG} \geq \text{Min safe value}$ && $V_{SIG\ HHG} \geq \text{Min safe value}$ && $I_{SIG\ HG} \geq \text{Min safe value}$ && $I_{SIG\ HHG} \geq \text{Min safe value}$]		
11	Sig No. --- HHG Aspect failed. Reasons not diagnosed.	[HR ↑ && HHR ↑ && DR ↓ && (HECR ↓ HHECR ↓) && (RECR ↑ HECR ↑ HHECR ↑) If any of the remaining logic at sr 8, 9 & 10 does not qualify]	SIG HHG UN- KOWN	

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12	Sig No. --- DG Aspect failed. Voltage of DPR OK at Loc. But Voltage or Current of DG aspect failed.	[HR ↑ && HHR ↑ && DR ↑ && DECR ↓ && (RECR ↑ HECR ↑ HHECR ↑) && $V_{SIG\ DPR} \geq \text{Min safe value}$ && $V_{IPS\ SIG-1\ 110\ AC} \geq \text{Min fail value}$ && ($V_{SIG\ DG} < \text{Min Fail value}$ $I_{SIG\ DG} < \text{Min Fail value}$)]	SIG DG VOLT/CURR FAIL	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient settles.
13	Sig No. --- DG Aspect failed. Voltage of DPR failed.	[HR ↑ && HHR ↑ && DR ↑ && DECR ↓ &&	SIG DG DPR VOLT FAIL	[] Logic is checked for 10 sec. Voltage shall be taken after transient settles.

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		(RECR ↑ HECR ↑ HHECR ↑) && $V_{IPS\ DC\ R\ EXT} \geq \text{Min fail value}$ && $V_{SIG\ DPR} < \text{Min Fail value}$]		
14	Sig No. --- DG Aspect failed. DECR Relay may be defective.	[HR ↑ && HHR ↑ && DR ↑ && DECR ↓ && (RECR ↑ HECR ↑ HHECR ↑) && $V_{SIG\ DPR} \geq \text{Min safe value}$ && $V_{SIG\ DG} \geq \text{Min safe value}$ && $I_{SIG\ DG} \geq \text{Min safe value}$]	SIG DG DECR RE- LAY DEFECT	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient settles.
15	Sig No. --- DG Aspect failed. Reasons not diagnosed.	[HR ↑ && HHR ↑ && DR ↑	SIG DG UN- KOWN	

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		&& DECR ↓ && (RECR ↑ HECR ↑ HHECR ↑) If any of the remaining logic at sr 12, 13 & 14 does not qualify.]		
16	Sig No. --- Signal Blank in OFF position.	[HR ↑ && RECR ↓ && HECR ↓ && HHECR ↓ && DECR ↓]	SIG BLANK	[] Logic is checked for 10 sec.

2.5 (a) Prediction Logic for Calling ON Signal

Note:

1. These logics are primitive logics for Prediction of failures. However, in addition to these primitive logics, these primitive logics, various AI/ML Techniques and learning models need to be developed and applied by Vendors for prediction of failures. The models used by them shall be shared with Zonal Railways/RDSO.
2. Comparisons in logics have been made with deviation in parameter value in terms of certain % of avg value OR deviation from Min/Max safe. Although % deviation has been given for predictive logics for reference, however, % may be modified as decided by Zonal Railway.

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This % is being denoted by “LD” for lower side deviation and “HD” for higher side deviation. For Calling ON Signal, LD=80 and HD=120.

- For Predictive logic, the analog continuous deviation (say < than 80% of avg value) should occur for at least certain duration to avoid predictive alerts for short transients. Indicative timing has been mentioned against logics. However, time duration can be decided by Zonal Railways as per site requirement.
- The deviation should occur for entire duration of at least 1 operation in discrete cases like voltage and current of point machine.
- Prediction Logics have been given based on defined parameters. However, Vendors may add new logics and modify given logics based on field experience OR otherwise also under approval of Railways and under intimation to RDSO so that suitable amendment may be issued in Specification after review.

	cause_detail	Logic	cause_code	Remarks on Logics
1	CO-Sig predictive Alert for CO---: Voltage or Current of Calling ON aspect Low.	[CO-HECR ↑ && VIPS SIG-1 110 AC ≥ Min safe value && (V _{COSIG} < LD % of avg value OR Min safe value I _{COSIG} < LD % of avg value OR Min safe value)]	COSIG ASPECT VOLT/CURR LOW	[] Logic is checked for 15 sec. Voltage and Current shall be taken after transient settles.
2	CO-Sig predictive Alert for CO---: Current high for Calling ON aspect	[CO-HECR ↑ && I _{COSIG} > HD % of avg value OR Max safe value]	COSIG ASPECT CURR HIGH	[] Logic is checked for 15 sec. Current shall be taken after transient settles.

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3	CO-Sig predictive Alert for CO---: Voltage of Co-HPR is Low in Loc Box	[CO-HECR $\uparrow$ && $V_{IPS\ DC\ R\ EXT} \geq \text{Min safe value}$ $V_{COSIG\ HPR} < \text{LD \% of avg value OR Min safe value}$]	COSIG HPR VOLT LOW	[] Logic is checked for 15 sec. Voltage shall be taken after transient settles.
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2.5 (b) Failure Logic for Calling ON Signal				
Note: Failure Logics have been given based on defined parameters. However, Vendors may add new logics and modify given logics based on field experience OR otherwise also under approval of Railways and under intimation to RDSO so that suitable amendment may be issued in Specification after review.				
	cause_detail	Logic	cause_code	Remarks on logics
1	CO-Sig No --- failed: Voltage of CO-HPR OK at Loc. But Voltage or Current of CO-aspect failed.	[CO-HR ↑ && CO-HECR ↓ && $V_{\text{COSIG HPR}} \geq \text{Min safe value}$ && $V_{\text{IPS SIG-1 110 AC}} \geq \text{Min fail value}$ && ($V_{\text{COSIG}} < \text{Min Fail value}$ $I_{\text{COSIG}} < \text{Min Fail value}$)]	COSIG ASPECT VOLT/CURR FAIL	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient settles.
2	CO-Sig No. --- failed. Voltage of CO-HPR failed.	[CO-HR ↑ && CO-HECR ↓ && $V_{\text{IPS DC R EXT}} \geq \text{Min fail value}$ && $V_{\text{COSIG HPR}} < \text{Min Fail value}$]	COSIG HPR VOLT FAIL	[] Logic is checked for 10 sec. Voltage shall be taken after transient settles.
3	CO-Sig No. --- failed. CO-HECR Relay may be defective.	[CO-HR ↑	COSIG HECR RELAY DEFECT	[] Logic is checked for 10 sec.

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		&& CO-HECR ↓ && $V_{\text{COSIG HPR}} \geq \text{Min safe value}$ && $V_{\text{COSIG}} \geq \text{Min safe value}$ && $I_{\text{COSIG}} \geq \text{Min safe value}$]		Voltage and Current shall be taken after transient settles.
4	CO-Sig No. --- failed. Reasons not diagnosed.	[CO-HR ↑ && CO-HECR ↓ If any of the remaining logic above does not qualify.]	COSIG REASON UNKNOWN	

2.6 (a) Prediction Logic for Route Signal
Note: 1. These logics are primitive logics for Prediction of failures. However, in addition to these primitive logics, these primitive logics, various AI/ML Techniques and learning models need to be developed and applied by Vendors for prediction of failures. The models used by them shall be shared with Zonal Railways/RDSO. 2. Comparisons in logics have been made with deviation in parameter value in terms of certain % of avg value OR deviation from Min/Max safe. Although % deviation has been given for predictive logics for reference, however, % may be modified as decided by Zonal Railway.

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This % is being denoted by “LD” for lower side deviation and “HD” for higher side deviation. For Route Signal, LD=80 and HD=120.

- For Predictive logic, the analog continuous deviation (say < than 80% of avg value) should occur for at least certain duration to avoid predictive alerts for short transients. Indicative timing has been mentioned against logics. However, time duration can be decided by Zonal Railways as per site requirement.
- The deviation should occur for entire duration of at least 1 operation in discrete cases like voltage and current of point machine.
- Prediction Logics have been given based on defined parameters. However, Vendors may add new logics and modify given logics based on field experience OR otherwise also under approval of Railways and under intimation to RDSO so that suitable amendment may be issued in Specification after review.

	cause_detail	Logic	cause_code	Remarks on Logics
1	RO-Sig predictive Alert for RO---: Voltage or Current of Route Aspect low.	[UECR ↑ && VIPS SIG-1 110 AC ≥ Min safe value && (V _{ROSIG} < LD % of avg value OR Min safe value I _{ROSIG} < LD % of avg value OR Min safe value)]	ROSIG ASPECT VOLT/CURR LOW	[] Logic is checked for 15 sec. Voltage and Current shall be taken after transient settles.
2	RO-Sig predictive Alert for RO---: Current high for Route aspect.	[UECR ↑ && I _{ROSIG} > HD % of avg value OR Max safe value]	ROSIG ASPECT CURR HIGH	[] Logic is checked for 15 sec. Current shall be taken after transient settles.
3	RO-Sig predictive Alert for RO---: Voltage of RO-HPR is Low in Loc Box	[UECR ↑ && VIPS DC R EXT ≥ Min safe value && V _{ROSIG HPR} < LD % of avg value OR Min safe value]	ROSIG HPR VOLT LOW	[] Logic is checked for 15 sec. Voltage shall be taken after transient settles.

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2.6 (b) Failure Logic for Route Signal				
Note: Failure Logics have been given based on defined parameters. However, Vendors may add new logics and modify given logics based on field experience OR otherwise also under approval of Railways and under intimation to RDSO so that suitable amendment may be issued in Specification after review.				
	cause_detail	Logic	cause_code	Remarks on logics
1	RO-Sig No --- failed: Voltage of RO-HPR OK at Loc. But Voltage or Current of Ro-aspect failed.	[UHR ↑ && UECR ↓ && $V_{ROSIG\ HPR} \geq \text{Min safe value}$ && $V_{IPS\ SIG-1\ 110\ AC} \geq \text{Min fail value}$ && ($V_{ROSIG} < \text{Min Fail value}$ $I_{ROSIG} < \text{Min Fail value}$)]	ROSIG AS- PECT VOLT/CURR FAIL	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient settles.
2	RO-Sig No. --- failed. Voltage of RO-HPR failed.	[UHR ↑ && UECR ↓ && $V_{IPS\ DC\ R\ EXT} \geq \text{Min fail value}$ && $V_{ROSIG\ HPR} < \text{Min Fail value}$]	ROSIG HPR VOLT FAIL	[] Logic is checked for 10 sec. Voltage shall be taken after transient settles.

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3	RO-Sig No. --- failed. UECR Relay may be defective.	[UHR ↑ && UECR ↓ && $V_{ROSIG\ HPR} \geq \text{Min safe value}$ && $V_{ROSIG} \geq \text{Min safe value}$ && $I_{ROSIG} \geq \text{Min safe value}$]	ROSIG UECR RELAY DE- FECT	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient set- tles.
4	RO-Sig No. --- failed. Reasons not diag- nosed.	[UHR ↑ && UECR ↓ If any of the remaining logic above does not qualify.]	ROSIG REASON UNKOWN	

2.7 (a) Prediction Logic for Shunt Signal

Note:

1. These logics are primitive logics for Prediction of failures. However, in addition to these primitive logics, these primitive logics, various AI/ML Techniques and learning models need to be developed and applied by Vendors for prediction of failures. The models used by them shall be shared with Zonal Railways/RDSO.
2. Comparisons in logics have been made with deviation in parameter value in terms of certain % of avg value OR deviation from Min/Max safe. Although % deviation has been given for predictive logics for reference, however, % may be modified as decided by Zonal Railway.

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This % is being denoted by “LD” for lower side deviation and “HD” for higher side deviation. For Shunt Signal, LD=80 and HD=120.

- For Predictive logic, the analog continuous deviation (say < than 80% of avg value) should occur for at least certain duration to avoid predictive alerts for short transients. Indicative timing has been mentioned against logics. However, time duration can be decided by Zonal Railways as per site requirement.
- The deviation should occur for entire duration of at least 1 operation in discrete cases like voltage and current of point machine.
- Prediction Logics have been given based on defined parameters. However, Vendors may add new logics and modify given logics based on field experience OR otherwise also under approval of Railways and under intimation to RDSO so that suitable amendment may be issued in Specification after review.
- These logics have been designed for independent Shunt Signal. Suitable modifications shall be done by Zonal railways for Shunt below Starter.

	cause_detail	Logic	cause_code	Remarks on Logics
1	SH-Sig predictive Alert for SH---: Voltage or Current of Shunt ON Aspect low.	[SH-HR ↓ && SH-ECR _{ON} ↑ && V _{IPS SIG-1 110 AC} ≥ Min safe value && (V _{SHSIG ON} < LD % of avg value OR Min safe value I _{SHSIG ON} < LD % of avg value OR Min safe value I _{SHSIG PILOT} < LD % of avg value OR Min safe value)]	SHSIG ON ASPECT VOLT/CURR LOW	[] Logic is checked for 15 sec. Voltage and Current shall be taken after transient settles.
2	SH-Sig predictive Alert for SH---: Current high for Shunt ON aspect	[SH-HR ↓ && SH-ECR _{ON} ↑	SHSIG ON ASPECT CURR HIGH	[] Logic is checked for 15 sec. Current shall be taken after transient settles.

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		&& ($I_{SHSIG\ ON} > \text{HD \% of avg value OR Max safe value}$ $I_{SHSIG\ PILOT} > \text{HD \% of avg value OR Max}$)]		
3	SH-Sig predictive Alert for SH---: Voltage or Current of Shunt OFF Aspect low.	[SH-HR ↑ && SH-ECR _{OFF} ↑ && $V_{IPS\ SIG-1\ 110\ AC} \geq \text{Min safe value}$ && ($V_{SHSIG\ OFF} < \text{LD \% of avg value OR Min}$ $I_{SHSIG\ OFF} < \text{LD \% of avg value OR Min safe value}$ $I_{SHSIG\ PILOT} < \text{LD \% of avg value OR Min safe value}$)]	SHSIG OFF ASPECT VOLT/CURR LOW	[] Logic is checked for 15 sec. Voltage and Current shall be taken after transient settles.
4	SH-Sig predictive Alert for SH---: Current high for Shunt OFF aspect	[SH-HR ↑ && SH-ECR _{OFF} ↑ && ($I_{SHSIG\ OFF} > \text{HD \% of avg value OR Max safe value}$ $I_{SHSIG\ PILOT} > \text{HD \% of avg value OR Max}$)]	SHSIG OFF ASPECT CURR HIGH	[] Logic is checked for 15 sec. Current shall be taken after transient settles.

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		)]		
5	SH-Sig predictive Alert for SH---: Voltage of SH-HPR is Low in Loc Box	[SH-HR ↑ && SH-ECR _{OFF} ↑ && $V_{IPS\ DC\ R\ EXT} \geq \text{Min safe value}$ $V_{SHSIG\ HPR} < \text{LD \% of avg value OR Min safe value}$]	SHSIG OFF ASPECT HPR VOLT LOW	[] Logic is checked for 15 sec. Voltage shall be taken after transient settles.

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2.7 (b) Failure Logic for Shunt Signal				
Note: 1. Failure Logics have been given based on defined parameters. However, Vendors may add new logics and modify given logics based on field experience OR otherwise also under approval of Railways and under intimation to RDSO so that suitable amendment may be issued in Specification after review. 2. These logics have been designed for independent Shunt Signal. Suitable modifications shall be done by Zonal railways for Shunt below Starter.				
	cause_detail	Logic	cause_code	Remarks on logics
1	SH-Sig No --- ON aspect failed: Voltage or Current of Shunt ON aspect failed.	[SH-HR ↓ && SH-ECR _{ON} ↓ && $V_{IPS\ SIG-1\ 110\ AC} \geq \text{Min fail value}$ && ($V_{SHSIG\ ON} < \text{Min Fail value}$ $(I_{SHSIG\ ON} + I_{SHSIG\ PILOT}) < \text{Min Fail value}$)]	SHSIG ON ASPECT VOLT/CURR FAIL	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient settles.
2	SH-Sig No --- ON aspect failed: Voltage or Current of Shunt ON aspect ok but ECR _{ON} may be defective.	[SH-HR ↓ && SH-ECR _{ON} ↓ && ($V_{SHSIG\ ON} \geq \text{Min safe value}$ &&]	SHSIG ON ASPECT ECR RELAY DEFECT	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient settles.

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		$I_{SHSIG\ ON} \geq \text{Min safe value}$ && $I_{SHSIG\ PILOT} \geq \text{Min safe value}$)]		
3	SH-Sig No --- ON aspect failed: Voltage or Current of Shunt ON aspect failed.	[SH-HR ↓ && SH-ECR _{ON} ↓ && If any of the remaining logic above does not qualify.]	SHSIG ON ASPECT REASON UNKNOWN	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient settles.
4	SH-Sig No --- OFF aspect failed: Voltage of HPR OK at Loc. But Voltage or Current of Shunt OFF aspect failed.	[SH-HR ↑ && SH-ECR _{OFF} ↓ && $V_{SHSIG\ HPR} \geq \text{Min safe value}$ && $V_{IPS\ SIG-1\ 110\ AC} \geq \text{Min fail value}$ && ($V_{SHSIG\ OFF} < \text{Min Fail value}$ $(I_{SHSIG\ OFF} + I_{SHSIG\ PILOT}) < \text{Min Fail value}$)]	SHSIG OFF ASPECT VOLT/CURR FAIL	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient settles.
5	SH-Sig No. --- OFF aspect failed. Voltage of HPR failed.	[SH-HR ↑ &&	SHSIG OFF ASPECT HPR VOLT FAIL	[] Logic is checked for 10 sec.

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		SH-ECR _{OFF} ↓ && $V_{IPS\ DC\ R\ EXT} \geq \text{Min fail value}$ && $V_{SHSIG\ HPR} < \text{Min Fail value}$]		Voltage shall be taken after transient settles.
6	SH-Sig No. --- OFF aspect failed. Sh-ECR _{OFF} Relay may be defective.	[SH-HR ↑ && SH-ECR _{OFF} ↓ && $V_{SHSIG\ HPR} \geq \text{Min safe value}$ && $V_{SHSIG\ OFF} \geq \text{Min safe value}$ && $I_{SHSIG\ OFF} \geq \text{Min safe value}$ && $I_{SHSIG\ PILOT} \geq \text{Min safe value}$]	SHSIG OFF ASPECT ECR RELAY DEFECT	[] Logic is checked for 10 sec. Voltage and Current shall be taken after transient settles.
7	SH-Sig No. --- failed. Reasons not diagnosed.	[SH-HR ↑ && SH-ECR _{OFF} ↓ If any of the remaining logic above does not qualify.]	SHSIG OFF ASPECT REASON UNKNOWN	

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Annexure – D

Alert mechanism and other processes for RDPMS Application

RDPMS Application processes logics and AI/ML techniques using data received from station gateway to generate various types of Alerts which are sent to mobile. The details of Alerts and other processes are being given below.

1. **Alert Types:** There shall be three types of Alerts i.e. Signalling assets Failure Alert/Predictive Alert, RDPMS health Alert and Maintenance mode active Alert. On clicking of the particular alert, the detailed technical information should come. It should be possible to give alerts to higher hierarchy.

1.1 Failure /Predictive Alert for Signalling Asset:

There shall be two types of alerts i.e Signalling assets Failure Alert and Signalling Assets Predictive Alert. These alerts shall give likely cause for each type of alert based on standard logics and AI/ML techniques. These alerts shall be sent to maintainer and higher hierarchy.

Alert type is denoted as:

Alert Type	alert_type_code
Failure	"Failure"
Predictive	"Predictive"

alert_type_code should be used in UIs and APIs where Alert Type is mentioned.

1.2 RDPMS Health Alert:

These alerts shall provide the health of IoT, Sensors, Station Gateway and Network. These alerts shall be sent to Railway officials (Maintainer and higher hierarchy) and vendor representative. The health of sensors and IoT devices shall be communicated from station gateway to RDPMS application software using standard data format. (annexure B) (Annexure A also to be referred for Standard Nomenclature).

1.3 Maintenance Mode Active Alert: This Alert shall be given as detailed in para 5 of this Annexure D.

2. The non-availability of data due to failure of IoT device or normal train operation should not cause false alerts.

3. Alert standard message format:

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Alert message format for various types of Alerts are given below.

For example,

3.1 Message format for Signal Failures/Predictive Alert (string under quotes ie “xyz” used is as per Annexure A and C):

Sig failure Alert:

“sc”, 15:35/13.12.25, “asset_number_code”, “cause_code”

Sig predictive Alert:

“sc”, 16:35/14.12.25, “asset_number_code”, “cause_code”

3.2 Message format for RDPMS health Alert:

Message format for Sensor failure Alert (string under quotes ie “xyz” used is as per Annexure A):

Sensor failure Alert:

“sc”, 16:35/14.12.25, “asset_number_code”, para:,

Message format for IoT failure Alert (string under quotes ie “xyz” used is as per Annexure A):

IoT failure Alert:

“sc”, 16:35/14.12.25, “asset_number_code”

Message format for station gateway failure Alert (string under quotes ie “xyz” used is as per Annexure A):

Stngw failure Alert:

“sc”, 16:35/14.12.25

Message format for Network failure Alert (string under quotes ie “xyz” used is as per Annexure A):

Network failure Alert:

“sc”, 16:35/14.12.25, “net_des”

3.3 Message format for Maintenance mode Active Alert:

Maintenance mode Active Alert:

“sc”, 16:35/14.12.25, “asset_number_code”

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4. Cause of Asset failure: it is used in Standard API, UI & reports. Denoted as: Details of cause_code are given in Annexure C.

CAUSES	cause_detail	cause_code
Point Machine failure	Details given in logics for understanding purpose	Given in logics in Annexure C and to be used in UI, reports and APIs.
Point Machine Pre-dictive	Details given in logics for understanding purpose	Given in logics in Annexure C and to be used in UI, reports and APIs.

cause_code should be used in Reports in UIs and APIs where Cause is mentioned.

5. Maintenance mode for Alerts:

1. There should be provision in UI (as shown in Annexure E & G) for activating maintenance mode to stop Alerts during maintenance.
2. There should be an option to select asset and enter “from date & time” and “to date & time” of maintenance active period before clicking activate tab.
3. It should also be possible to modify the date and time by re-entering and re-clicking activate tab.
4. Maintenance mode will be cleared automatically as per end time.
5. Table in the page shall display all active maintenance mode in UI.
6. There should be icon for clearing maintenance mode, if maintainer wants to clear it before end time.
7. In case, ESM forgets to clear maintenance mode and end time given is high, therefore, then it shall generate regular Alert after standard interval as defined below advising Railway staff that that maintenance mode is active till maintenance mode is cleared.
8. In case, Alert gets generated because of maintenance, ESM can press “M” button at the time of feedback of alert to exclude it in counting.

Time after which regular Alert shall be generated that maintenance mode is active.		
Sr	Name of Asset	Duration
1	Track Circuit	60 min
2	Point Machine	60 min
3	Signal	45 min

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Note: Above time duration are indicative. Zonal Railways can decide depending on standard disconnection/block duration for all equipments which shall be configurable in RDPMS Application.

- 6. Alert feedback:** Feedback should be provided by Railway employee (Maintainer) whether generated alert was True ("T"), Partially True ("PT"), False ("F") OR Maintenance ("M") after attending the failure/prediction. "M" feedback option will exclude Alerts generated during maintenance for analysis and counting. Failure should be attended and feedback should be given by maintainer in definite time window (configurable and to be decided by Zonal Railway).

Denoted as:

Alert Feedback	alert_feedback_code	Duration within which feedback should be given by maintainer after Alert is received (in hours) (configurable)	
		Failure Alert	Predictive Alert
True	"T"	*	*
Partially True	"PT"	*	*
False	"F"	*	*
Maintenance	"M"	*	*

" * " To be decided by Zonal Railway.

alert_feedback_code should be used in repots, UIs and APIs where Alert feedback is mentioned.

6.1 Alert feedback for failure: True (T) feedback should be given when failure has actually occurred and corresponding cause found at site is same as given by RDPMS failure alert. Partially True (PT) feedback should be given when failure has actually occurred and corresponding cause found at site is not same as given by RDPMS failure alert. False (F) feedback should be given when failure has not actually occurred at site. Maintenance (M) feedback should be given, if failure alert has been given due to maintenance activity at site. After T/PT/F/M feedback, it should not be possible to modify the feedback by the maintainer. However, it should be possible to modify the feedback by higher authority (JE/SSE) with remarks.

False (F) and Partially True (PT) Alert notifications shall be given to Vendors for their remarks within a specified time span (configurable). In such cases, higher authority (JE/SSE) shall decide the final Alert feedback with remarks after examining

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feedback/remarks from maintainer and vendor both. System shall take the Alert feedback as given by the higher authority (JE/SSE) with remark as final feedback.

6.2 Alert feedback for Prediction: True (T) feedback should be given if corresponding cause found at site is same as given by RDPMS predictive alert. Partially True (PT) feedback should be given if corresponding cause found at site is partially correct as given by RDPMS predictive alert. False (F) feedback should be given if corresponding cause found at site does not match with cause given by RDPMS predictive alert Or there is no cause found. Maintenance (M) feedback should be given, if prediction alert has been given due to maintenance activity at site. After T/PT/F/M feedback, it should not be possible to modify the feedback by the maintainer. However, it should be possible to modify the feedback by higher authority (JE/SSE) with remarks.

False (F) and Partially True (PT) Alert notifications shall be given to Vendors for their remarks within a specified time span (configurable). In such cases, higher authority (JE/SSE) shall decide the final Alert feedback with remarks after examining feedback/remarks from maintainer and vendor both. System shall take the Alert feedback as given by the higher authority (JE/SSE) with remark as final feedback.

7. Remarks: The Railway Person shall give remarks after attending the failure/prediction in this head. The box for remarks shall open after pressing “T”, “PT”, “F” OR “M” Tab for feedback to ensure that feedback is not missed. Remarks column shall have option to enter date and time of rectification and remarks of failure/prediction which can be captured by RDPMS Application to reflect in Alert history. After T/PT/F/M feedback and remarks, it should not be possible to re-enter the feedback and remarks in UI by maintainer. However, it should be possible to modify the remarks (given by maintainer) by higher authority (JE/SSE) with his remarks.

Denoted as “remarks”. It is to be used in UI, API & Reports.

8. Alert Status:

“cleared” Alerts denotes that failure/prediction has been attended and feedback (T, PT, F & M) with remarks have been given by maintainer. If any modification of alert feedback/remarks is pending on account of JE/SSE as described in above paras, the feedback and remarks given by maintainer shall prevail and will be treated as “cleared” for all purposes. After feedback modification by JE/SSE, feedback and remarks shall be updated accordingly. “pending” Alerts denotes that Alert feedback has not been given even if failure/prediction may have been attended at site.

Alert Status	alert_status_code
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Cleared	"cleared"
Pending	"pending"

alert_status_code should be used in UIs and APIs where Alert status is mentioned

9. Incidence Date & Time: It is actual time of Occurrence of Failure/Prediction.
(However, Alert may take some time to generate and reach after occurrence).

Denoted as "incidence_date_time" For ex 18/12/2023 14:40:20

incidence_date_time should be used in UIs and APIs where Incidence Date & Time is mentioned

10. Rectification Date & Time: It is actual time of rectification of Failure/prediction. If rectification has not taken place, it will show "-----".

Denoted as "rectification_date_time" For example "18/12/2023 16:40:20"

rectification_date_time should be used in UIs and APIs where Rectification Date & Time is mentioned. Actual date and time of rectification can be found out by two methods, one is to find it using appropriate logic through RDPMS Application and other is to get it entered by maintainer during remarks entry.

11. Incidence duration: It is actual duration of Failure/prediction.

Denoted as: "incidence_duration": "01:40:20"

incidence_duration should be used in UIs and APIs where Incidence duration is mentioned

12. Alert feedback Date & Time: It is time of providing feedback of (T, PT, F & M) & Remarks of Alert.

Denoted as "alert_feedback_date_time" For example "18/12/2023 16:40:20"

alert_feedback_date_time should be used in UIs and APIs where Alert feedback Date & Time is mentioned

13. ALERT MECHANISM:

13.1 Selection of Alert for pushing when generated through multiple logics: To avoid multiple Alerts generated through multiple logics for one asset, a scheme should be

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followed to select one Alert for pushing. For this, Alert pushing flow has been described as below:

- (i) Once first predictive Alert is pushed for any particular asset say PT-201, It will not be eligible for other predictive Alerts (due to other predictive logics) for same asset i.e. PT-201 till Alert feedback is given with remarks. However, data and Alerts generated during this intervening period may be kept for records to show whenever required and for developing AI/ML models.

However, if Failure Alert is generated during the intervening period between First Predictive Alert and before feedback with remarks, Failure Alert shall be pushed.

- (ii) Once First Failure Alert is pushed for any particular gear say PT-201 OR Failure Alert is pushed after one Predictive Alert as mentioned in sr (i) above, It will not be eligible for other Failure & Predictive Alerts for same asset ie PT-201 till failure Alert feedback is given with remarks. However, data and Alerts generated during this intervening period may be kept for records to show whenever required and for developing AI/ML models.

13.2 Repeat Frequency of same Alert to maintainer: Failure and predictive Alerts shall be generated as per the logics and AI/ML models and as per 13.1 above and sent to maintainer immediately once. However, the repetition of same Alerts will be sent to maintainer only after duration as mentioned below OR as decided by Zonal Railway (configurable) to avoid excessive Alerts till the cause of failure/prediction is rectified and alert feedback with remarks are given. Repeat Alerts should not be counted in total Alerts for analysis. However, it can be separately analyzed for the purpose of checking promptness of staff.

Sr no	Gear	Designation for pushing	Failure Alert Freq (in hrs)	Predictive Alert Freq (in hrs)
1	Point Machine	ESM	2	12
2	LC gate	ESM	2	12
3	IPS	ESM	1	12
4	Signal	ESM	3	12
5	Track circuit /Axle Counter	ESM	3	12

13.3 Repeat Frequency of same Alert to higher hierarchy: Alerts should be pushed to higher hierarchy (JE/SE) 30 min (configurable) after occurrence. However, repetition of same alerts should be as below.

Sr no	Gear	Designation for pushing	Failure Alert Freq (in hrs)	Predictive Alert Freq (in hrs)
1	Point Machine	JE/SE	3	18

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2	LC gate	JE/SE	3	18
3	IPS	JE/SE	1	18
4	Signal	JE/SE	4	18
5	Track circuit /Axle Counter	JE/SE	4	18

Alerts should be pushed to higher hierarchy (SSE) 1 hour (configurable) after occurrence. However, repetition of same alerts should be as below.

Sr no	Gear	Designation for pushing	Failure Alert Freq (in hrs)	Predictive Alert Freq (in hrs)
1	Point Machine	SSE	6	24
2	LC gate	SSE	6	24
3	IPS	SSE	1	24
4	Signal	SSE	8	24
5	Track circuit	SSE	8	24

Alerts should be pushed to higher hierarchy (ASTE/DSTE) 2 hours (configurable) after occurrence. However, repetition of same alerts should be as below.

Sr no	Gear	Designation for pushing	Failure Alert Freq (in hrs)	Predictive Alert Freq (in hrs)
1	Point Machine	ASTE/DSTE	9	36
2	LC gate	ASTE/DSTE	9	36
3	IPS	ASTE/DSTE	2	36
4	Signal	ASTE/DSTE	12	36
5	Track circuit	ASTE/DSTE	12	36

Note: All durations mentioned above are indicative. Zonal Railway may decide the durations.

14. Type of Equipment Room in S&T installation: There are various types of equipment rooms in S&T installation such as Relay room, IPS room etc. Equipment room code for different type of rooms have been given in Annexure A. This shall be used to display Temperature, humidity and Relay room door open/close in Equipment Live and history page of UI as shown in Annexure E.

15. Parameter for performance as per para 14 of Main specification document:

(i) % of (True + Partially True) Failure Alerts out of total Failure Alerts generated is denoted as "fail_alert_per". This shall be used in UI, APIs. This shall be float value between 0-100.

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- (ii) % of (True + Partially True) Predictive Alerts out of total Predictive Alerts generated is denoted as “pred_alert_per”. This shall be used in UI, APIs. This shall be float value between 0-100.
- (iii) % of (True + Partially True) Failure Alerts by RDPMS out of total actual failures occurred at station for the assets being monitored by RDPMS is denoted as “actual_fail_alert_per”. This shall be used in UI, APIs. This shall be float value between 0-100.

- 16.** RDPMS health shall be monitored through two ways. One is to give Alerts for health issue of sensor, IoT, Gateway and Network as per the Annexure A & B data format. Other is to give overall performance of health of sensors etc in any given duration in terms of availability. The formula for finding out availability of Sensors shall be as following:

$$\text{Availability of Sensors in 24 hours} = \frac{\text{Sum of healthy time of each sensor in 24 hours}}{\text{Number of total Sensors X 24 hours}}$$

Similarly, availability of IoT, Station gateway and network can be calculated for any given time duration.

- 17. Configurable parameters in RDPMS Application:** As mentioned above in this annexure and other places, following are configurable parameters. It should be possible to configure them by Admin in user Interface as decided by Zonal Railways. If Zonal Railway does not want to give any duration, it should be possible to configure it as “No time limit”

sr	Configuration details	Configuration value (in hours)
1	Frequency for repeat Alerts to advise maintainer staff that Maintenance mode is active. (asset type wise)	
2	Failure Alert feedback duration by Maintainer (asset type wise)	
3	Predictive Alert feedback duration by Maintainer (asset type wise)	
4	Failure Alert comment duration by Vendor after PT/F feedback given by Maintainer (asset type wise)	
5	Predictive Alert comment duration by Vendor after PT/F feedback given by Maintainer (asset type wise)	

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6	Failure Alert feedback modification duration by JE/SSE after Vendor comment for PT/F feedback given by Maintainer (asset type wise)	
7	Predictive Alert feedback modification duration by JE/SSE after Vendor comment for PT/F feedback given by Maintainer (asset type wise)	
8	Duration after which first Failure Alert shall be pushed to higher hierarchy (JE/SSE/Officers) asset wise	
9	Duration after which first Predictive Alert shall be pushed to higher hierarchy (JE/SSE/Officers) asset wise	
10	Repeat Failure Alert frequency to Maintainer (asset type wise)	
11	Repeat Failure Alert frequency to JE/SE (asset type wise)	
12	Repeat Failure Alert frequency to SSE (asset type wise)	
13	Repeat Failure Alert frequency to ASTE/DSTE (asset type wise)	
14	Repeat Failure Alert frequency to higher Officer (asset type wise)	
15	Repeat Predictive Alert frequency to Maintainer (asset type wise)	
16	Repeat Predictive Alert frequency to JE/SE (asset type wise)	
17	Repeat Predictive Alert frequency to SSE (asset type wise)	
18	Repeat Predictive Alert frequency to ASTE/DSTE (asset type wise)	
19	Repeat Predictive Alert frequency to higher Officer (asset type wise)	
20	Number of days for taking average of parameter (avg operating parameter) value which has been taken as 15 days in Annexure A and used in logics in Annexure C.	
21	Time synch data packet sending frequency as per para 1.1.3 of Annexure B.	7 (days)
22	Information request data packet sending frequency as per para 1.1.5 of Annexure B.	30 (days) 21
23	Image request data packet sending frequency as per para 1.1.6 of Annexure B.	7 (days)

18. The system shall be designed to achieve the objective of providing real-time information on the parameters of signaling assets, as proposed in this Specification. The maximum time for updating an event at application shall normally be within one minute as per existing packet sending scheme, however it may vary, if packet sending scheme is changed.

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19. Vendor shall maintain audit log that contain unauthorize access to IoT Station Gate-way and RDPMS application and the audit report can be accessed by admin or as desired by Railway authority in RDPMS application.

Summary of nomenclatures

Sr	Description	Short name used	Remarks	Use case
1	Alert type code	alert_type_code	string	APPLICATION, API
2	Alert feedback code	alert_feedback_code	string	APPLICATION, API
3	Alert status code	alert_status_code	string	APPLICATION
4	Cause detail	cause_detail		
5	Cause code	cause_code	string	APPLICATION, API
6	Incidence date time	incidence_date_time	Date & time format as described	APPLICATION, API
7	Rectification date time	rectification_date_time	Date & time format as described	APPLICATION, API
8	Incidence duration	incidence_duration	Date & time format as described	APPLICATION, API
9	Alert feedback date time	alert_feedback_date_time	Date & time format as described	APPLICATION, API
10	remarks	remarks	string	APPLICATION, API
11	Total count	total_count	hexadecimal	APPLICATION, API
12	True count	true_count	hexadecimal	APPLICATION, API
13	Maintainer name	maintainer_name	string	APPLICATION, API
14	Maintainer designation	maintainer_designation	string	APPLICATION, API
15	Maintainer mobile	maintainer_mobile	string	APPLICATION, API

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Annexure E

Standard User Interface of RDPMS Application for Laptop/desktop

1. The RDPMS application supports different user roles with specific permissions. Head Quarter and Division Level Users can assign users, monitor, and generate reports but lack control functions. Station Level Users can monitor, generate reports, and control functions but cannot assign users. Guest Users may also be defined for users other than three users defined. Guest users have limited monitoring access only. UI template may be filtered to show only relevant icons to different users as per their role.
2. Standard User interface (UI) template has been described in this annexure to keep it similar across all vendors. Changes in standard UI template can be done during the course of/ or after the vendor development as per requirement. Vendors shall modify the UI as per changes made in this Specification in future.
Standard UI in this annexure is template only. Vendors may further add functions/features to make it more informative and useful. They may also augment each function with suitable analysis/reports. Shortcut icons may also be added to switch between different pages/functions. This addition should be done keeping in view that over all UI should not be too complicated for a user to understand.
3. **Landing page:** It has been defined where user id and password can be entered.
4. **Sidebar:** Various functions of Applications like Dashboard, Alerts, Telemetry RDPMS health etc. have been mentioned at side bar icons. Each function icon shall show further sub functions in drop down menu. Sidebar shall appear at every page. Each sub function can be selected by clicking the icon. Various sub functions have been defined below: Arrow mark in icons indicate that it his drop-down menu.
5. **Icons in each sub function/page:** In each sub function/page, several icon shall appear horizontally with several field in dropdown menu as shown in UI template ahead. Fields can be selected in all icons and search icon in the end can be clicked to display the result accordingly. The result table shall have columns with different desired fields There are following options in while selecting fields in icons.
 - (i) One or more field can be selected. Result table shall show details for each selected field.
 - (ii) "All " dropdown shall appear to select all fields. Result table shall show details for each individual field for all fields.
 - (iii) "-----" drop down shall appear to select all fields (same as selecting no field). Result table shall show details for all fields (cumulative) but table shall not show them individually and will display "-----" in table against that field.

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- (iv) Searchable fields shall have flexibility of selecting fields in such a way that all possible analysis can be done.
- (v) “View” icon has been given options like table, bar chart, circuit etc for indicative purpose. Vendors can decide appropriate presentable options.

6. **Dashboard:** After entering user id and pass word, it lands on Dashboard page where live status of asset types has been displayed whether how many are healthy, predictive and failure. It also displays healthy and failure status of Sensors and IoTs. It also displays % utilization of CPU, RAM & Storage of RDPMS Application Server. This page has searchable fields to select Zone, Division or station with a drop-down menu. After selecting various options, search icon should be clicked to implement the search.

7. **Alerts:** This icon shall have 3 dropdown menus.

7.1 **Alert Live:** Alert Live Status gives details of each “pending” alert (which are not yet attended or feedback with remarks not yet given, if attended). If failure has been attended, date & time of rectification shall appear else “---” shall appear in table shown in respective column. It has searchable fields with dropdown options as shown. Right side icon will display only those fields in drop down which are pertaining to fields selected in its left side icon. Result shall display only after search icon is clicked after selecting all dropdown fields. Feedback can be given by clicking “T”, “PT”, “F” or “M” icon after failure/prediction is attended. Remarks box shall appear after clicking “T”, “PT”, “F” or “M” icon to ensure feedback is given. If “PT” feedback is being given then drop down should also appear with all causes so that correct cause can be selected at the time of feedback along with text remarks. This page for JE/SSE should also display “PT” or “F” given by maintainer for modification of feedback as described in Annexure D.

7.2 **Alert History:** Clicking this icon shall open an Alert History page. It shall have searchable icons with dropdown menu to select various options to generate Alert detailed report. It gives details of each alert for selected duration and for selected fields for all Alerts in different presentable form. Right side icon will display only those fields in drop down which are pertaining to fields selected in its left side icon. Result shall display only after search tab is clicked (shown ahead). The fields shall show “---” where data is not yet available. Additional icons can be added to present the report in different form like month wise analysis etc.

7.3 **Alert Summary:** Clicking this icon shall open an Alert Summary page. It shall have searchable icons with dropdown menu to select various options. Alert summary report

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gives count of total and true Alerts for selected duration and for selected fields for all cleared Alerts as per definition in different presentable form. Right side icon will display only those fields in drop down which are pertaining to fields selected in its left side icon. Result shall display only after search tab is clicked. (shown ahead). Additional icons can be added to present the report in different form like month wise analysis etc.

7.4 Maintenance mode: Clicking this icon shall open maintenance mode page to activate and clear maintenance mode to stop Alerts during maintenance. It shall have searchable options with dropdown menu to select asset and enter “from date & time” and “to date & time” of maintenance active period before clicking activate icon. It should also be possible to modify the date and time by re-entering and re-clicking activate icon. Table in the page shall display all active maintenance mode. Maintenance mode shall clear at given end time, however, if maintainer wants to clear it before end time, there should be icon for clearing maintenance mode in the table. (Please refer Annexure D)

Note: icon can be given at suitable place in Alert function to show comments of Vendor on PT and F Alerts and review by SSE as per para 6 of Annexure D.

8. Telemetry: This icon shall have 2 dropdown menus.

8.1 Telemetry Live: Clicking this icon shall open a Telemetry Live page. It shall have searchable icons with dropdown menu to select various options. Telemetry Live gives Live parameters of various assets for selected fields in different presentable form. Right side icon will display only those fields in drop down which are pertaining to fields selected in its left side icon. Result shall display only after search tab is clicked. (shown ahead)

8.2 Telemetry History: Clicking this icon shall open a Telemetry History page. It shall have searchable icons with dropdown menu to select various options. Telemetry History gives parameters of various assets for selected fields for selected duration in different presentable form. Right side icon will display only those fields in drop down which are pertaining to fields selected in its left side icon. Result shall display only after search tab is clicked. (shown ahead). Additional icons can be added to present the report in different form like month wise analysis etc.

9. RDPMS Health: This icon shall have 2 dropdown menus.

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9.1 RDPMS Health Live: Clicking this icon shall open a RDPMS health Live page. It shall have searchable icons with dropdown menu to select various options. RDPMS health Live gives the health of Sensor, IoT, Station Gateway & Network for various assets for selected fields in different presentable form. Right side icon will display only those fields in drop down which are pertaining to fields selected in its left side icon. Result shall display only after search tab is clicked. (shown ahead).

Pictorial form shall represent Sensor, IoT and station gateway in pictorial form with green as working and Red as fail. Generally, IoTs can be shown in one screen for a station with very small icon/dot for connected Sensors with IoTs. For a bigger scenario like Division, only Station gateways can be shown in one screen with small icon/dot for connected IoTs.

9.2 RDPMS Health Summary: Clicking this icon shall open an RDPMS health Summary report page. It shall have searchable icons with dropdown menu to select various options. RDPMS health detail report gives the status of Sensor, IoT, Station Gateway & Network for various assets for selected fields for selected duration in different presentable form. Right side icon will display only those fields in drop down which are pertaining to fields selected in its left side icon. Result shall display only after search tab is clicked. (shown ahead). Additional icons can be added to present the report in different form like month wise analysis etc.

10. Equipment Room: This icon shall have 2 dropdown menus.

10.1 Equipment room Live: Clicking this icon shall open Equipment room Live page. It shall have searchable icons with dropdown menu to select various options. It gives the Live values of equipment room parameters such as Temperature, Humidity and Relay room door open/close for various types of rooms for selected fields in different presentable form. Right side icon will display only those fields in drop down which are pertaining to fields selected in its left side icon. Result shall display only after search tab is clicked. (shown ahead).

10.2 Equipment room History: Clicking this icon shall open equipment room history page. It shall have searchable icons with dropdown menu to select various options. It gives temperature, humidity and door close/open position of various types of room in past for selected duration in different presentable form. Right side icon will display only those fields in drop down which are pertaining to fields selected in its left side icon. Result shall display only after search tab is clicked. (shown ahead). Additional icons can be added to present the report in different form like month wise analysis etc.

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11. Asset: This icon shall have 2 dropdown menus

11.1 Asset details: Clicking this icon shall open an Asset detail page. It shall have searchable icons with dropdown menu to select various options. Asset detail gives count of total assets for each type of asset with its details as on present date for selected fields in different presentable form. Right side icon will display only those fields in drop down which are pertaining to fields selected in its left side icon. Result shall display only after search tab is clicked. (shown ahead)

11.2 Asset utilization: Clicking this icon shall open an Asset utilization page. It shall have searchable icons with dropdown menu to select various options. Asset utilization gives utilization of various assets for selected fields for selected duration in different presentable form. Right side icon will display only those fields in drop down which are pertaining to fields selected in its left side icon. Result shall display only after search tab is clicked. (shown ahead). Additional icons can be added to present the report in different form like month wise analysis etc.

12. Performance: Clicking this icon shall open performance page. It shall have searchable icons with dropdown menu to select various options. It gives performance as per para 14 of main document for selected duration in different presentable form. Right side icon will display only those fields in drop down which are pertaining to fields selected in its left side icon. Result shall display only after search tab is clicked. (shown ahead). Additional icons can be added to present the report in different form like month wise analysis etc.

An additional icon shall be provided under this section to enable the Railway Personnel (Nominated by Zonal Railways) to enter the number of actual signal failures of the assets being monitored by RDPMS at the station. This entry shall be done asset wise either daily or at definite interval which can be decided by Zonal Railway. This entry shall be used by RDPMS vendor to calculate the % as per para 14 (iii) of performance criteria of main specification.

13. Admin: It gives details of admin and icons for admin related actions. Vendor may design as per requirement.

14. User Profile: It gives details of user and icons for user related actions. Vendor may design as per requirement.

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Landing page

Remote Diagnostic and Predictive Maintenance System
Vendor name

User id

XYZ

Password

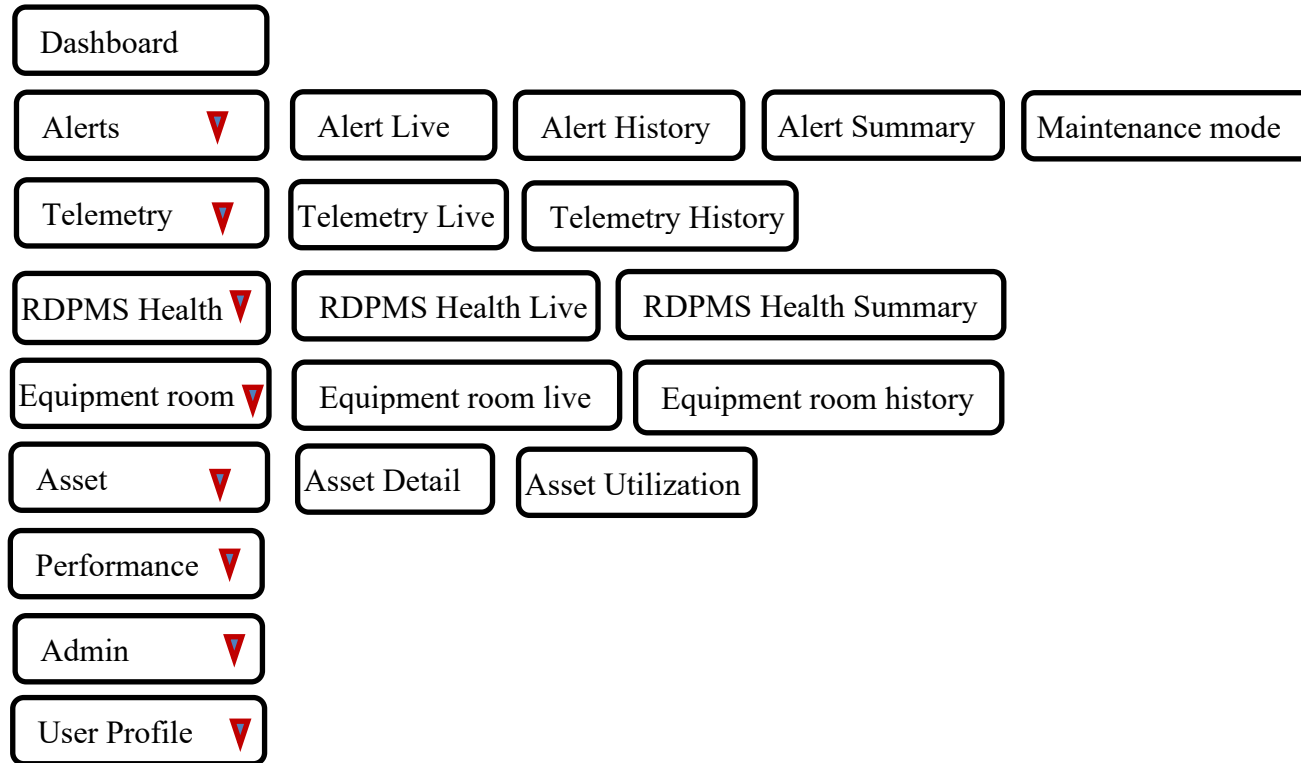
OTP based

Sign in for New User

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Side Bar



Note:

1. This side bar shall appear in all pages for accessing various functions.
2. Sub-tabs have been shown horizontally in front of main tab for the purpose of clarity. However, it shall actually appear as a drop-down menu vertically (or in any suitable manner).

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Dashboard

Alerts ▼

Telemetry ▼

RDPMS Health ▼

Equipment room ▼

Asset ▼

Performance ▼

Admin ▼

User Profile ▼

Dashboard

Zone ▼

Division ▼

Station ▼

% utilization of CPU, RAM & Storage of RDPMS Application should be displayed

DC Track Circuit

Healthy – 60/75
Predictive – 10/75
Failure – 5/75

Main Signal

Healthy – 34/42
Predictive – 6/42
Failure – 2/42

Axle Counter

Healthy – 34/42
Predictive – 6/42
Failure – 2/42

LC Gate

Healthy – 34/42
Predictive – 6/42
Failure – 2/42

Point Machine

Healthy – 34/42
Predictive – 6/42
Failure – 2/42

Suitable chart may be placed here

Total Assets

Healthy – 170/200
Predictive – 20/200
Failure – 10/200

Sensor

Healthy – 170/200
Failure – 10/200

IoT

Healthy – 170/200
Failure – 10/200

Note: This is template only. Designs may be improved further.

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Alert Live

Zone ▼

Division ▼

Station ▼

Alert Type ▼

Asset Type ▼



ALL

ALL

Predictive

Point Machine

Failure

Track Ckt.

Predictive	20
------------	----

Failure	5
---------	---

Total	25
-------	----

Sr no.	Zone	Division	Station	Alert Type	Asset Type	Asset Number	Incidence Date & Time	Duration	Rectification date & Time	cause of As-set fail-ure	Feedback & Remarks
1											T PT F M
2											T PT F M

Alert History

Zone▼

Division▼

Station▼

Alert type▼

Asset Type▼

Asset number▼

Cause▼

Alert Feedback▼

From date▼

From time▼

To date▼

To time▼

ALL

ALL

ALL

ALL

ALL

ALL

ALL

ALL

NR

LKO

NDLS

Predictive

As per SMM

As per SMM

PT-OBS

T

NER

Failure

As per SMM

As per SMM

F

M

From :-

Date

Time

To :-

Date

Time

Sr no	Zone	Division	Station	Alert Type	Asset Type	Asset Number	Alert status	Cause of Asset failure	Alert Feedback (T/PT/M/F)	Incidence Date & Time	Rectification Date& Time	Incidence Duration	Alert feedback date & Time	Maintainer Name	Maintainer designation	Maintainer Mobile Number	Maintainer Re-marks

Alert summary

Zone ▼

Division ▼

Station ▼

Alert Type ▼

Asset Type ▼

Asset Number ▼

Cause ▼

From date ▼

From time ▼

To date ▼

To time ▼

View ▼

Table

Pie

Bar

Graph

From :-

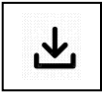
Date

Time

To :-

Date

Time



Sr no.	Zone	Division	Station	Alert Type	Asset Type	Asset Number	Cause of As- set failure	Total Alert Count	True Alert Count	Partially True Alert Count	% of (True + Partially True) Alerts out of To- tal Alerts

Maintenance mode

Zone ▼

Division ▼

Station ▼

Asset Type ▼

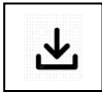
Asset Number ▼

From date ▼

From time ▼

To date ▼

To time ▼



Sr no.	Zone	Division	Station	Asset Type	Asset Number	Date and Time of Maintenance mode to activate	Date and Time of Maintenance mode to clear	Click icon to clear Maintenance mode
								Clear
								Clear

Telemetry Live

Zone ▼

Division ▼

Station ▼

Asset Type ▼
DCT

Asset Number ▼
TC-12

View. ▼
Table
Graph
Circuit



NCR	PRYG	Track ckt	TC-12	Present Time
-----	------	-----------	-------	--------------

Parameters as per annexures											

Telemetry History

Zone ▼

Division ▼

Station ▼

Asset Type ▼

Asset No ▼

From date ▼

From time ▼

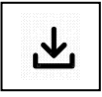
To date ▼

To time ▼

View ▼



NCR	PRYG	Track Circuit	TC-12
-----	------	---------------	-------



Date & Time	Parameters as per annexures											

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RDPMS Health Live

Zone ▼

Division ▼

Station ▼

Asset Type ▼

View ▼

Table

Pictorial

Note: Description of Faulty Sensor/IoT should be given in a way so that it can be identified. It may be given using parameter name for that sensor and asset number for IoT.

Total Sensors	Faulty Sen-sors	Total IoTs	Faulty IoTs	Total Network	Faulty Network	Total Stn gateway	Faulty Stn gateway
500	10	50	2	50	2	2	1

Sr no	Zone	Division	Station	Asset Type	Asset Number	Faulty Sensor description	Faulty IoT description	Faulty Net-work descrip-tion	Faulty Station gateway

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RDPMS Health Summary

Zone ▼

Division ▼

Station ▼

Asset Type ▼

From date ▼

From time ▼

To date ▼

ToTime ▼

View ▼



Table

Pie

Bar

Graph

From

Date

Time

To

Date

Time



Sr no	Zone	Division	Station	Asset Type	Total number of Sensors	% Availability of Sensors	Total number of IoTs	% Availability of IoTs	Total number of Network	% Availability of Network	Total number of Station Gateway	% Availability of station gateway

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Equipment Room Live

Zone ▼

Division ▼

Station ▼

View ▼



Table

Pictorial

Sr no	Zone	Division	Station	Equipment Room Type	Temperature	Humidity	Relay Room opening (Open/Close)
				RR			
				IPS			

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Equipment Room History

Note: Some basis may be worked out to display history of parameters at previous date and time like at each 30 min or for certain change etc.

Zone ▼

Division ▼

Station ▼

Equipment Room ▼

From date ▼

From Time ▼

To Date ▼

To time ▼

View ▼



RR

IPS

BATT

Table

Pie

Bar

Graph

From

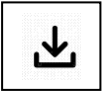
Date

Time

To

Date

Time



Sr no.	Zone	Division	Station	Date & Time	Equipment Room Type	Temperature	Humidity	Relay Room opening (Open/Close)
					RR			
					IPS			

Asset detail

Zone ▼

Division ▼

Station ▼

Asset Type ▼

Asset Make ▼

ALL

XYZ

View ▼

Table

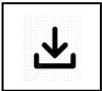
Pie

Bar

Graph



As on “current date”



Sr no.	Zone	Division	Station	Asset Type	Asset make	Count of Asset

Asset utilization

Zone ▼

Division ▼

Station ▼

Asset Type ▼

Asset No ▼

From date ▼

To date ▼

View ▼

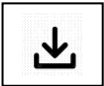


From:

Date

To:

Date



Sr No.	Zone	Division	Station	Asset Type	Asset number	Number of operations

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Performance of RDPMS

Zone ▼

Division ▼

Station ▼

From date ▼

To date ▼

View ▼



Actual Failure entry

Table

Bar

Graph

Pie

From

Date

Time

To

Date

Time



Sr no.	Zone	Division	Station	% of (True + Partially True) Failure Alerts out of total Failure Alerts generated	% of (True + Partially True) Predictive Alerts out of total Predictive Alerts generated	% of (True + Partially True) Failure Alerts by RDPMS out of total actual failures occurred at station

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Annexure F

Standard APIs for common Dashboard

There are three types of reports which are to be generated by common dashboard application (to be used by Railway management). In order to generate these reports, standard APIs (Application Programming Interface) have been designed to access these reports from Application software of various Vendors. Vendors shall incorporate this in their application. Four types of reports have been given below, however, additional reports may added in future:

1. Alert Summary report
2. Alert History report
3. Telemetry History report
4. Asset detail report
5. Performance report

1.(a) Alert Summary Report request structure

Summary Report

Summary Report outlines the count by selecting various fields.

Here is sample report format:

Zone	Division	Station	Vendor Name	Alert Type	Asset Type	Asset Number	Cause of Asset failure	Total Alert Count	True Alert Count	Partially True Alert Count	% of (True +Partially True) Alerts out of total Alert

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URL Structure

```
base_address/vc_alert_summary
```

In the URL address vcc will be replaced by the actual value. Similarly, the base address will be replaced with hostname/address along with the path to the API end-point.

Request Structure

The request is composed of following items:

- Start date – provided in “start_date” key in JSON request payload
- Start time – provided in “start_time” key in JSON request payload
- End date – provided in “end_date” key in JSON request payload
- End time – provided in “end_time” key in JSON request payload
- Zone array – provided in “zone” key enclosed in request object in JSON request payload. zc is passed in array.
- Division array – provided in “division” key enclosed in request object in JSON request payload. dc is passed in array.
- Station array – provided in “station” key enclosed in request object in JSON. sc is passed in array.
- Alert Type array – provided in “alert_type” key enclosed in request object in JSON. alert_type_code is passed in array.
- Asset Type array – provided in “asset_type” key enclosed in request object in JSON. asset_type_code is passed in array.
- Asset Number array – Asset number cannot be provided alone as there may be asset with same name across stations. Hence Asset Number array should be in Key Value pairs where key should be the station code and value should be the asset number. This should also be there enclosed in request object. sc and asset_number_code is passed in array.
- Cause array – provided in “cause” key enclosed in request object in JSON. cause_code is passed in array.
- Page Number – provided in “page_number” key in JSON
- page size count – provided in “page_size” key in JSON
- API Key – to be provided as a part of request header

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Suppose we want to list all stations in NR zone, and cause if Obstruction, then the payload be like:

```
{
  "start_date": "12/11/2023",
  "start_time": "12:40:20",
  "end_date": "20/11/2023",
  "end_time": "13:35:30",
  "request": {
    "zone": [
      "NR"
    ],
    "cause": [
      "OBS"
    ]
  }
}
```

If we don't pass Zone array then the API would assume that all zone data is required.

```
{
  "start_date": "12/11/2023",
  "start_time": "12:40:20",
  "end_date": "20/11/2023",
  "end_time": "13:35:30",
  "request": {
    "cause": [
      "OBS"
    ]
  }
}
```

If we want the data for station from multiple zone/division/section, we can skip the zone, division, section array and pass the station array only.

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```
{
  "start_date": "12/11/2023",
  "start_time": "12:40:20",
  "end_date": "20/11/2023",
  "end_time": "13:35:30",
  "request": {
    "station": [
      "DHN", "MAS", "NDLS", "JAT", "MMCT"
    ]
  }
}
```

Since absence or presence of multiple keys can increase validation complexities, so API should always take the lowest level of the keys and ignore higher level if present. For example, consider following request:

```
{
  "start_date": "12/11/2023",
  "start_time": "12:40:20",
  "end_date": "20/11/2023",
  "end_time": "13:35:30",
  "request": {
    "zone": [
      "NR"
    ],
    "station": [
      "DHN"
    ]
  }
}
```

Now, since DHN station is not part of NR zone and both keys present here (zone and station), the API will ignore higher key and process based on lower key. This behaviour should be only applicable with – zone, division, station and asset number related keys. For alert type, asset type, because it will do a cartesian product or the array with the lowest keys of station hierarchy.

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Here is an example of how to use asset number:

```
{
  "start_date": "12/11/2023",
  "start_time": "12:40:20",
  "end_date": "20/11/2023",
  "end_time": "13:35:30",
  "request": {
    "asset Number": [
      {
        "sc": "DHN",
        "asset_number_code": "P-101"
      }
    ]
  },
  "page_number": 1,
  "page_size": 20
}
```

In above example, if zone or division or station array are also provided, they will be ignored.

Key Details

- start_date** – Compulsory. Represents start date.
- start_time** – Compulsory. Represents start time.
- end_date** – Optional. Represents end date, if not provided then current date is assumed as end date.
- end_time** - Optional. Represents end time, if not provided then current time is assumed as end time.
- request** – Compulsory. This key encapsulates all other keys except the two date keys and two pagination related keys.
- zone** – Optional. Represents selected zones. If not given, it will assume that data is required for all Zones individually.
- division** – Optional. Represents selected divisions. Zones if present with this key will be ignored. If not given, it will assume that data is required for all Divisions individually for given Zones.
- station** – Optional. Represents selected stations. Zones, Divisions, Sections if present with this key will be ignored. If not given, it will assume that data is required for all stations individually for given Divisions & Zones.
- alert_type** – Optional. Represents specific alert type. This key can exist with station hierarchy keys and asset type key. Same cartesian product rule is applied. If this key is present alone, then by default all station across all zones and applicable asset type is assumed.

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- i) **asset_type** – Optional. Represents specific asset type (point machine, track circuit, etc.). This key can exist with station hierarchy keys and will perform cartesian product with lowest key. If this key is present alone then by default all station across all zones will be assumed.
- j) **asset_number** – Optional. Represents specific asset at a given station. If this key is present, zones, divisions, sections, stations, asset type keys are ignored.
- k) **cause** – Optional. Represents specific cause for an alert type. This key can exist with station hierarchy keys and alert type key as well. Same cartesian product rule is applied. If this key is present alone, then by default all station across all zones and applicable asset type, applicable alert type is assumed.
- l) **Total count** – **total_count** of alerts for given cause code (see API below).
- m) **True count** – **true_count** feedback of alerts for given cause code (see API below).
- n) **Partially True count** – **partiallytrue_count** feedback of alerts for given cause code (see API below).
- o) **page** – Optional. Represents the **page_number** in record set. Must be provide with **page_size** key. If absent then complete record will be sent at once.
- p) **Page_size** – Optional. Represents how many records should be there in one page. Must be provided with **page** key.

1.(b) Alert Summary Report response **Sample:** It shall provide count information for fields as per request.

```
{
  "vcc": "XYZ",
  "vcn": "XYZ TECHNOLOGIES",
  "start_date": "1/11/2023",
  "start_time": "13:40:20",
  "end_date": "15/11/2023",
  "end_time": "14:40:20",
  "zone": [
    {
      "zc": "NCR",
      "division": [
        {
          "dc": "PRYJ",
          "station": [
            {
              "sc": "MJA",
              "alert_type": [
                {
                  "alert_type_code": "Failure",
                  "asset_type": [
```

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```

{
  "asset_type_code": "PM",
  "asset_number": [
    {
      "asset_number_code": "PT-101",
      "cause": [
        {
          "cause_code": "PT-OBS"
          "total_count": 8
          "true_count": 2
          "partiallytrue_count": 1
        }
      ]
    }
  ]
}

```

2. (a) Alert history report Request structure:

Detailed Report

Zone	Division	Station	Vendor Name	Alert Type	Asset Type	Asset Number	Alert Status	Cause of Asset failure	Alert Feedback (T/F/PT/M)	incidence Date & Time	Rectification Date& Time ration	Incidence Duration	Alert feedback Date & Time	Maintainer Name	Maintainer Designation	Maintainer Mobile Number	Maintainer Remarks

URL Structure

```
base_address/vc_alert_history
```

In the URL address vcc will be replaced by the actual value. Similarly, the base address will be replaced with hostname/address along with the path to the API end-point.

Request Structure

The request is composed of following items:

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- Start date – provided in “start_date” key in JSON request payload
- Start time – provided in “start_time” key in JSON request payload
- End date – provided in “end_date” key in JSON request payload
- End time – provided in “end_time” key in JSON request payload
- Zone array – provided in “zone” key enclosed in request object in JSON request payload. *zc* is passed in array.
- Division array – provided in “division” key enclosed in request object in JSON request payload. *dc* is passed in array.
- Station array – provided in “station” key enclosed in request object in JSON. *sc* is passed in array.
- Alert Type array – provided in “alert_type” key enclosed in request object in JSON. *alert_type_code* is passed in array.
- Asset Type array – provided in “asset_type” key enclosed in request object in JSON. *asset_type_code* is passed in array.
- Asset Number array – Asset number cannot be provided alone as there may be asset with same name across stations. Hence Asset Number array should be in Key Value pairs where key should be the station code and value should be the asset number. This should also be there enclosed in request object. *sc* and *asset_number_code* is passed in array.
- Cause array – provided in “cause” key enclosed in request object in JSON. *cause_code* is passed in array.
- Alert feedback - given as T, *PT*, M or F
- Page Number – provided in “page_number” key in JSON
- page size count – provided in “page_size” key in JSON
- API Key – to be provided as a part of request header

Suppose we want to list all stations in NR zone, and cause if Obstruction, then the payload be like:

```
{
  "start_date": "12/11/2023",
  "start_time": "12:40:20",
  "end_date": "20/11/2023",
  "end_time": "13:35:30",
  "request": {
    "zone": [
      "NR"
    ],
    "cause": [
```

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```

        "OBS"
    ]
}

```

If we don't pass Zone array then the API would assume that all zone data is required.

```

{
  "start_date": "12/11/2023",
  "start_time": "12:40:20",
  "end_date": "20/11/2023",
  "end_time": "13:35:30",
  "request": {
    "cause": [
      "OBS"
    ]
  }
}

```

If we want the data for station from multiple zone/division/section, we can skip the zone, division, section array and pass the station array only.

```

{
  "start_date": "12/11/2023",
  "start_time": "12:40:20",
  "end_date": "20/11/2023",
  "end_time": "13:35:30",
  "request": {
    "station": [
      "DHN", "MAS", "NDLS", "JAT", "MMCT"
    ]
  }
}

```

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Since absence or presence of multiple keys can increase validation complexities, so API should always take the lowest level of the keys and ignore higher level if present. For example, consider following request:

```
{
  "start_date": "12/11/2023",
  "start_time": "12:40:20",
  "end_date": "20/11/2023",
  "end_time": "13:35:30",
  "request": {
    "zone": [
      "NR"
    ],
    "station": [
      "DHN"
    ]
  }
}
```

Now, since DHN station is not part of NR zone and both keys present here (zone and station), the API will ignore higher key and process based on lower key. This behaviour should be only applicable with – zone, division, section, station and asset number related keys. For alert type, asset type, because it will do a cartesian product or the array with the lowest keys of station hierarchy.

Here is an example of how to use asset number:

```
{
  "start_date": "12/11/2023",
  "start_time": "12:40:20",
  "end_date": "20/11/2023",
  "end_time": "13:35:30",
  "request": {
    "asset Number": [
      {
        "sc": "DHN",
        "asset_number_code": "P-101"
      }
    ]
  }
}
```

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```

    },
    "page_number":1,
    "page_size":20
  }

```

In above example, if zone or division or section or station array are also provided, they will be ignored.

Key Details

- a) start_date – Compulsory. Represents start date.
- b) start_time – Compulsory. Represents start time.
- c) end_date – Optional. Represents end date, if not provided then current date is assumed as end date.
- d) end_time - Optional. Represents end time, if not provided then current time is assumed as end time.
- e) request – Compulsory. This key encapsulates all other keys except the two date keys and two pagination related keys.
- f) zone – Optional. Represents selected zones.
- g) division – Optional. Represents selected divisions. Zones if present with this key will be ignored.
- h) station – Optional. Represents selected stations. Zones, Divisions, Sections if present with this key will be ignored.
- i) alert_type – Optional. Represents specific alert type. This key can exist with station hierarchy keys and asset type key. Same cartesian product rule is applied. If this key is present alone, then by default all station across all zones and applicable asset type is assumed.
- j) asset_type – Optional. Represents specific asset type (point machine, track circuit, etc.). This key can exist with station hierarchy keys and will perform cartesian product with lowest key. If this key is present alone then by default all station across all zones will be assumed.
- k) Asset Number – Optional. Represents specific asset at a given station. If this key is present, zones, divisions, sections, stations, asset type keys are ignored.
- l) cause – Optional. Represents specific cause for an alert type. This key can exist with station hierarchy keys and alert type key as well. Same cartesian product rule is applied. If this key is present alone, then by default all station across all zones and applicable asset type, applicable alert type is assumed.
- m) Alert feedback - given as T, M, PT or F
- n) page – Optional. Represents the page_number in record set. Must be provide with page_size key. If absent then complete record will be sent at once.
- o) Page_size – Optional. Represents how many records should be there in one page. Must be provided with page key.

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2. (b) Alert detailed Report response sample :It shall provide following information for following fields as per request.

alert_status_code
 incidence_date_time
 rectification_date_time
 incidence_duration
 alert_feedback_date_time
 maintainer_name
 maintainer_designation
 maintainer_mobile
 remarks

Sample:

```

{
  "vcc": "XYZ",
  "vcn": "XYZ TECHNOLOGIES",
  "start_date": "1/11/2023",
  "start_time": "13:40:20",
  "end_date": "15/11/2023",
  "end_time": "14:40:20",
  "zone": [
    {
      "zc": "NCR",
      "division": [
        {
          "dc": "PRYJ",
          "station": [
            {
              "sc": "MJA",
              "alert_type": [
                {
                  "alert_type_code": "FLR",
                  "asset_type": [
                    {
                      "asset_type_code": "PM",

```

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]
}

3.(a) Telemetry History Report request structure

Telemetry history Report

Telemetry history Report outlines the parameters of asset by selecting various fields.

Here is sample report format:

Zone	Division	Station	Vendor Name	Asset Type	Asset Number	Date & Time	Parameters as per Annexure			

URL Structure

```
base_address/vc_telemetry_history
```

In the URL address vcc will be replaced by the actual value. Similarly, the base address will be replaced with hostname/address along with the path to the API end-point.

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Request Structure

The request is composed of following items:

- Start date – provided in “start_date” key in JSON request payload
- Start time – provided in “start_time” key in JSON request payload
- End date – provided in “end_date” key in JSON request payload
- End time – provided in “end_time” key in JSON request payload
- Zone array – provided in “zone” key enclosed in request object in JSON request payload. zc is passed in array.
- Division array – provided in “division” key enclosed in request object in JSON request payload. dc is passed in array.
- Station array – provided in “station” key enclosed in request object in JSON. sc is passed in array.
- Asset Type array – provided in “asset_type” key enclosed in request object in JSON. asset_type_code is passed in array.
- Asset Number array – Asset number cannot be provided alone as there may be asset with same name across stations. Hence Asset Number array should be in Key Value pairs where key should be the station code and value should be the asset number. This should also be there enclosed in request object. sc and asset_number_code is passed in array.
- Page Number – provided in “page_number” key in JSON
- page size count – provided in “page_size” key in JSON
- API Key – to be provided as a part of request header

Suppose we want to list all stations in NR zone, then the payload be like:

```
{
  "start_date": "12/11/2023",
  "start_time": "12:40:20",
  "end_date": "20/11/2023",
  "end_time": "13:35:30",
  "request": {
    "zone": [
      "NR"
    ],
  },
}
```

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```
}
```

If we don't pass Zone array then the API would assume that all zone data is required.

```
{
  "start_date": "12/11/2023",
  "start_time": "12:40:20",
  "end_date": "20/11/2023",
  "end_time": "13:35:30",
  "request": {
  }
```

If we want the data for station from multiple zone/division/section, we can skip the zone, division, section array and pass the station array only.

```
{
  "start_date": "12/11/2023",
  "start_time": "12:40:20",
  "end_date": "20/11/2023",
  "end_time": "13:35:30",
  "request": {
    "station": [
      "DHN", "MAS", "NDLS", "JAT", "MMCT"
    ]
  }
}
```

Since absence or presence of multiple keys can increase validation complexities, so API should always take the lowest level of the keys and ignore higher level if present. For example, consider following request:

```
{
  "start_date": "12/11/2023",
  "start_time": "12:40:20",
```

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```
{
  "end_date": "20/11/2023",
  "end_time": "13:35:30",
  "request": {
    "zone": [
      "NR"
    ],
    "station": [
      "DHN"
    ]
  }
}
```

Now, since DHN station is not part of NR zone and both keys present here (zone and station), the API will ignore higher key and process based on lower key. This behaviour should be only applicable with – zone, division, station and asset number related keys. For alert type, asset type, because it will do a cartesian product or the array with the lowest keys of station hierarchy.

Here is an example of how to use asset number:

```
{
  "start_date": "12/11/2023",
  "start_time": "12:40:20",
  "end_date": "20/11/2023",
  "end_time": "13:35:30",
  "request": {
    "asset Number": [
      {
        "sc": "DHN",
        "asset_number_code": "P-101"
      }
    ]
  },
  "page_number": 1,
  "page_size": 20
}
```

In above example, if zone or division or station array are also provided, they will be ignored.

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Key Details

- i) start_date – Compulsory. Represents start date.
- j) start_time – Compulsory. Represents start time.
- k) end_date – Optional. Represents end date, if not provided then current date is assumed as end date.
- l) end_time - Optional. Represents end time, if not provided then current time is assumed as end time.
- m) request – Compulsory. This key encapsulates all other keys except the two date keys and two pagination related keys.
- n) zone – Optional. Represents selected zones. If not given, it will assume that data is required for all Zones individually.
- o) division – Optional. Represents selected divisions. Zones if present with this key will be ignored. If not given, it will assume that data is required for all Divisions individually for given Zones.
- p) station – Optional. Represents selected stations. Zones, Divisions, Sections if present with this key will be ignored. If not given, it will assume that data is required for all stations individually for given Divisions & Zones.
- q) asset_type – Optional. Represents specific asset type (point machine, track circuit, etc.). This key can exist with station hierarchy keys and will perform cartesian product with lowest key. If this key is present alone then by default all station across all zones will be assumed.
- r) asset_number – Optional. Represents specific asset at a given station. If this key is present, zones, divisions, sections, stations, asset type keys are ignored.
- s) page – Optional. Represents the page_number in record set. Must be provide with page_size key. If absent then complete record will be sent at once.
- t) Page_size – Optional. Represents how many records should be there in one page. Must be provided with page key.

3.(b) Telemetry History Report response Sample: It shall provide value of parameters as per desired field as per request.

```
{
  "vcc": "XYZ",
  "vcn": "XYZ TECHNOLOGIES",
  "start_date": "1/11/2023",
  "start_time": "13:40:20",
  "end_date": "15/11/2023",
  "end_time": "14:40:20",
  "zone": [
    {
      "zc": "NCR",
      "division": [
        {
```

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```

    "dc": "PRYJ",
      "station": [
        {
          "sc": "MJA",
          "asset_type": [
            {
              "asset_type_code": "PM",
              "asset_number": [
                {
                  "asset_number_code": "PT-101",
                  "parameter code as per annexure)": ".....",
                  "prv": [... , ... , ...],
                  "prt": [... , ... , ...]
                }
              ]
            }
          ]
        }
      ]
    }
  ]
}

```

4. (a) Asset report request structure:

Asset Report

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Zone	Division	Station	Vendor Name	Asset Type	Make of Asset	Count

URL Structure

```
base_address/vc_asset_detail
```

In the URL address vcc will be replaced by the actual value. Similarly, the base address will be replaced with hostname/address along with the path to the API end-point.

Request Structure

The request is composed of following items:

- Zone array – provided in “zone” key enclosed in request object in JSON request payload. zc is passed in array.
- Division array – provided in “division” key enclosed in request object in JSON request payload. dc is passed in array.
- Station array – provided in “station” key enclosed in request object in JSON. sc is passed in array.
- Asset Type array – provided in “asset_type” key enclosed in request object in JSON. asset_type_code is passed in array.
- Page Number – provided in “page_number” key in JSON
- page size count – provided in “page_size” key in JSON
- API Key – to be provided as a part of request header

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If we want the data for station from multiple zone/division/section, we can skip the zone, division, section array and pass the station array only.

```
{
  "request": {
    "station": [
      "DHN", "MAS", "NDLS", "JAT", "MMCT"
    ]
  }
}
```

Since absence or presence of multiple keys can increase validation complexities, so API should always take the lowest level of the keys and ignore higher level if present. For example, consider following request:

```
{
  "request": {
    "zone": [
      "NR"
    ],
    "station": [
      "DHN"
    ]
  }
}
```

Now, since DHN station is not part of NR zone and both keys present here (zone and station), the API will ignore higher key and process based on lower key.

In above example, if zone or division or section or station array are also provided, they will be ignored.

Key Details

- request – Compulsory. This key encapsulates all other keys except the two date keys and two pagination related keys.
- zone – Optional. Represents selected zones.
- division – Optional. Represents selected divisions. Zones if present with this key will be ignored.

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- d) station – Optional. Represents selected stations. Zones, Divisions, Sections if present with this key will be ignored.
- e) asset_type – Optional. Represents specific asset type (point machine, track circuit, etc.). This key can exist with station hierarchy keys and will perform cartesian product with lowest key. If this key is present alone then by default all station across all zones will be assumed.
- f) page – Optional. Represents the page_number in record set. Must be provide with page_size key. If absent then complete record will be sent at once.
- g) Page_size – Optional. Represents how many records should be there in one page. Must be provided with page key.

4. (b) Asset Report response Sample: It shall provide make and count of each asset type for fields as per request as on date status.

```
{
  "vcc": "BT",
  "vnapp": "XYZ TECHNOLOGIES",
  "zone": [
    {
      "zc": "NCR",
      "division": [
        {
          "dc": "PRYJ",
          "station": [
            {
              "sc": "MJA",
              "asset_type": [
                {
                  "asset_type_code": "M-Sig",
                  "asset_make": [
                    {
                      "make": "wbg",
                      "total_count": 10
                    },
                    {
                      "make": "xvz",
                      "total_count": 10
                    }
                  ]
                }
              ]
            }
          ]
        }
      ]
    }
  ]
}
```

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```
    ]
    }
  ]
}
]
```

5.(a) Performance Report request structure

Performance Report

Performance Report outlines the performance parameters.

Here is sample report format:

Zone	Division	Station	Vendor Name	% of (True + Partially True) Failure Alerts out of total Failure Alerts generated	% of (True + Partially True) Predictive Alerts out of total Predictive Alerts gener-ated	% of (True + Partially True) Failure Alerts by RDPMS out of total actual failures oc-curred at sta-tion

URL Structure

base_address/vc_performance

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In the URL address vcc will be replaced by the actual value. Similarly, the base address will be replaced with hostname/address along with the path to the API end-point.

Request Structure

The request is composed of following items:

- Start date – provided in “start_date” key in JSON request payload
- Start time – provided in “start_time” key in JSON request payload
- End date – provided in “end_date” key in JSON request payload
- End time – provided in “end_time” key in JSON request payload
- Zone array – provided in “zone” key enclosed in request object in JSON request payload. zc is passed in array.
- Division array – provided in “division” key enclosed in request object in JSON request payload. dc is passed in array.
- Station array – provided in “station” key enclosed in request object in JSON. sc is passed in array.
- Page Number – provided in “page_number” key in JSON
- page size count – provided in “page_size” key in JSON
- API Key – to be provided as a part of request header

Suppose we want to list all stations in NR zone, then the payload be like:

```
{
  "start_date": "12/11/2023",
  "start_time": "12:40:20",
  "end_date": "20/11/2023",
  "end_time": "13:35:30",
  "request": {
    "zone": [
      "NR"
    ],
  },
}
```

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```
}
}
```

If we don't pass Zone array then the API would assume that all zone data is required.

```
{
  "start_date": "12/11/2023",
  "start_time": "12:40:20",
  "end_date": "20/11/2023",
  "end_time": "13:35:30",
  "request": {
  }
}
```

If we want the data for station from multiple zone/division/section, we can skip the zone, division, section array and pass the station array only.

```
{
  "start_date": "12/11/2023",
  "start_time": "12:40:20",
  "end_date": "20/11/2023",
  "end_time": "13:35:30",
  "request": {
    "station": [
      "DHN", "MAS", "NDLS", "JAT", "MMCT"
    ]
  }
}
```

Since absence or presence of multiple keys can increase validation complexities, so API should always take the lowest level of the keys and ignore higher level if present. For example, consider following request:

```
{
  "start_date": "12/11/2023",
```

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```

"start_time": "12:40:20",
"end_date": "20/11/2023",
"end_time": "13:35:30",
"request": {
  "zone": [
    "NR"
  ],
  "station": [
    "DHN"
  ]
}
}

```

Now, since DHN station is not part of NR zone and both keys present here (zone and station), the API will ignore higher key and process based on lower key. This behaviour should be only applicable with – zone, division, station.

Here is an example of how to use equipment room:

In above example, if zone or division or station array are also provided, they will be ignored.

Key Details

- u) start_date – Compulsory. Represents start date.
- v) start_time – Compulsory. Represents start time.
- w) end_date – Optional. Represents end date, if not provided then current date is assumed as end date.
- x) end_time - Optional. Represents end time, if not provided then current time is assumed as end time.
- y) request – Compulsory. This key encapsulates all other keys except the two date keys and two pagination related keys.
- z) zone – Optional. Represents selected zones. If not given, it will assume that data is required for all Zones individually.
- aa) division – Optional. Represents selected divisions. Zones if present with this key will be ignored. If not given, it will assume that data is required for all Divisions individually for given Zones.
- bb) station – Optional. Represents selected stations. Zones, Divisions, Sections if present with this key will be ignored. If not given, it will assume that data is required for all stations individually for given Divisions & Zones.
- cc) page – Optional. Represents the page_number in record set. Must be provide with page_size key. If absent then complete record will be sent at once.
- dd) Page_size – Optional. Represents how many records should be there in one page. Must be provided with page key.

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5.(b) Performance Report response Sample: It shall provide value of parameters as per desired field as per request.

```
{
  "vcc": "XYZ",
  "vcn": "XYZ TECHNOLOGIES",
  "start_date": "1/11/2023",
  "start_time": "13:40:20",
  "end_date": "15/11/2023",
  "end_time": "14:40:20",
  "zone": [
    {
      "zc": "NCR",
      "division": [
        {
          "dc": "PRYJ",
          "station": [
            {
              "sc": "MJA",
              "fail_alert_per": "75.78",
              "pred_alert_per": "77.60",
              "actual_fail_alert_per": "79.60"
            }
          ]
        }
      ]
    }
  ]
}
```

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Annexure G

Standard User Interface of RDPMS Application for mobile

Note:

1. Para 1 to 12 of Annexure E applies to Mobile App also in principal.
2. Backend admin shall allow Div/Zone icon to staff as per their role and jurisdiction.
3. Sub functions which have not been shown here for UI template, may be designed by Vendors in similar fashion.

Landing page

Remote Diagnostic and Predictive Maintenance System
vendor name

user id

xyz

Password

Otp based

Sign in for New User

Dashboard

Zone ▼

Division ▼

Station ▼

🔍

Others ▼

Admin

User

Perform

PT-M/C
Normal - 50
Failed - 5
Predicted - 10

Track Circuit
Normal - 40
Failed- 4
Predicted- 7

Signal
Normal - 45
Failed - 6
Predicted -10

Axle-Counter
Normal - 55
Failed -8
Predicted - 10

LC-Gate
Normal - 10
Failed- 1
Predicted- 2

Other Gears
Normal - 30
Failed- 2
Predicted- 5

Sensors
Normal - 55
Failed -8

IoT
Normal - 55
Failed -8

Alert Hist

Alert Live

Dashboard

Alert ▼

Telemetry ▼

RDPMS health ▼

Equipment R ▼

Alert Live

user ▼

Zone ▼

Div ▼

Stns ▼

Alert Type ▼

Asset Type ▼

Normal - 50
Predictive- 10
Failure- 5

PRYG asset_number_code

02-12-2022 19:34

cause_code.....

Ack

T

PT

F

M

Remark

Failure

PRYG asset_number_code

02-12-2023 18:36

cause_code.....

Ack

T

PT

F

M

Remark

Predictive

Dashboard

Alert ▼

Telemetry ▼

RDPMS health ▼

Equipment R ▼

Maintenance Mode

Zone ▼

Div ▼

Stns ▼

Asset Type ▼

Asset no ▼

From date ▼

From Time ▼

To date ▼

To Time ▼

Zone
XYZ
Division
ABC

Station	Asset Type	Asset Number	Date & time to activate	Date & time to clear	Click to clear
					Clear
					Clear

Dashboard

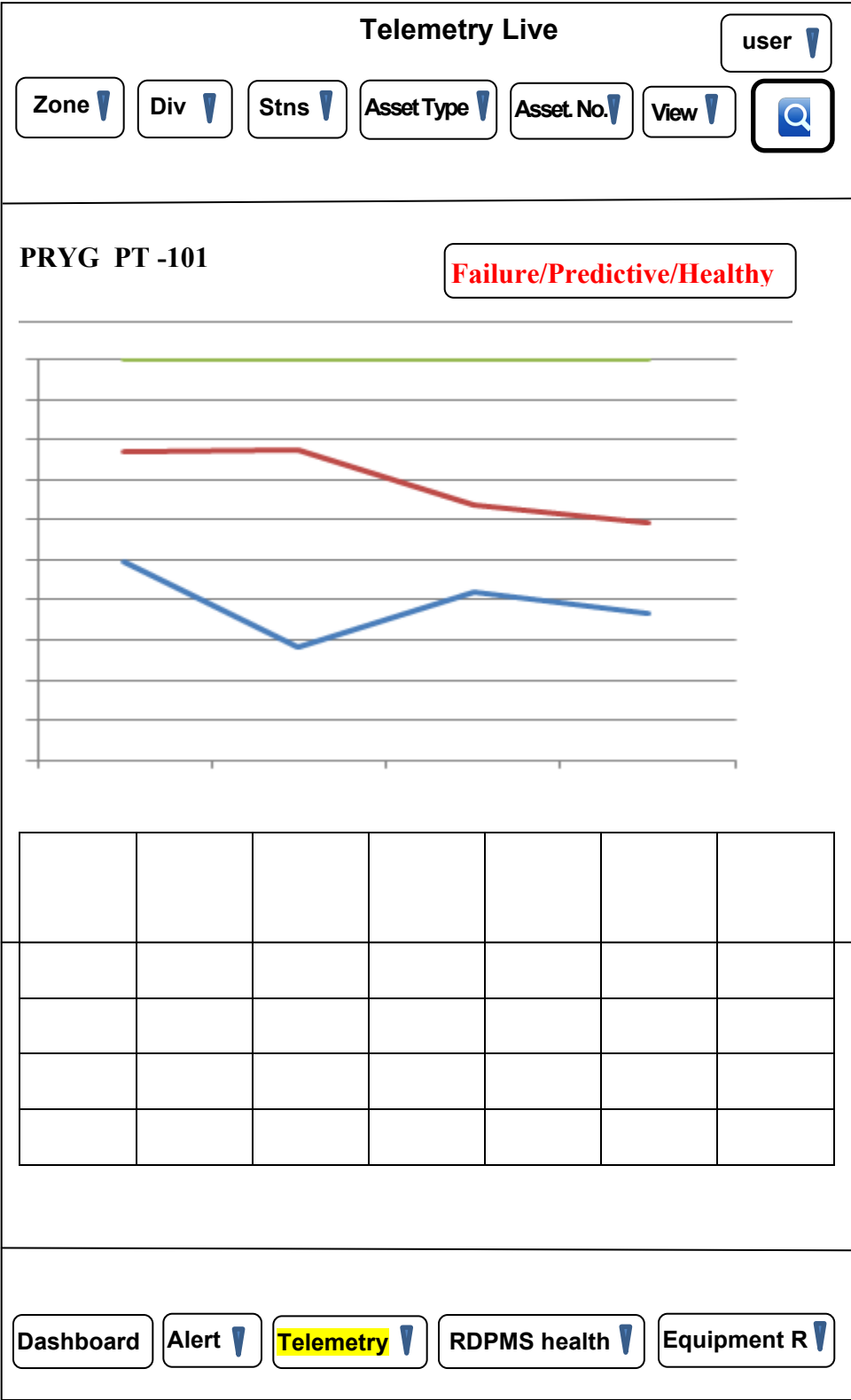
Alert ▼

Telemetry ▼

RDPMS health ▼

Equipment R ▼

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RDPMS Health Live

user ▼

Zone ▼

Div ▼

Stns ▼

Asset Type ▼

View ▼

Table

Pic

Zone	XYZ	Division	ABC
------	-----	----------	-----

Total sen- sors	500	Total IoTs	50	Total Net- work	50	Total Stn gateway	2
Faulty sen- sors	20	Faulty IoTs	2	Faulty Network	2	Faulty Stn gateway	1

Station	Asset Type	Asset Number	Faulty Sensor	Faulty IoT	Faulty Network	Faulty Stn Gateway

Dashboard

Alert ▼

Telemetry ▼

RDPMS health ▼

Equipment R ▼

Equipment Room Live

User

Zone

Div

Stns

View

Table

Pic

Zone	XYZ	Division	ABC
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Station	Equipment Room	Temp	Humdity	Open/ Close

Dashboard

Alert

Telemetry

RDPMS health

Equipment R

Performance

user

Perform

Zone

Div

Stns

From date

To date

View

Zone	XYZ	Division	ABC
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Sta-tion	% of (True + Partially True) Failure Alerts out of total Failure Alerts generated	% of (True + Partially True) Predictive Alerts out of total Predictive Alerts generated	% of (True + Partially True) Failure Alerts by RDPMS out of total actual failures occurred at station

Dashboard

Alert

Telemetry

RDPMS health

Equipment R